



République Tunisienne
Ministère de l'Enseignement Supérieur et de la Recherche Scientifique
Université de Tunis El Manar
Ecole Nationale d'Ingénieurs de Tunis



ELECTRICAL ENGINEERING DEPARTMENT

End of Studies Project

Presented by

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For obtaining

National Research Master's Degree in Automatics

Design of a Pneumatic Vehicle

Project's Host

AKER Services

Project's defense

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Academic Year 2019 - 2020

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Abstract

This work describes the project realized for obtaining National Research Master's Degree in Automatics.

This project's objective is designing and realizing a Compressed-Air Vehicle as an alternative to the use of fossil fuels in transportation.

Along this work, the needed background, in terms of pneumatics, mechanics and electricity, to design such a vehicle are presented followed by the steps taken in the realization, to end with perspectives to work on in future upgrades.

Key Words : Pneumatics, Mechanics, Compressed-Air, Transport, Automobile Engineering, Electrical Engineering, Compressed-Air Vehicle.

Résumé

Ce rapport décrit les démarches réalisées, dans le cadre d'un projet, pour l'obtention du Diplôme National de Mastère de Recherche en Automatique.

L'objectif de ce projet est de concevoir et de réaliser un véhicule à air comprimé comme alternative à l'utilisation de combustibles fossiles dans le transport.

Tout au long de ce travail, les bases nécessaires, en termes de pneumatique, de mécanique et d'électricité, pour concevoir un tel véhicule sont présentées, suivies des étapes de la réalisation, pour finir par les perspectives sur lesquelles des améliorations auront lieu dans de futures prototypes.

Mots clés : Pneumatique, mécanique, air comprimé, transport, construction automobile, génie électrique, véhicule à air comprimé.

摘要

这项工作描述了为获得国家自动研究硕士学位而实现的项目。

气动、机械、压缩空气、运输、汽车工程、电气工程、压缩空气汽车。

在这项工作中，介绍了设计此类车辆所需的气动，机械和电力方面的背景知识，随后介绍了实现过程中要采取的步骤，最后提出了未来升级工作的观点。

关键词：气动、机械、压缩空气、运输、汽车工程、压缩空气汽车。

Dedication

I dedicate this work to

My parents, Sami HAMMAMI and Haifa Ben AICHA, whose words of encouragement and push for tenacity still ring in my ears and for supporting me with whatever help they could offer. God Bless You

My sister, Isra HAMMAMI, whom I appreciate her stand beside me up and for always cheering me. May God guide you within the righteous path.

My brother Souraw HAMMAMI, my source of inspiration whom I owe becoming the person I am today. I hope you get well.

My Friends, Firas KHLIF, Oussema BEDOUI and Luigi TUCCI, for supporting me and the enormous help they offered during the realization of this work.

All the Tunisian Red Crescent's volunteers, to whom I offer my sincere gratitude.

My Supervisor and Teacher, M. Férid KOURDA, for pushing me and being there for me in time of need, whom I considered more of a parent to me than a teacher.

Rouslen HAMMAMI ATTOUCHI

Acknowledgment

First, I would thank God, who gave me the power to realize such a humble work.

I offer my sincere gratitude for AKER Services, specially, M. Sofiane KOURDA, for being the host of this project, for guiding me throughout this period and for financing this realization. May this work be a testimony to my highest consideration.

I express my gratitude to M. Férid KOURDA, who always managed to find time to help me within this project, for correcting my mistakes and for supporting me morally. May he find here the expression of my high consideration.

I thank particularly, MECANORM and the Local Committee of the Tunisian Red Crescent in Menzel Bouzelfa, for opening their doors and helping the realization of this project within their work-spaces.

My sincere gratitude for my Friends, Firas KHLIF, Oussema BEDOUI and Lugui TUCCI, for their help in the realization of the project.

Finally, I would like to thank whoever contributed to this work.

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Acronyms

CAV Compressed-Air Vehicle
IEA International Energy Agency
TFC Total Final Consumption
MDI Motor Development International
RATP Régie autonome des transports parisiens
PLC Programmable Logic Controller
IT Information Technology
ANR Atmosphère Normalisée de Référence
TND Tunisian Dinar

Glossary

CATIA computer-aided three-dimensional interactive application FOR computer-aided engineering
Continuum a continuous nonspatial whole or extent or succession in which no part or portion is distinct or distinguishable from adjacent parts
MATLAB (matrix laboratory), a proprietary multi-paradigm programming language and numeric computing environment developed by MathWorks
SMC eTOOLS online simulator developed by Sintered Metal Corporation
Tinkercad free, online 3D modeling program

Units

rpm Revolutions per minute (Rotational Speed)
W Watt (Power)
Wh Watt-hour (Energy = 3600 Joules)
Mt of CO₂ MegaTon of Carbon dioxide
ktoe kiloTon of oil equivalent (Energy = 11630 Wh)
m/s Meters per Second (Speed)
m Meters (Length)
m² Square meter (Area)
m³ Cubic meter (Volume)
bar, Pa, N/m², psi Pressure 1 bar = 100000 Pa = 14,5038 psi = 100000 N/m²
K, °C Kelvin, Celsius (Temperature 273,15 K = 0°C)
kg kilogram (Mass) **kg/m³** kilogram per cubic metre (Density)
N Newton (Force)
N.m Newton meter (Torque)

General Introduction

As the world is pressed by the energy and fuel crises and compounded by pollution, any technologies that bring out some solutions to these problems is considered as a bounty.

In fact, the transportation sector faces some great challenges, including climate impacts of greenhouse gas emissions, local health impacts of criteria pollutants, and political as well as economic impacts of petroleum dependence as most of the energy utilized in the vehicles for the transportation purposes comes from burning of fossil fuels such as natural gas, diesel, and petrol etc. This kind of combustion releases harmful gases hazardous for the environment and for the health of the human beings.

Due to hyper motorization in the developing world, these problems surely will be magnified sooner or later. Although several evolutionary solutions are being developed to reduce these impacts such as increased fuel economy standards and accelerated adoption of hybrid vehicles, revolutionizing new concept might be crucial.

Compressed-Air vehicles, also known as air cars, in which a Pneumatic motor is powered by compressed air stored in an on board pressurized tank, might server as one of the new approaches.

It is claimed that CAVs offer a greener yet cheaper alternative to operate since there is no emission of pollutants or consumption of hydrocarbons when traveling, but indirectly during the production of compressed air, and produce zero tailpipe emissions while they might reach the power and performance needed for light-duty vehicle use.

This approach is not fully green as filling the CAVs air containers needs electricity where fossil fuels might be used to produce, yet it is still greener.

This work, throughout its chapters, represents one way to realize a CAV. The first chapter presents the problematic through statics for which, CAVs are being produced, followed by a brief history of this type of vehicles and its state of the art upgrades, finishing by the advantages and disadvantages for their use.

The second chapter represents the bibliographical study for this work. It should contain all the explication, mathematical and physical background used for the design and the realization later.

The third chapter is dedicated to the electrical part of the CAV, regarding its power autonomy, source, and all electrical the vehicle functionalities.

This work ends with the last chapter containing all the design and realisation of the vehicle as well as different approaches and problems encountered, finishing with displaying the capabilities of the designed vehicle, its limits and perspective ideas to be working on in the next upgrades.

Chapter 1

Project Study

1.1 Introduction

Pneumatics is the discipline that applies the principles of mechanical properties of gases such as pressure and density to use compressed gas, the most widely is air, as a source of power to overcome engineering problems. As mainly air is has been used, its use has become synonymous with the term pneumatics^[1].

The use of air as an energy transfer medium can be traced back more than 2000 years, and the varying areas of application reflect the changes in technology since then. The industrial use on a larger scale began 1888 when a 1,500-kW central compressor station was installed in Paris to supply the city with compressed air [17].

As electricity stated evolving, this form of energy became obsolete. Around the 1900, the most often used pneumatic components were pneumatic hammers. As technology kept evolving, riveting has been replaced by welding and pneumatic hammers are mainly to be seen in construction sites. As adequate materials for valve and piston seals became available, the demand of automation in industrial production has risen, therefore, around 1950, humanity has seen the first application of pneumatics in this field.

As automation requires some control logic. In the 1960s, fluidic elements were developed to give Boolean AND or OR functions to be the pneumatic equivalent of electronic control [26] [5]. Nowadays, valve logic^[2] is mostly restricted to machines operating in explosive environments. Typically, programmable logic controller, PLC, with magnetic sensors for piston position and electrically operated valves does the adequate work.

Today, the simple conversion of pressure to force and translational displacement. These actuators are simple to design, can be quite fast and do not overheat even. Its forces can be controlled by a pressure regulator.

¹the term used for liquids is Hydraulics

²Control logic when applied to pneumatics

1.2 Project's Host : AKER Services

The project's host is an energy and IT services company. AKER Services was founded in January 2013 and based at 18bis, Ibn Khaldoun Street, 1001 Tunis.

In terms of energy, AKER Services offers energy audit, energy demand management, the development of renewable energies and the exploitation of lost energy. From an IT point of view, AKER Services offers the development of all types of web and mobile applications as well as the design of smartphones and tablets.

AKER services is divided into several departments, including the energy services department, called AKER ENERGIE abbreviated, AKER NRJ. AKER NRJ offers services dedicated to energy savings and renewable and renewed energies. In the field of eco-efficiency, AKER NRJ is able to identify, design and implement energy upgrading solutions without impacting the process of its customers. In this sense, AKER NRJ offers various services such as: feasibility study of projects of renewable origin, sizing of equipment, estimation of the deliverable, assistance on technical choices, drawing up of specifications, the provision of information on the various grants and financial aid, assistance with project management and coordination with the different actors, etc.

AKER Services is formed by a team of four people including Mr. Sofiane KOURDA, the Director General of AKER Services and responsible for the development of the activities of AKER NRJ, Mr. Sofiane EL KAROUI member of the Board of Directors and responsible for the development of the activities of AKER IT, Mr. Mehdi EL KAROUI is a member of the Board of Directors of AKER and responsible for monitoring operations and Ms. Khouloud BACHROUCH AYARI is the hierarchical leader of the operational teams.

This team has established within AKER Services various work values mainly trust, respect and commitment as well as innovation and partnership.



Figure 1.1: Logo of AKER Services

1.3 Presentation of the Project

1.3.1 Context

In this race against time towards limiting the effects of the enormous use of fossil fuels, many alternatives that have been offered to the public are considered to be a major change in the world. In this section and to put the reader in the context of this project, some statistics about the CO₂ emission in Tunisia are presented.

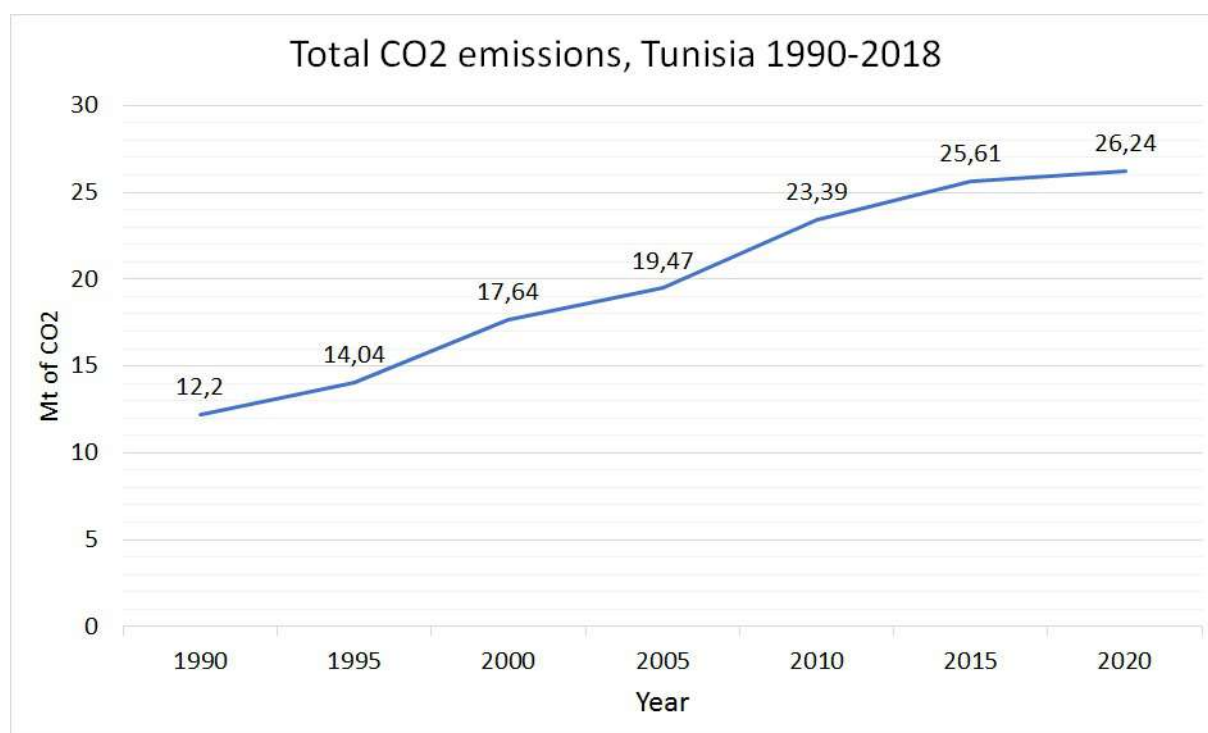


Figure 1.2: Total CO₂ emission in Tunisia between 1990 and 2018 (in Mt of CO₂) [6]

According to this research, Tunisia alone contributes with 26.2 Mt CO₂ [6], which is the gas released after burning fossil fuels to harvest its energy, compared to 12 Mt of CO₂ in 1990. It's obvious that emissions are skyrocketing rapidly meaning the demand of energy has been met with more fossil fuel use.

The International Energy Agency³ (IEA) has shared other statistics peeling this CO₂ emissions value into its contributing sectors.

From the figure below, by comparing the values between 1990 and 2018, the transport sector's emission seem to be increasing to surpass those of the industry.

³Established in 1974 as an autonomous intergovernmental organisation, dedicated to responding to physical disruptions in the supply of oil and offer information about the energy market

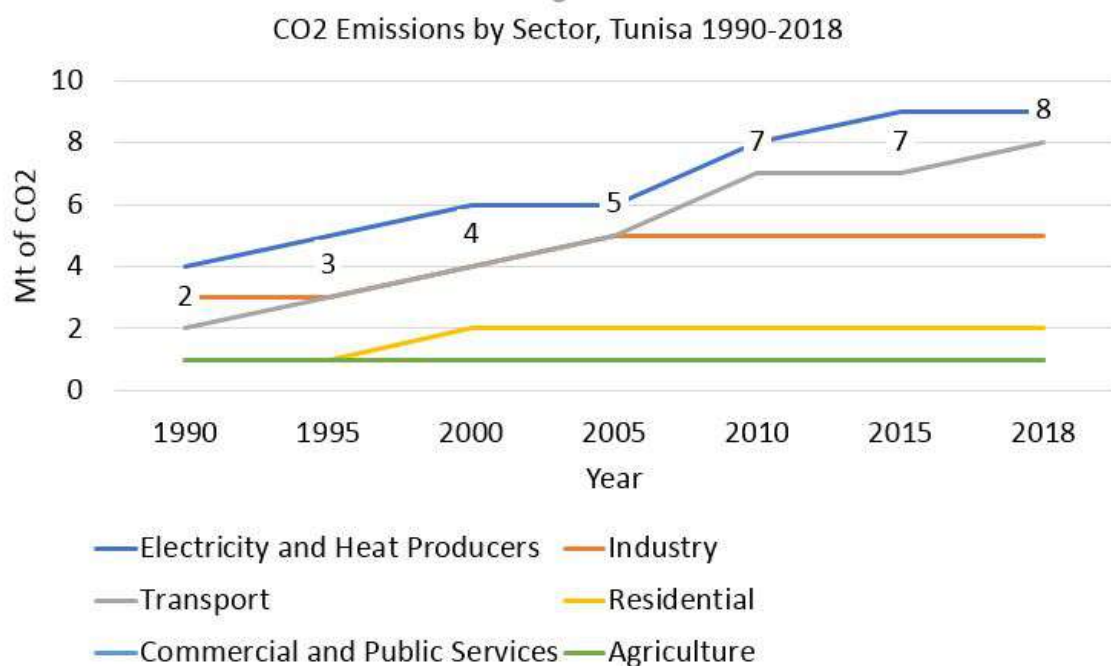


Figure 1.3: CO2 emissions by sector in Tunisia between 1990 and 2018 (Mt of CO2) [6]

Assuming there will be no law put to limit these emissions, the transport sector will be the first contributor to the release of these gases.

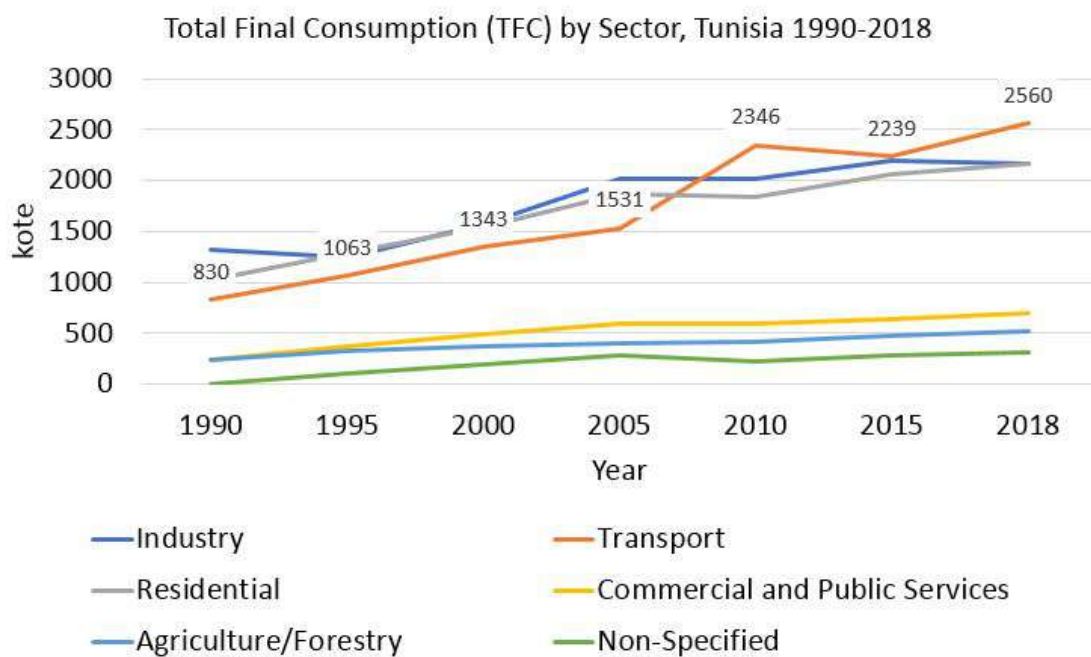


Figure 1.4: Total Final Consumption by Sector in Tunisia between 1990 2018 (ktoe) [6]

This conclusion is referenced to the figure above showing that the energy consumption

Chapter 1. Project Study

is really higher than the other sectors when it comes to transport (in kt_{oe}⁴).

The CO₂ emissions coming from the electricity and heat producers are decelerating due to Tunisia's policies and encouragements towards the use of green energy for such use e.g Solar and Wind power.

Unfortunately, a small number of policies were put to use when it comes to transport. In fact, Tunisia has only 7 charging stations for electrical cars placing it in the list of the countries least concerned about this problem, even though, on a global scale, many other alternatives were taken in consideration and being commercialized, also known as alternative fuel vehicles.



Figure 1.5: Charging Stations in Tunisia (2021 Update)^[2]

To put Tunisia's contribution to the global CO₂ to perspective, the IEA offers different data for several countries. This figure represents a comparison between the contributions of the United States of America, Europe, China and Tunisia between 1990 and 2019.

⁴kilo tons of oil equivalent, equal to 11630 kwh

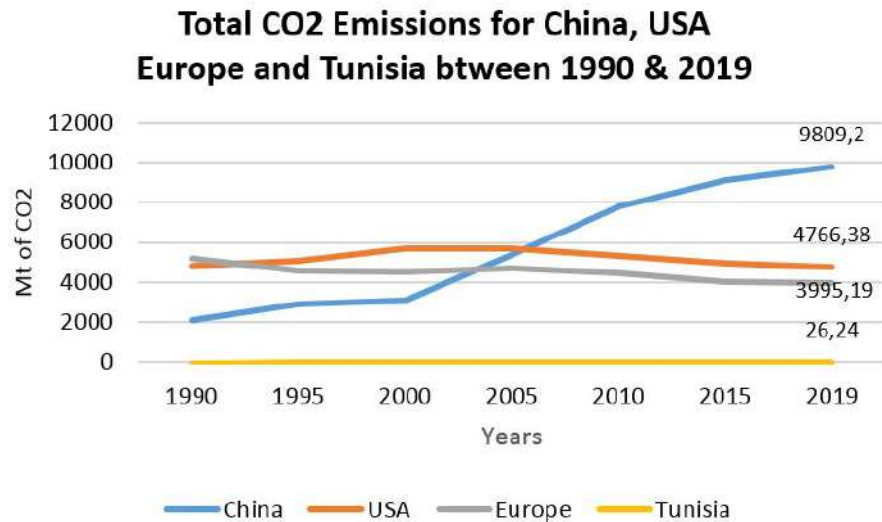


Figure 1.6: Total CO2 Emissions for China, USA, Europe and Tunisia between 1990 and 2019

From this figure, Tunisia's emissions are negligible when compared to China, The USA and Europe.

If 26.4 Mt of CO2 looks enormous as it is, it surely looks nothing like the 9809.2 Mt of CO2 emitted by China alone, that's why alternative solutions are to be taken seriously.

1.3.2 Existing Solutions

Many factors contributed to the development of these alternatives and has become a major priority for many governments. In fact, some statistics have shown that there are 116 million alternative fuel and advanced technology vehicles have been sold worldwide by the end of 2016, underlining the demand for such a solution. There are many alternatives varying from the use of less pollutant materials to completely other kind of technologies.

1. **Electric vehicles** : Electric vehicles can be divided into 3 subcategories :

- Electric, fed by external source ; like the light metro of Tunis



Figure 1.7: Light Metro of Tunis

- Battery-electric ; whose main energy storage is in the chemical energy of batteries. This kind is the most commercialized type.



Figure 1.8: Tesla Roadster (Electrical Car)

- Solar ; using a solar panel. Unfortunately, solar panels alone are not sufficient enough to power a full vehicle so this type is used mainly to extend the range of electrical vehicles.

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2. **Hydrogen Vehicles** : Hydrogen Vehicles use hydrogen as their primary source of energy.



Figure 1.9: Audi A7 h-tron quattro (Hydrogen Car)

3. **Hybrid Vehicles** : The term Hybrid refers to the use of two different types of energy, Electrical and old school fossil based fuel.



Figure 1.10: Toyota Prius (Hybrid Car)

4. **Compressed Air Vehicles** : also known as CAV. Explanation in the next section.

1.4 Compressed Air Vehicles

In the past, air Locomotives played a major role in transportation due to its obvious advantages: simplicity, safety, economy and hygiene. Air engines were first designed during the 1880s and 1890s by inventors and advocates of air vehicles such as General Hermann Haupt⁵.

In New York City, a 1,500 horsepower building-sized pressure station was installed to use the transit locomotives tested on daily roads. In addition, New York City hosted the first installation of pressure station with about 1,500 horsepower for urban transit via air locomotives. The history also noted the use of air locomotives in mining generally manufactured by the same vehicles responsible for the accomplishment already mentioned. Until the 1930s and 1940s, air vehicles surpassed both electric and combustion engines in this field due to fact that mining prohibits using engine that can cause sparks or heat. .

1.4.1 History of CAV [1] [27]

None can claim that the use of compressed air as an alternative energy is a new technology. Records have shown that Denis Papin⁶ was the first ever to come up with such an idea in 1687 . However, it is with the end of the 19th century that the first schemes of such a locomotive were applied in reality, as in 1872, the Mekarski⁷ air engine was used for urban transportation with a simple single stage engine.

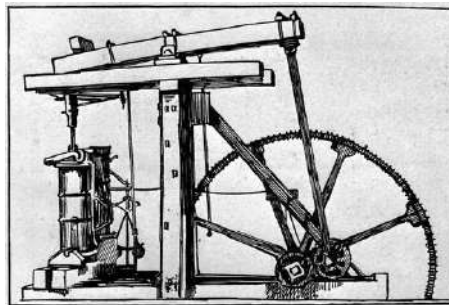


Figure 1.11: Denis Papin's Invention

This event was a breakthrough in the world of pneumatics. Mekarski used basic thermodynamics' laws by heating up the air in pipes using boiling water to apply his idea.

After that, the Mekarski air engine was put to use in urban transit. A single-stage engine that represented an advance in air engine technology that made air cars feasible: after leaving the tank and before entering the engine, the air was reheated through the use of a hot water tank where compressed air picks up hot water vapor which improved

⁵1817-1905, American civil engineer and railroad construction engineer.

⁶1647-1713, French physicist, mathematician and inventor, known for working on the first steam engines

⁷1843-1923, French engineer and inventor of the compressed-air tram

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the engine's range and make one fill up last longer than non heated compressed air, as shown in the figure.

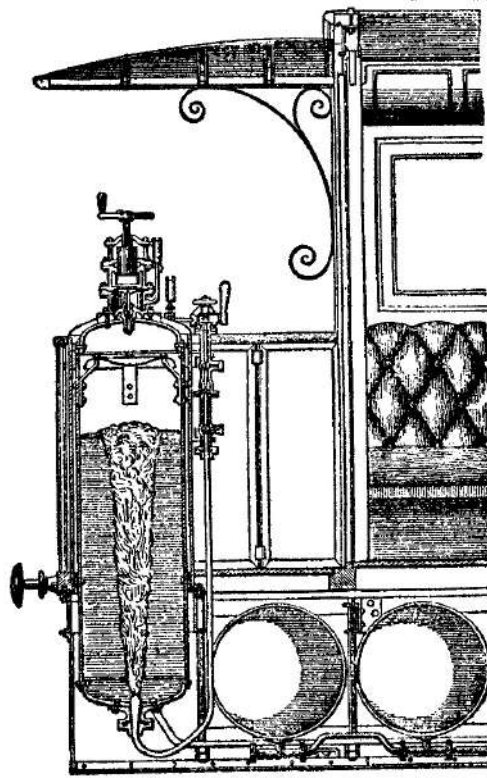


Figure 1.12: Meksarski Street Transit

Moreover, another compressed air locomotive was designed by Robert Hardie⁸ in the late 1900s, and used in street transit in New York City. Air car advocate General Herman Haupt, a civil engineer, has published a lot of work involving this domain especially on the advantages of the use of such an energy, based on Hardie's work, to gain New York's support for the implementation and use of air locomotives for urban transit.

The engine was a like the Meksarski engine, but use a more sophisticated and more adequate re-heater. The new feature of this locomotive was using the engine as an air compressor during deceleration therefore, the vehicle gained more air an heat to be stored in the tank giving a chance for the vehicle to longer its fill-up ranges.

A 1500 horsepower steam air compressor station was realized in New York City to apply the Hardie compressed air locomotives as well as the Hoadley-Knight pneumatic locomotives.

⁸1865-1949, Canadian physicist

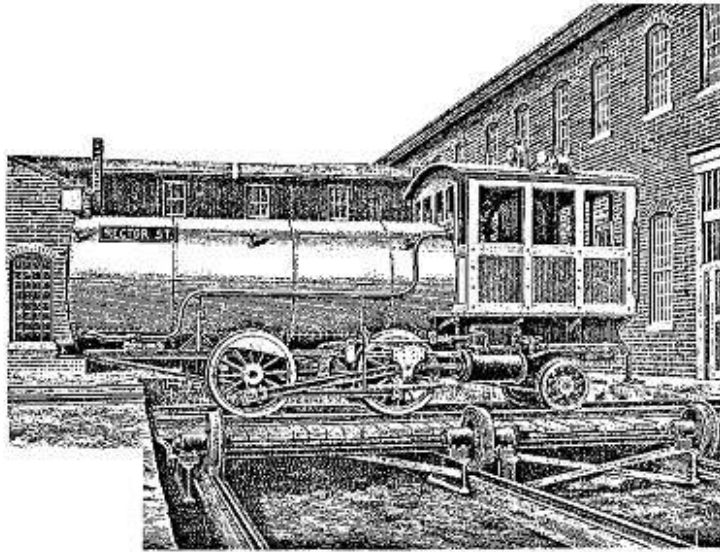


Figure 1.13: The Hardie Compressed Air Locomotive

By the beginning of the 20th century, pneumatic vehicles entered a new era. Humanity has seen the dawn of air cars commercialization which its credits go to Charles B. Hodges, first and only air car inventor to live the air vehicle's commercial prosperity. With a two-stage engine, an in-between piston re-heater, this design was a great success.

A substantial gain in the range between full strokes was thus achievable without cost for the additional energy, which was provided by the sun. The design was so clever that the actual realization achieved a significant gain for its range between fill-ups using the sun's energy. The compressed air locomotive was used a lot in mines around the world because it air would not create neither heat nor sparks protecting from the explosion hazards that electric and combustion engines could lead to in this environment. Therefore, The HK Porter Company of Pittsburgh⁹ sold hundreds of these locomotives to mining companies in the eastern U.S.A.

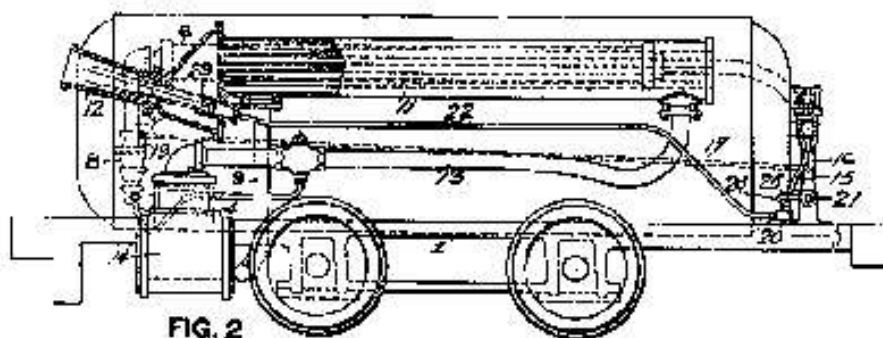


Figure 1.14: H. K. Porter Compound Air Locomotive

⁹1866, american Railroad Company

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Later, European Engineers referred to Charles B.'s design to add more stages to the engine and used inter-heaters. An other solution was created, is letting the locomotives absorb heat in mines transferring it to the air engine which allows the range between fill-ups to rise about 60 percent. That's why in countries like France, Belgium and Germany used only air locomotives in their mines.

In 1930, on a 1200 horsepower full-size locomotive had been developed in Germany.

The locomotive used a diesel compressor on-board which filled its tank to let the air engine drive the vehicle. The heated produced by the compressor was also put to use by transferring it to the air engine increasing its capacity 26 per cent.

However, after world war two air engines where not used again due to the excessive use of petroleum fuel for economic reasons. Researches in this field were forced to stop and are generally concluded by the inefficiency of this energy.

In 1932, the physicist Robert C. Burt PhD designed " The Burt Air Drive " used a gas engine compressor with 4 cylinders, re-compressing the exhaust air so it was both hybrid and closed-cycle. Unfortunately, this modal was forced to extinct for no obvious reasons in 1980s.

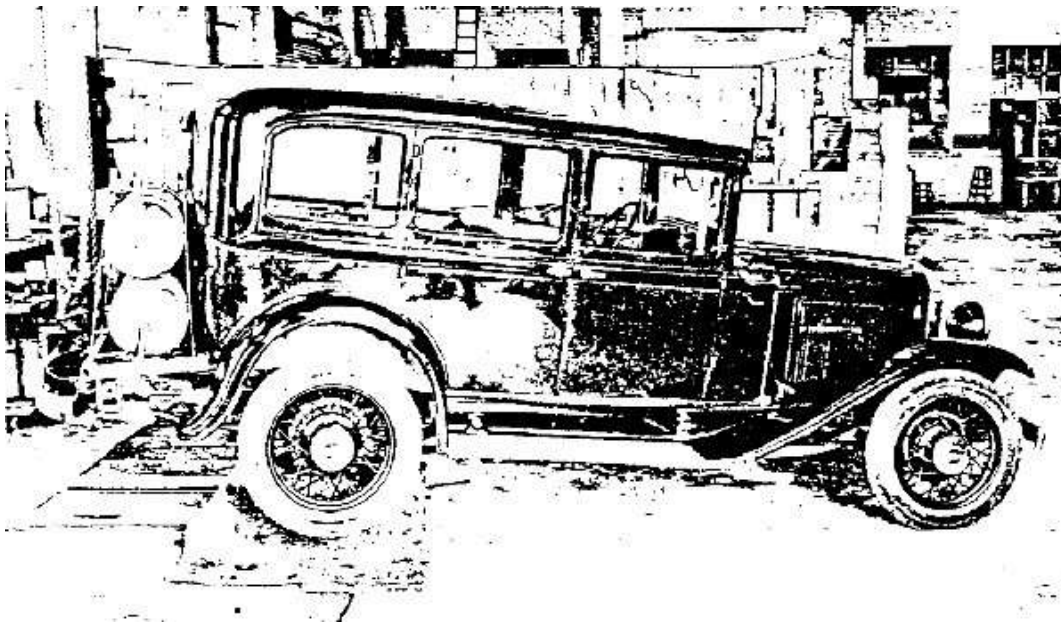


Figure 1.15: The Burt Air Drive

It was until 1979, when Terry Miller¹⁰ designed spring-powered car and announced that compressed air was the perfect solution for alternative cars as air never wears off.

In 1993, Miller restarted the project with the support of Toby Butterfield of Joplin, Missouri. They developed the Spirit of Joplin air car with parts donated by manufactur-

¹⁰1911-1989, English railway engineer

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ers. Terry's air motors have shown the feasibility of realizing air motors with spare parts on a small budget.



Figure 1.16: Terry Miller's air car

1.4.2 Present-Day Air Vehicles

Since the early 1990s, the french inventor Guy Negre¹¹ tried to set up an organization to market his air car designs in several countries. A web search for air cars will find thousands of references to his company, Motor Development International (MDI). Mr. Negre holds patents on his unique air motor in several countries. Along with other inventors such as Armando Regusci, Angelo Di Pietro, Terry Miller and others, it was Guy Negre and his son Cyril Negre who put air cars back on the map.

After that, every so often an established car company pretends it's going to put out a compressed air car, and the ruse lasts long enough to keep the public confused, discourage inventors from trying to go into competition with them, and make air cars in general look like a "no go". A couple years ago, for example, a big car company came out with a fairy-tale concept involving flying windmills to operate compressor stations, but still there are few real project that got some development to create real air cars.

¹¹1941-2016, French motor engineer

1.4.3 Developers and Manufacturers

There are quite a few developers and manufacturers of such a technology mentioning :

- Motor Development International (MDI), Luxembourg
Price : 26 266 TND



Figure 1.17: AirPod 2.0

- Tata Motors, India
Price : 22 209 TND



Figure 1.18: Tata's Air Car

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- Engineair Pty Ltd, Australia



Figure 1.19: EngineAir CAV

- Association de Promotion des Usages de la Quasiturbine (APUQ), Canada



Figure 1.20: APUQ's Quasi-turbine air car

1.4.4 Various Possible Applications

The compressed air engine can be used in many vehicles. Some of its applications to be used as engine are:

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a) Mopeds

Jem Stansfield, an English inventor has been able to convert a regular scooter to a compressed air moped. This has been done by equipping the scooter with a compressed air engine and air tank.

b) Buses

MDI makes MultiCATs vehicle that can be used as buses or trucks. RATP has also already expressed an interest in the compressed-air pollution-free bus.

c) Locomotives

Compressed air locomotives have been historically used as mining locomotives and in various areas.

d) Trams

Various compressed-air-powered trams were trailed, starting in 1876 and has been successfully implemented in some cases.

e) Watercraft and aircraft

Currently, no water or air vehicles exist that make use of the air engine. Historically compressed air engines propelled certain torpedoes.



Figure 1.21: Compressed Air Moped

1.4.4.1 Advantages

It does not use fossil fuels or other bio-carbon based fuel i.e. zero emission so this might offer a solution to the climate crisis.

There is no over heating problems, also Compressed air produces no sparks, meaning no explosive danger to users.

Zero fuel cost (the cost is involved only in the compression of air).

1.4.4.2 Disadvantages

The main disadvantages are the stages of energy conversion and transmission, as each has its own loss. For cars with combustion engines, energy is lost when the chemical energy contained in fossil fuels is converted into mechanical energy by the engine.

For electric cars, electricity from a power station (from any source) is transmitted

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to the car's batteries, which then transmit it to the car's engine, which converts it into mechanical energy.

For compressed-air cars, electricity from the power plant is transmitted to a compressor, which mechanically compresses the air in the car's tank. The car's engine then converts the compressed air into mechanical energy.

1.5 Conclusion

This chapter highlighted a recent problematic that currently dominates the present, which is pollution. Therefore, offering alternative existing solution either, electric or other type to reduce the impacts of such a phenomenon.

Taking the solution one step forward and proposing one that has been considered obsolete which is **Compressed air Vehicles**, this part showed the origins and history of such an idea, present day manufacturers and developers and possible uses underlining its advantages and drawbacks.

Chapter 2

Bibliographical Study

2.1 Introduction

This chapter presents the physical background needed to comprehend the given idea in later chapters. This work offers different aspects defining Air from a thermodynamics and mechanical point of view, furthermore, giving mathematical modeling used to do the necessary calculations.

2.2 Compressed Air and Pneumatics

Compressed air is a gas, or a combination of gases, that has been put under greater pressure than the air in the general environment. This gas has been put to use in numerous and diverse applications, including jackhammers, tire pumps, air rifles, and aerosol cheese.

Compressed air also has a great deal of potential as a clean, inexpensive, and infinitely renewable energy source. Its use is currently being explored as an alternative to fossil fuels.

2.2.1 Composition of Air

The clean dry air is composed of 78 % Nitrogen and 21% Oxygen in addition to other minor gases.

It has technically the same behaviour between sea level and an altitude of 20 km, but Pressure and temperature might vary its density.

At standard technical conditions, with a pressure of 1 *bar*, 20 °C of temperature and 65% of relative humidity, air's density is equal to 1.185 kg/m^3 .

At these same conditions, the mean Velocity of the air's molecules is of the order of 500 m/s

2.2.2 General Assumptions for Compressed Air

The effect of a change in temperature that has on air is modifying the value of the mean velocity. The resultant pressure therefore varies with the temperature. Similarly, any change in volume or mass effects the pressure. Thus pressure, temperature and volume are interrelated [13].

In this work, Compressed air is treated as a continuum. Thus, Pressure, Velocity and density are assumed to be defined everywhere in space and to vary continuously from point to point within a flow (This assumption are not valid in micro-techniques) [28].

To facilitate the understanding of the phenomenon and to easily model the compressed air, assumption that its molecules are perfectly elastic with a size negligible compared to the length of their free path and exert no forces on each other.

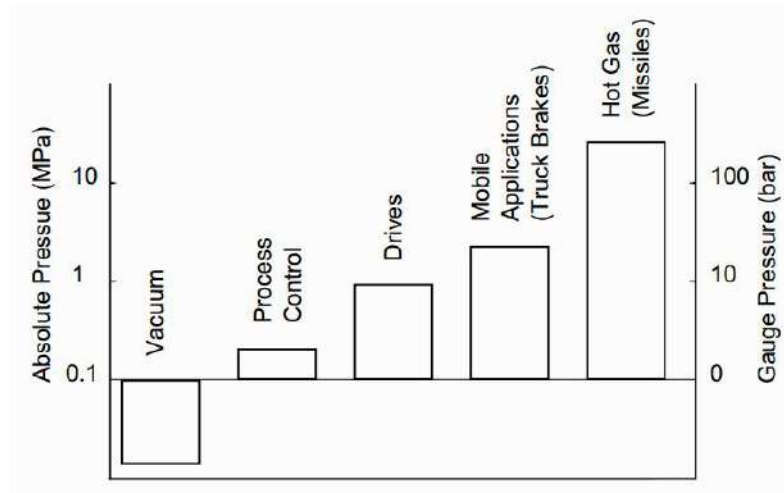


Figure 2.1: Pressure ranges of pneumatic systems

The figure below gives the pressure ranges of pneumatic systems. For example, Vacuum systems operate below atmospheric pressure.

2.2.3 Mathematical Modeling

There are many ways to model the relationship between the different parameters. In this project, the *Ideal Gas Law* is used based on the assumptions mentioned in the previous paragraph.

The *Ideal Gas Law* leads to the relation between mass, pressure, temperature and volume given by :

$$p \cdot V = m \cdot R \cdot T \quad (2.1)$$

where p absolute pressure in Pa,
 V volume in m^3 ,
 m gas mass in kg,

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R gas constant in J/(kg.K),

T absolute temperature in K

Relying on the comparison of the measured and calculated values, the accuracy of the model given by the data from [14] [11] [10] is determined calculating then, the relative error e_{rel} in percent between tabulated and computed values of density $\rho = m/V$.

$$e_{rel} = \frac{\rho_{tab} - \rho_{comp}}{\rho_{tab}} \cdot 100 \quad (2.2)$$

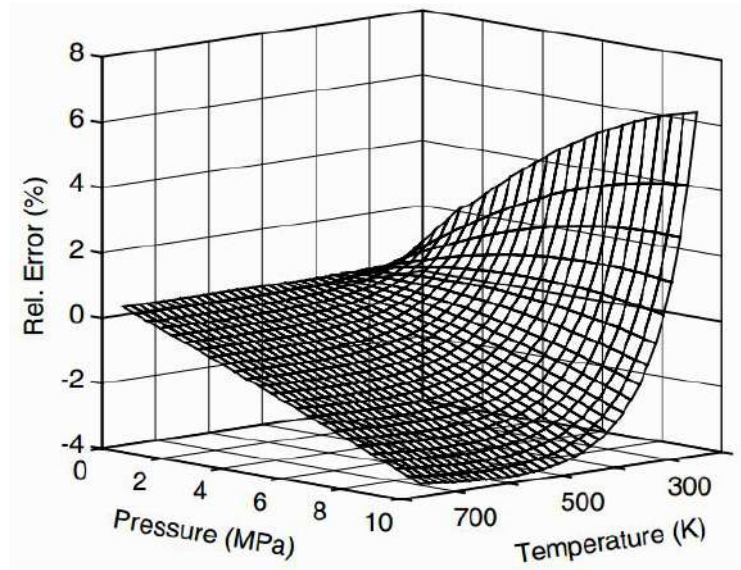


Figure 2.2: Relative error between computed and measured density

For accurate calculations in pneumatics, the error should be less than 5%.

Focusing on the region of $200 \text{ K} < T < 800 \text{ K}$ and $0.1 \text{ MPa} < P < 10 \text{ MPa}$, assuming, therefore, that the equation (1.1) represents an accurate model to be used for the calculations and simulations.

For more advanced models, see the *Van Der Waals*¹ equation, the *Beattie-Bridgeman* or the *Benedict-Webb-Rubin* equation of state [21] [36].

2.2.4 Atmospheric Air

The atmospheric air generally contains water vapor. It is important because to calculate the total Pressure, summing the dry air pressure and the water vapor is mandatory.

In case the pressures are equal, the air is said to be saturated where this saturation depends only the temperature. Water vapor may cause some issues when it gets

¹Johannes Diderik van der Waals, 1837-1923, Dutch theoretical physicist and thermodynamicist

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separated due to condensation only if the air is cooled to its *dew point* [4]. To overcome this phenomenon, air dryers and filter are applied, therefore, no moist will be found in the compressed air.

Based on the previous assumptions, and after referring to the ISO6356 standards for humidity of 65%, for a temperature $T=293.15$ K and a reference pressure $P_0 = 10^5$ Pa, the gas constant R is taken as :

$$R=288 \text{ J/Kg.K}^{-1}$$

Table 2.1: Definitions relating to pressure [7] [8]

Term	Definition
Absolute pressure	The pressure measured with reference to absolute zero, i.e. from an absolute vacuum. It is equal to the algebraic sum of the atmospheric pressure and the gauge pressure.
Atmospheric pressure	The absolute pressure of the atmosphere measured at the place under consideration.
Dynamic pressure	The dynamic pressure is that increase of pressure which would result if the kinetic energy of the uniform flow were converted to pressure completely and without any losses, i.e. isentropically. The pressure is equal to the product $\rho w^2 / 2$ where ρ is the fluid density and w is the flow velocity.
Gauge pressure	The pressure measured with reference to atmospheric pressure, also called effective pressure.
Static pressure	The pressure measured in a gas in such a manner that no effect on measurement is produced by the gas velocity
Total pressure	The sum of the static and dynamic pressures. It designates the gas condition at which the flow energy of the gas is converted into pressure without any losses. In a stationary body of gas the static pressure and total pressure are numerically equal.

The SI unit for pressure is $1 \text{ Pa} = 1 \text{ N/m}^2$ simply To avoid the use of scaling factors, this unit is be used for calculations. $1 \text{ bar} = 10^5 \text{ Pa}$ is used which is widely used in the compressed air world.

In this work, the unit Pa is used to denote absolute pressure, while the unit bar is used to denote gauge pressure

For the standard parameters of a given pneumatic system, this work refers to the standard ISO 8778 defining a *standard reference atmosphere* ” whose pressure has been

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approximated to be nearly that at sea level, whose temperature typically considered to be room temperature and whose relative humidity is arbitrarily established

-pressure 10^5 Pa

-temperature 293.15 K

-relative humidity 65%

2.2.5 Thermodynamic Process

Thermodynamics is the field of converting heat into other forms of energy and provides the theoretical background for calculating the conversion of heat into mechanical work in ideal processes. The laws of thermodynamics will be used to describe the behaviour of air in pneumatic components, the resulting changes in pressure and temperature, and the mechanical work required. Advanced level for the use of this discipline is beyond the scope of this work. Further knowledge is given in the introductory text books for thermodynamics for example [31] [36].

Based on the situation, 4 possible thermodynamic processes are distinguished.

2.2.5.1 Constant Volume Processes

A constant volume process is also called *isochoric*. From the thermodynamic point of view, this system is a closed system because heat is transferred from the outside into the system, but no mass transfer occurs.

This kind of process is the easiest way to increase pressure with a container by heating it up. It is identified that old compressed air vehicle inventors relied on this technique. meaning using boiling water to make the pressure rise.

The required energy, i.e. the supplied heat Q , is represented as :

$$Q = c_v \cdot m \cdot (T_2 - T_1) \quad (2.3)$$

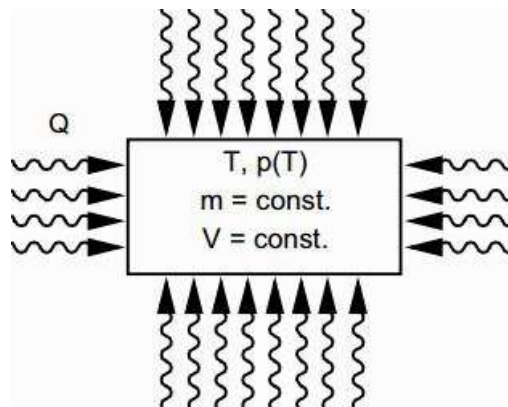


Figure 2.3: Constant volume process with heat transfer

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The constant C_v that relates the heat to the temperature differential is called *specific heat capacity at constant volume*, for air 717 J/(kg.K).

2.2.5.2 Constant Pressure Processes

A constant pressure process is also called *isobaric* e.g. a piston moves free from frictional forces. In this case the pressure in the chamber is constant (see figure below)

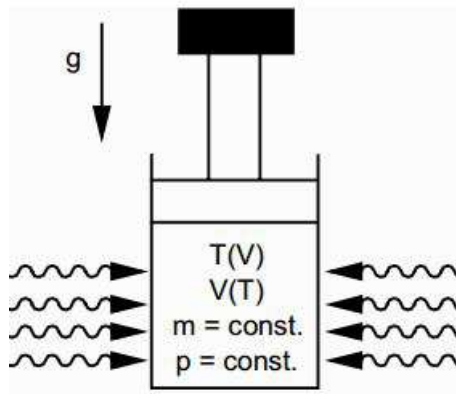


Figure 2.4: Constant Pressure system

2.2.5.3 Constant Temperature Processes

Considering the heat transfer from the air in the chamber to the environment is ideal, the process has a constant temperature and is called *isothermal*. In this case, the total work also represents the amount of heat extracted to maintain the constant temperature.

2.2.5.4 Reversible Processes without Heat Transfer

In this process, a perfect insulation is needed, the piston should freely, i.e. there are no friction forces, and the inertia forces are negligible. This means that there is no change in the total heat energy of the gas under compression. Pressure and volume of this *isentropic* process can be described by

$$p_1 \cdot V_1^\gamma = p_2 \cdot V_2^\gamma \quad (2.4)$$

Where γ represents the ratio of specific heat capacities or index of isentropic expansion or compression.

It is defined by

$$\gamma = \frac{c_p}{c_v} \quad (2.5)$$

Where c_p represents the *specific heat capacity at constant pressure* equals 1005 J/(kg.K) for air.

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Taking $\gamma = 1.4$ for air.

$$W_{12} = - \int_1^2 p dV = -p_1 \cdot V_1^\gamma \int_1^2 \frac{1}{V^\gamma} dV = \frac{p_1 \cdot V_1}{\gamma - 1} \left(\left(\frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma}} - 1 \right) = \frac{m \cdot R}{\gamma - 1} (T_2 - T_1) \quad (2.6)$$

The temperature will be described by

$$T_2 = T_1 \left(\frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma}} \quad (2.7)$$

This equations are used in later chapters.

2.2.6 Steady-State Losses

Like electricity, where components are connected through wires, in pneumatics, circuits are made of lines.

Nowadays they are made out of plastic tubes from nylon or polyurethane. This lines have a certain amount of characteristics that they should meet mentioning a required permissible pressure, bending radius and temperature. [\[13\]](#)

Copper tubes might be used by they are generally needed when it comes to heavy duty applications.

For plastic, a simple push-in lines can be used as no tools are required to install or remove.

However, when designing a pneumatic circuit, different problems may accrue specially after reaching a Steady-State.

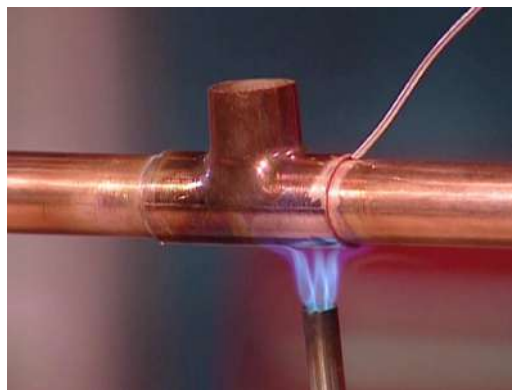


Figure 2.5: Soldering a copper line (example)

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2.2.6.1 Steady-State Losses of Lines

An important criteria in line design is their resistance. The ISO standard 4414 gives a table that bases the flow rates upon a 10 % pressure drop in 30 m of ISO 65 grade medium wrought iron pipe at 20 °C [9]. Typical air velocity is between 10 and 40 m/s [33] [23], typical Reynolds² numbers are between [19] $4 \cdot 10^3 \leq Re \leq 4 \cdot 10^6$, and the maximum pressure drop should be below 0.1 bar [15].

Table 2.2: Maximum recommended flow rates to be used in pneumatic system piping in l/s (ANR) [9]

Working pressure bar	Inside 6	diameter 9	of piping 13	(mm) 16
0.20	0.18	0.41	0.91	1.7
0.40	0.28	0.62	1.4	2.6
0.63	0.38	0.85	1.9	3.5
0.80	0.44	1.0	2.2	4.1
1.00	0.52	1.2	2.6	4.9
1.25	0.62	1.4	3.1	5.8
1.60	0.75	1.7	3.8	7.0
2.00	0.91	2.0	4.5	8.4
2.50	1.1	2.5	5.5	10
3.15	1.3	3.0	6.7	12
4.00	1.6	3.7	8.3	15
5.00	2.0	4.6	10	19
6.30	2.5	5.6	13	23
8.00	3.1	7.0	16	29
10.0	3.9	8.7	19	36
12.5	4.8	10	24	45
16.0	6.1	13	31	57

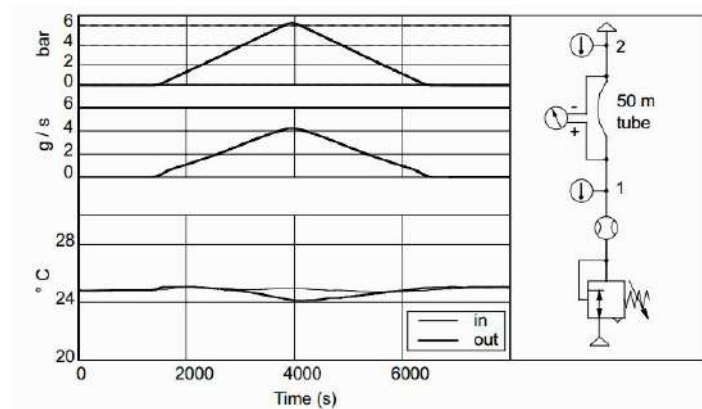


Figure 2.6: Pressure drop, mass flow rate and temperature as a function of time; and measurement set-up, line length $L = 50$ m, inner diameter $D = 5.7$ mm

²Osborne Reynolds, 1842-1912, Irish innovator in the understanding of fluid dynamics

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This simulation shows the pressure drop in a 50 m line with an inner diameter of 5.7mm.

Unless the lines are “long” they are typically not considered during the design or analysis of a pneumatic circuit.

2.2.6.2 Steady-State Losses of Fittings

Pneumatic circuits are connected with lines. However, the ends these lines have to be connected to other lines or components. First, push-on connectors with *barbs* were used before inventing *push-in connectors*.

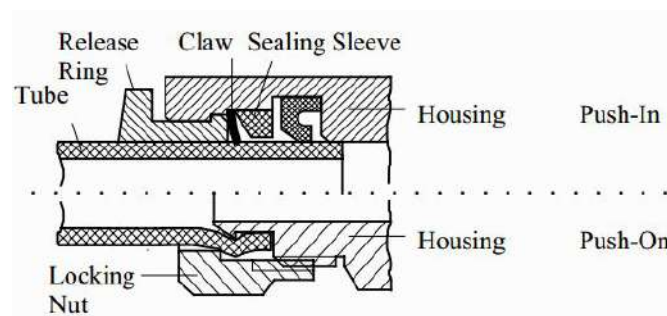


Figure 2.7: Cross-Sectional View of a Tube with Push-in Fitting (top) and Barbed Push-on Fitting (bottom).

For modelling purposes, the fittings may have to be taken into consideration because they cause higher losses than the tube. Unfortunately, there is no detailed information about their resistance.



Figure 2.8: Quick fittings for connecting two lines (or more)



Figure 2.9: Quick fittings for connecting a line to a component

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For this matter, this work refers to Fleischer's approximation [18] and giving the equivalent length L of a straight, unrestricted line:

$$L = n \cdot K \cdot d \quad (2.8)$$

where L equivalent length,

n number of similar fittings in the conduction leg,

K multiplying factor assigned to the device,

d smallest diameter in the grouping.

The K value of an integral conductor angle fitting having a bend radius of 10 times its inner diameter or better is essentially zero.

Table 2.3: Multiplying factors for various valves and fittings [18]

Component	K
In line ball non-return valve, fully open	12.5
90° narrow radius elbow	4
9 standard elbow	2.5
90° long radius elbow	1.5
45° narrow elbow	2
45° standard elbow	1.3
Reducer (1 size)	1.5
Enlarger (1 size)	2.0
Standard tee - flow through run	1.5
Standard tee - flow through branch	5

Bends in a line does not change the resistance significantly. However, if the line is spiral wound corrections must be made. Therefore, on-line normal lines are chosen.



Figure 2.10: Spiral line

2.2.7 Cylinders

A cylinder is a mechanical device that converts pneumatic energy to mechanical work. A cylinder is generally made of piston and a piston rod in-closed in a cylindrical bore. The following figure represents a Cut-away view of a cylinder:

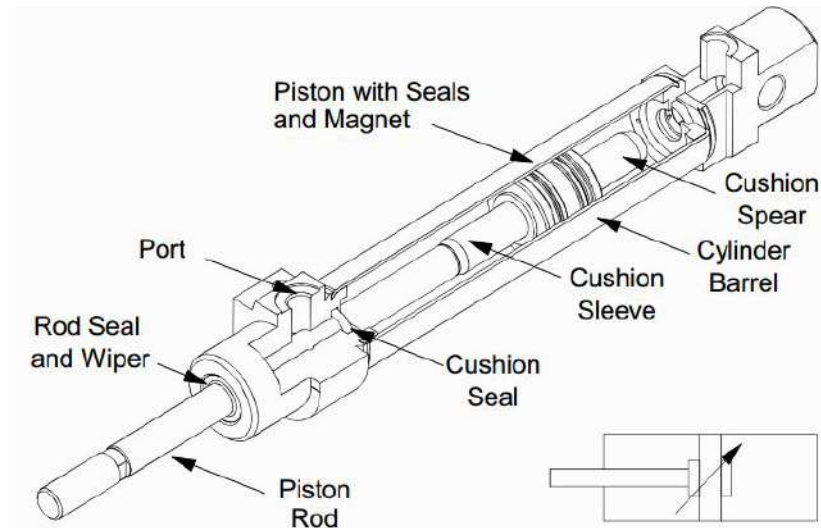


Figure 2.11: Cut-Away view of Single Rod Cylinder According to ISO 6432

2.2.7.1 Cylinder Parameters

There are many parameters for a cylinder that can be given in standards. However, the most important are maximum pressure, piston diameter, rod diameter and maximum stroke. Examples may be found the third chapter as well as the appendix.

2.2.7.2 Types of Air Cylinders

There are many types of cylinders varying from size to function, they are categorized into;

- **Single acting cylinders :** generally equipped with a spring return.

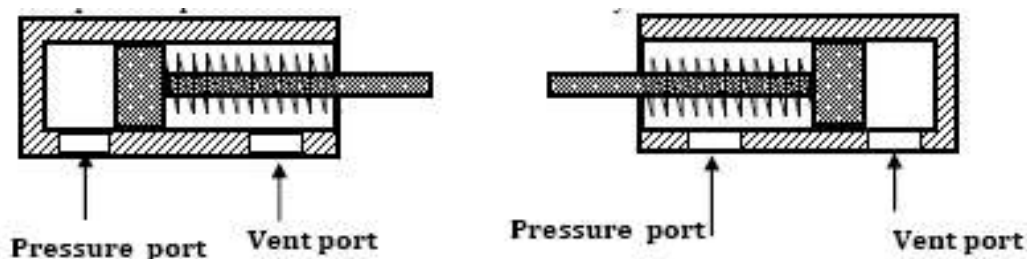


Figure 2.12: Schematic View of Push- and Pull-type Single Acting Cylinders

- **Double-acting cylinders :** This kind can be driven both ways extending and retracting strokes.

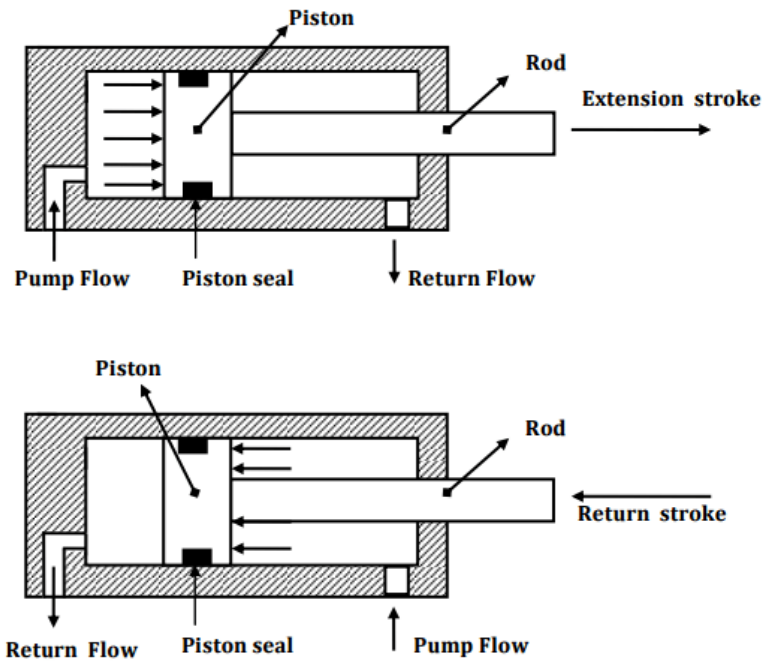


Figure 2.13: Schematic View of a Double Acting Cylinder (with directional valve)

- **Multi-stage Cylinders :** As it name indicates, theses cylinders have different levels of extension.

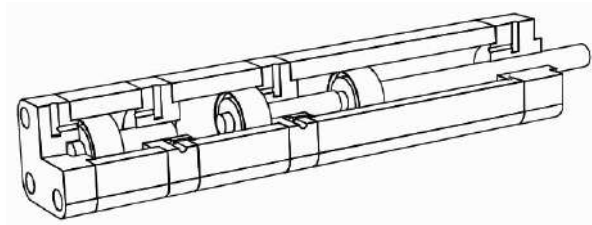


Figure 2.14: Cut-Away View of a Multi-Stage Cylinder

- **Special Types :** like Rotary air cylinders, rodless air cylinders, Impact air cylinder... etc.

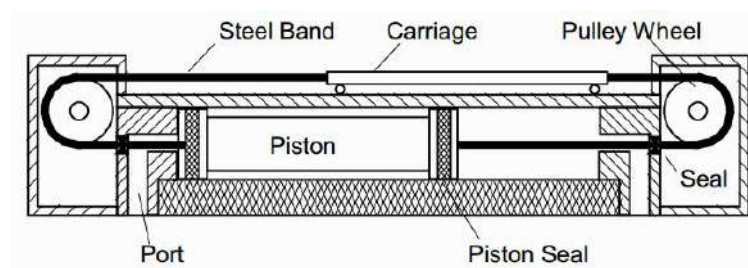


Figure 2.15: Schematic View of a Rodless Cylinder

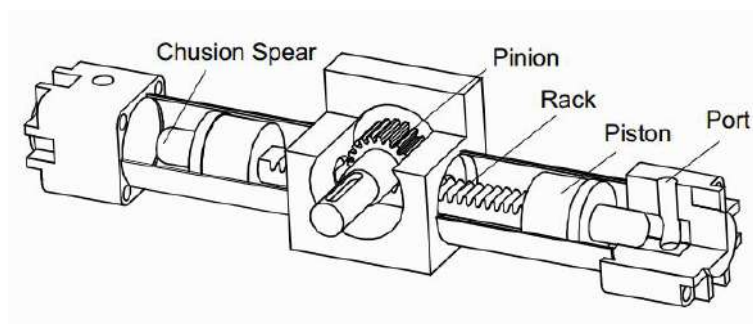


Figure 2.16: Cut-Away View of a Rotary Cylinder

2.2.8 Air Engines

Cylinders are generally used for translational.

However, electric drives are operated for continuous rotation even-though pneumatic cylinders are way more easier to make therefore cheaper. This is explained by the fact that electricity is almost ubiquitous while compressed air requires specified equipment.

Nevertheless, there are many applications where pneumatic motors offer unique advantages as they have high power-to-weight ratio, indifference to overload and stall, cool operation, indifference to dirty or explosive atmospheres, very high rotational speed and an easy energy storage: in a pressurised vessel. Their torque vs speed characteristic is very well suited for many applications and they are easily reversible. If necessary, a nozzle is a simple way to reduce speed.

Generally, pneumatic motors have power patterns similar to those found in series-wound DC motors [\[13\]](#)

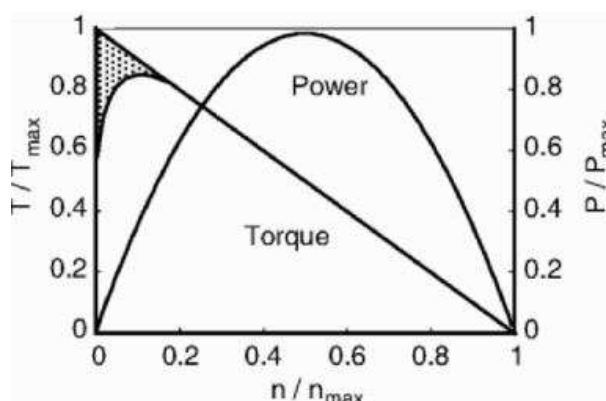


Figure 2.17: Power and Torque as a Function of Speed at Constant Supply Pressure

The figure above represents the power and torque as a function of speed at a given pressure. If the load torque increases, the speed decreases leading to a higher torque which balances the load.

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The figure shows that the motor should have its maximum power at 50 % of its maximum speed and torque.

There is always a minimum speed below which no stable operation is possible.

The starting and the form of the curves might look similar but it differs a bit regarding the type of motor and technology used.

To obtain a maximum service life, the load should be carefully aligned with the motor.

If the motor is coupled to an excessive axial or radial load, its performance can decrease significantly and permanent damage may be caused mentioning **Rod Buckling** (accrues when the stroke is relatively long regarding the rod diameter) and **Rod Bearing Failure**. Rod diameters have been standardized to be adequate to length and load. Please refer to the appendix.

While using air in these type of motors, the temperature drops causing condensation of water vapour. Water could cause rust or even ice in the exhaust ports.

When dealing with high pressure, Filters and regulators are a necessity.

2.2.8.1 Pneumatic Vane Motors

Vane motors are made of a cylinder called a stator enclosing a rotor with an eccentric form. The rotor has grooves letting the vanes move freely. The stator has two openings for exchanging air. The form and the position of the openings make the system start correctly as they let the air push into the interior of the rotor. This type of motors relies on the centrifugal force for its continuous rotation.

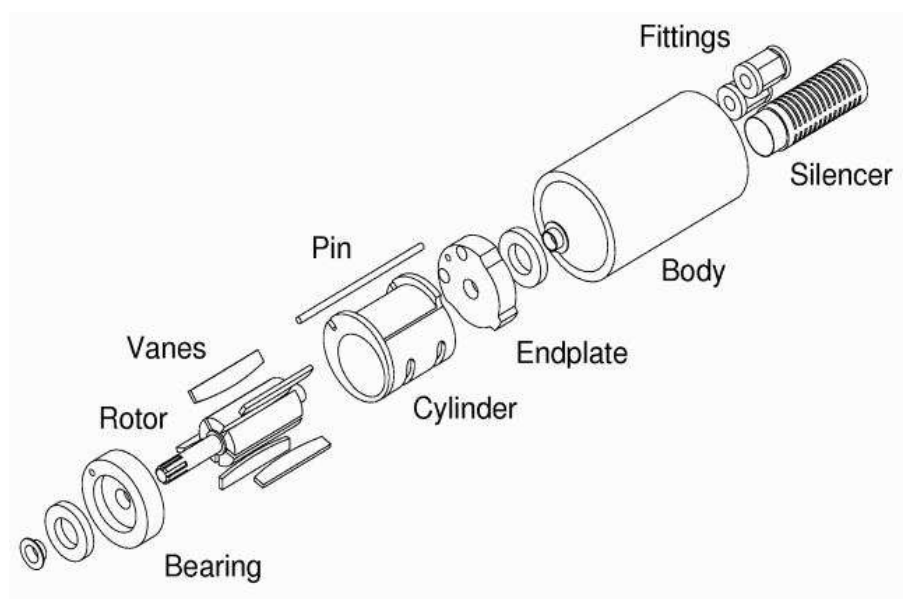


Figure 2.18: Arrangement of Reversible Vane Motor

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Its maintenance is considered to be easy as it is simple to disassemble. Vane motors are the most used type in industry. They run in grate speeds giving a huge advantage when it comes to power weight ratio. However, they offer a less effective starting torque. Vane motors are lighter and less expensive than other motors.

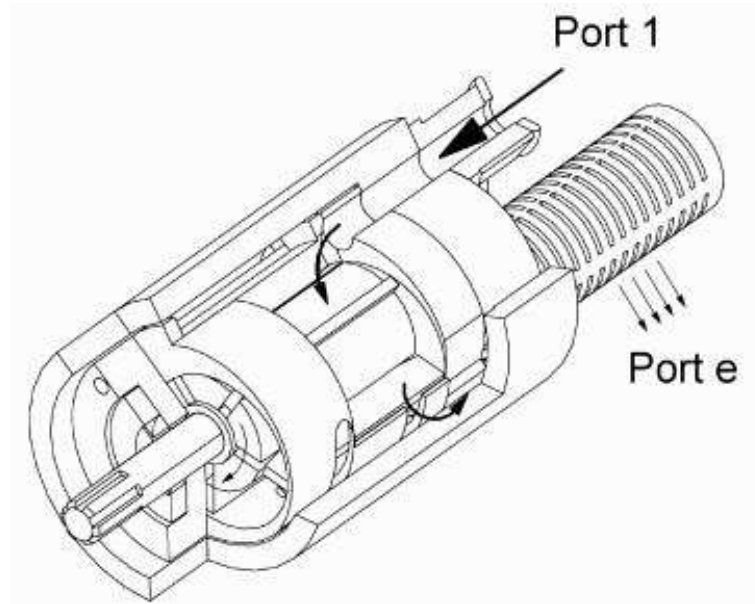


Figure 2.19: Cut-Away View of an Air Vane Motor

Vane motors can be both reversible and non-reversible depending on the interior form of its rotor. This figure offers an example of three types of air vane motors:

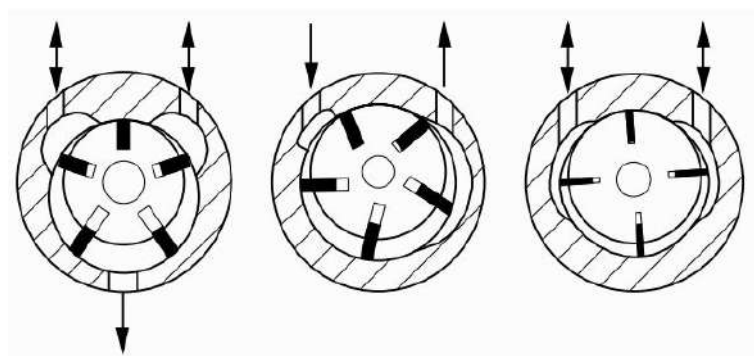


Figure 2.20: Schematic View of Three Vane Motors

The first one has three ports and is a reversible type (left) followed by the two-port vane motor in the middle, non-reversible. Finally, (right) this two-port model is reversible.

Full examples of calculations and mathematical modeling of such a motor are included in [13].

2.2.8.2 Piston Air Motors

This motor rely on the pressure held in its cylinder to generate the power. It rotates at small speed but offer high torque. They are adequate for working with heavy weight and they assure good control over their speed.

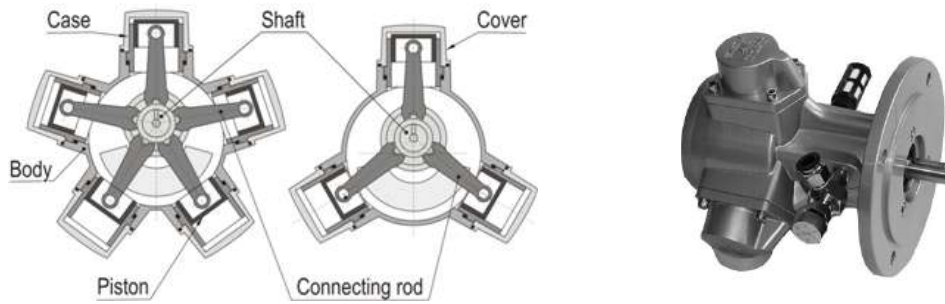


Figure 2.21: Two Types of Piston Motors (3D Model)

2.2.9 Directional Control Valves

Directional control valves are used to chose and control the direction of compressed air between components. There are many types of directional control valves that can be chosen according to :

- **Number of ports**
- **number of internal stable states**
- **Tupe of command** electrical, pneumatical or mechanical.

These valves are labeled by the number of ports and number of stable states. the ports are labeled 1 for supply, 2 and 4 for working ports and 3 and 5 for exhaust ports.

In addition, they are characterized by the state after supplying compressed air, meaning, whether the valve is Normally Open or Normally closed.

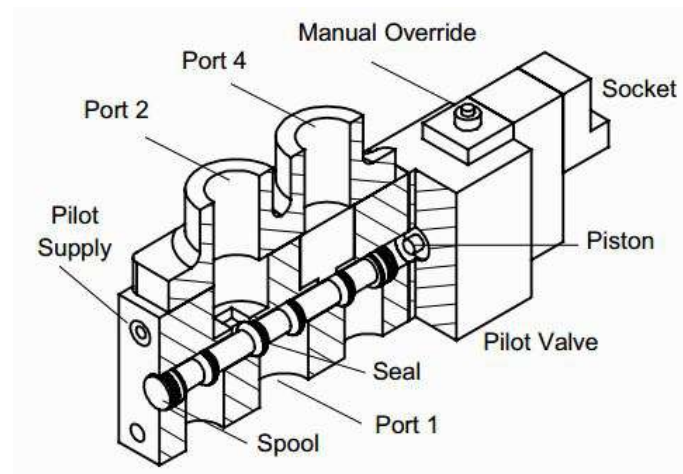


Figure 2.22: Cut-Away View of Directional Control Valve (5/2-way)

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Some times, valves can contain middle state in which nothing happens.

The choice of these characteristics depends on the use .i.e type of cylinder and the type of the pneumatic installation given.

The directional valves consist an important component in pneumatic logics. For further details, please refer to the appendix.

2.2.10 Pressure Control Valve

As the named indicates, these valves control the supply pressure generated by the compressor. There are also pneumatic relief valves which vent compressed air when reaching a critical pressure. They are used for safety.

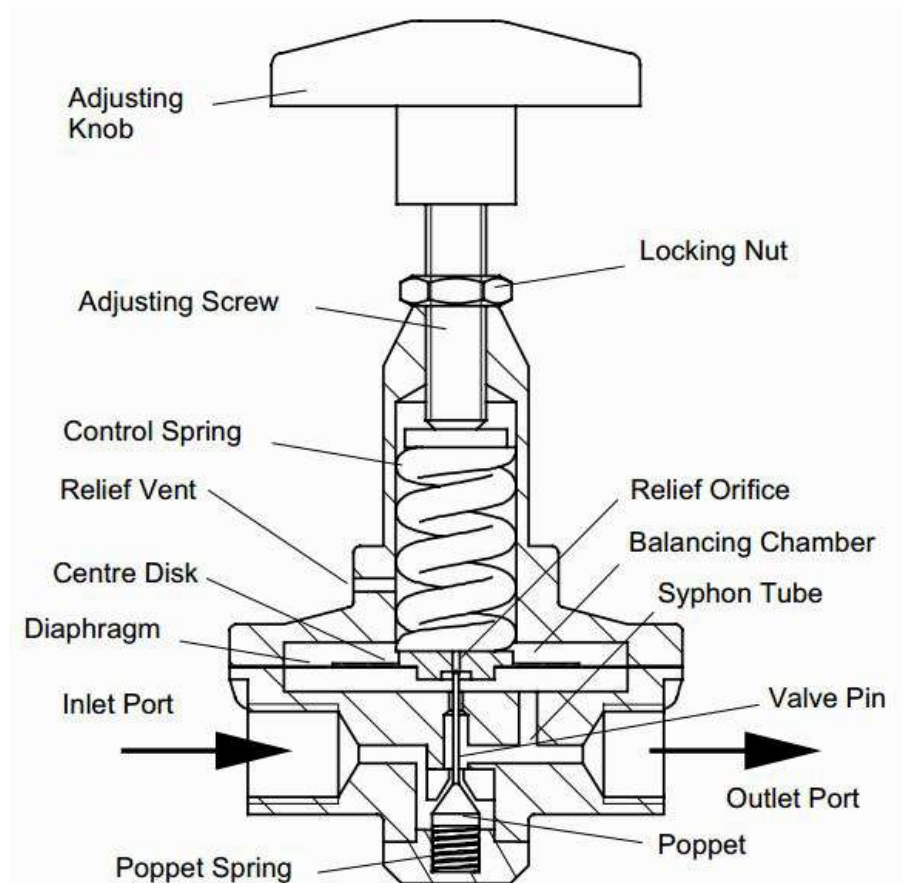


Figure 2.23: Cross-Sectional View of a Pressure Reducing Valve

2.2.10.1 Other Type of Valves

- **Shut-off Valves** Prevent flow from passing.
- **Non-return Valves** Prevent flow from returning.

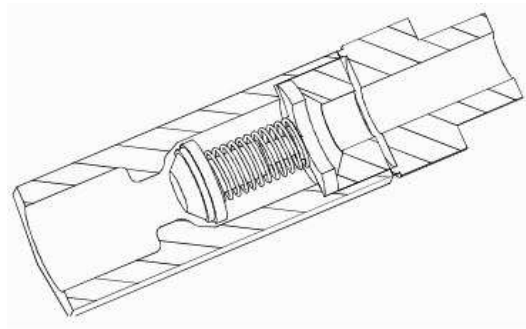


Figure 2.24: Non Return Valve

- **Shuttle Valves** compare pressure and lets only the higher pressure to pass.

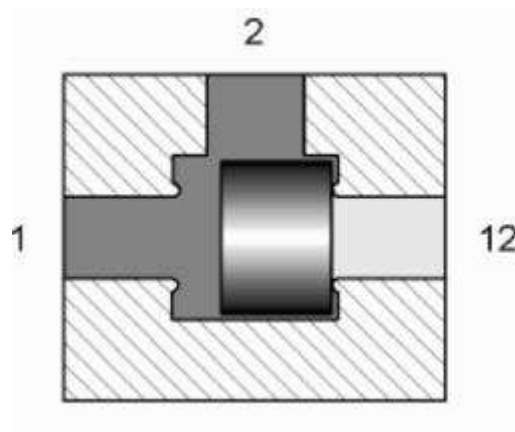


Figure 2.25: Shuttle Valve

- **Twin Pressure Valves** compare pressure and lets only the lower pressure to pass

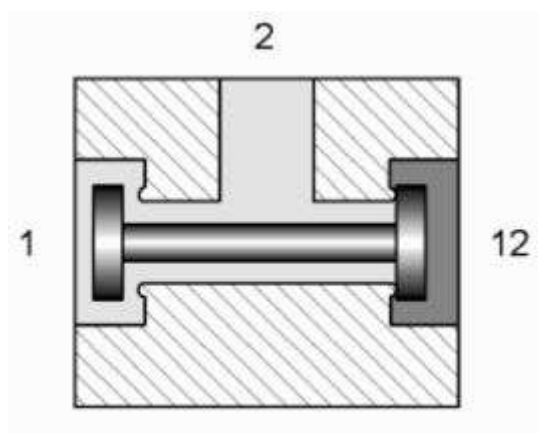


Figure 2.26: Twin Pressure Valve

- **Soft-Start Valves**

- **Delay Valves** make a delay

More pneumatic components as well as as their standardized symbols are to be found in the appendix. Pneumatic Logic control valves (OR - AND - NO ...etc) are also joined in the appendix.

2.3 Mechanics

The mechanical parts of this project focuses on the modelling background for the solutions given, later, in chapter 3.

2.3.1 Three-wheeled Vehicles

The work presented in this chapter includes the concept used in three-wheeled vehicles. There two configurations possible for this type of vehicles:

- **Tadpole** : Two front wheels and one rear wheel
- **Delta** : One front wheel and two rear wheels

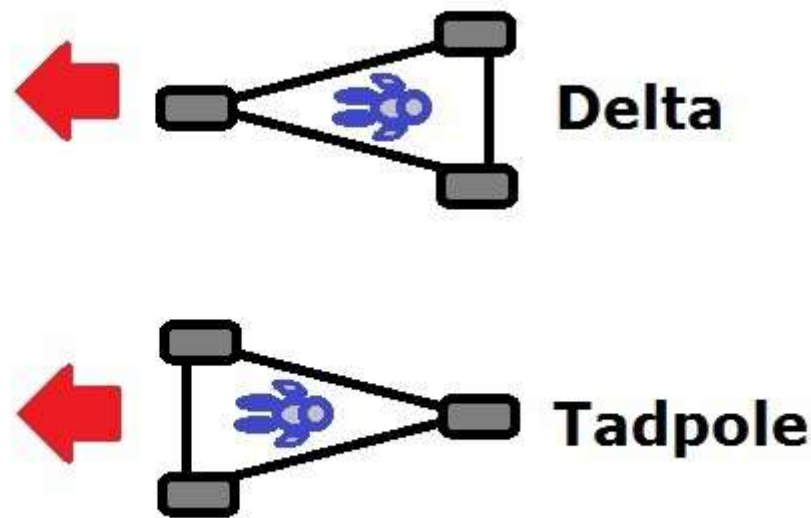


Figure 2.27: Delta VS Tadpole

2.3.1.1 Delta Trikes

A Delta trike has many advantages such as reduced cost, and complexity. However, it does not have the handling abilities of a Tadpole design. The Delta trike front-end shows a higher degree of momentary acceleration when compared to the rear resulting in over-steering and difficulties in handling turns.

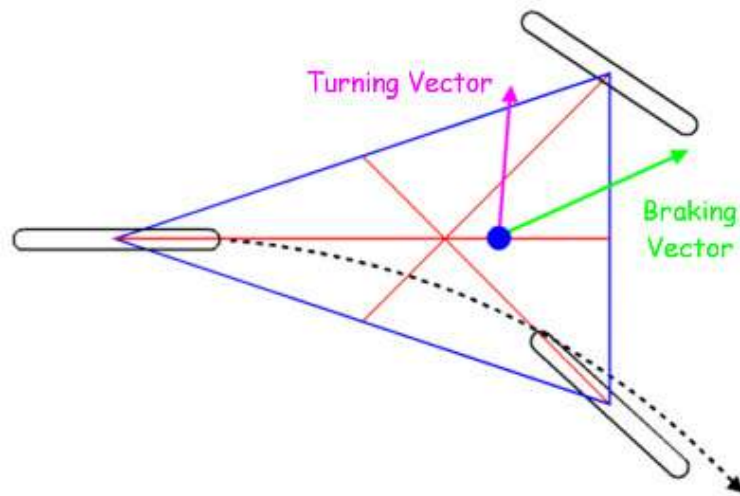


Figure 2.28: Tadpole Turning

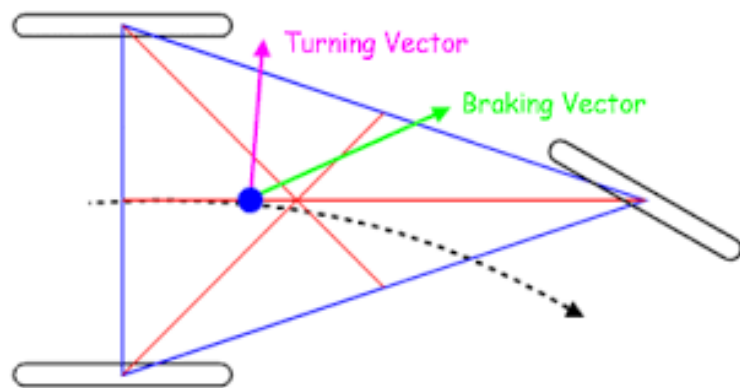


Figure 2.29: Delta Turning

Despite the use of simple steering design on the Delta Vehicles similar to bicycles, it is deemed to fail when applied on car designs. One of the designs that use the Delta design is **Reliant**'s³. This type of vehicles was considered to be unstable as any simple turn or brake could cause it to flip over.



Figure 2.30: Reliant Robin Drive Test

³Reliant Motor Company, 1935-2002, British car manufacturer

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A link of the full Drive Test made and published by **TOP GEAR**⁴ is included in the references [12].

2.3.1.2 Tadpole Trikes

As its front-end has less acceleration when compared to the rear end when turning, this exhibits a good stability and speed when it comes to taking corners. Despite the fact the handling should be fully dependent to the over-all design of the vehicle, the Tadpole design is the one to choose in this matter.

Due to the fact that the weight displacement and force goes forward while braking, the tadpole design has a big advantage because of the two wheels on front, rendering the vehicle more stable and effective. braking with two wheels is obviously more efficient.

2.3.2 Handling

Handling is the responds of the vehicle to the user's commands, taking in consideration the environment i.e. the road condition [34]. A good handling is defined when the vehicle and the steer wheel are easy to maneuver [24].

To provide a good handling, pitch and roll acceleration should be minimized. In addition, it is also mandatory to have a rigid chassis and suspension system. The under-steer coefficient K_{us} is used to determine the effect of the surrounding factors on the handling quality. This factor is expressed by [34]:

$$K_{us} = \frac{F_f}{C_{\alpha f}} - \frac{F_r}{C_{\alpha r}} \quad (2.9)$$

where F_f and F_r are the normal forces regarding the front and rear tires, and $C_{\alpha f}$ and $C_{\alpha r}$ are the stiffness.

If K_{us} is positive, the vehicle is under-steered and it deviate outside outside of the turn.

If K_{us} is negative, the vehicle is over-steered and it deviate inside of the turn.

If K_{us} is zero, the vehicle is neutral-steered and it stays on the desired path.

⁴British motoring magazine and television program

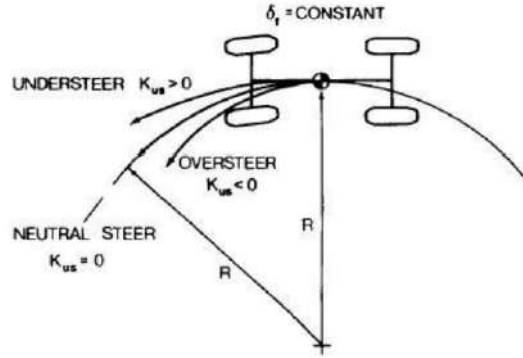


Figure 2.31: Curvature response of neutral steered, under-steered, and over-steered vehicles at a fixed steer angle [34]

For good handling, generally K_{us} is chosen between 2 and 6 degrees [35] but it varies regarding the vehicle and its maximum speed like shown in the figure below.

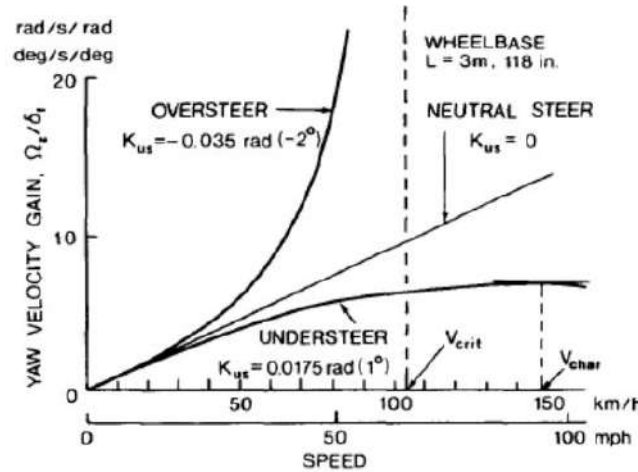


Figure 2.32: Steer angle vs. speed of neutral steer, under-steer, and over-steer [34]

2.3.2.1 Side slip angle

Side slip angle determines the lateral stability of a vehicle. The effect of side slip angle on lateral forces can be measured by the cornering force [35]

$$F_{y\alpha} = C_{\alpha}\alpha \quad (2.10)$$

Where C_{α} is the cornering coefficient [34].

2.3.2.2 Camber and tilt

To improve the performance and stability of a three-wheeled vehicle in harsh maneuvering, tilting the body or applying camber angle are effective ways in this matter.

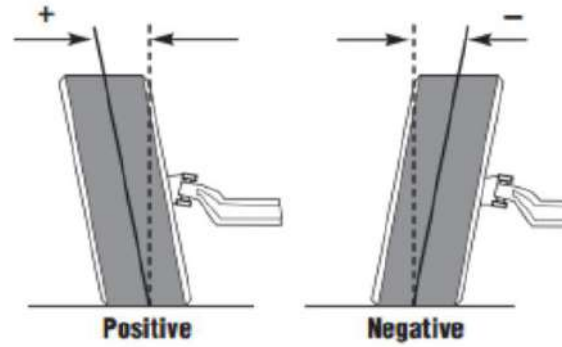


Figure 2.33: Schematic Demonstration of Negative and Positive Camber Angle

When applying camber angle, even a small angle, cornering becomes significantly shorter [29]. Using The Ackermann⁵ steering compensation provides a much effective way i.e. when the vehicle is steered in either direction, the inside wheel shall always turn sharper than the outside wheel [20].

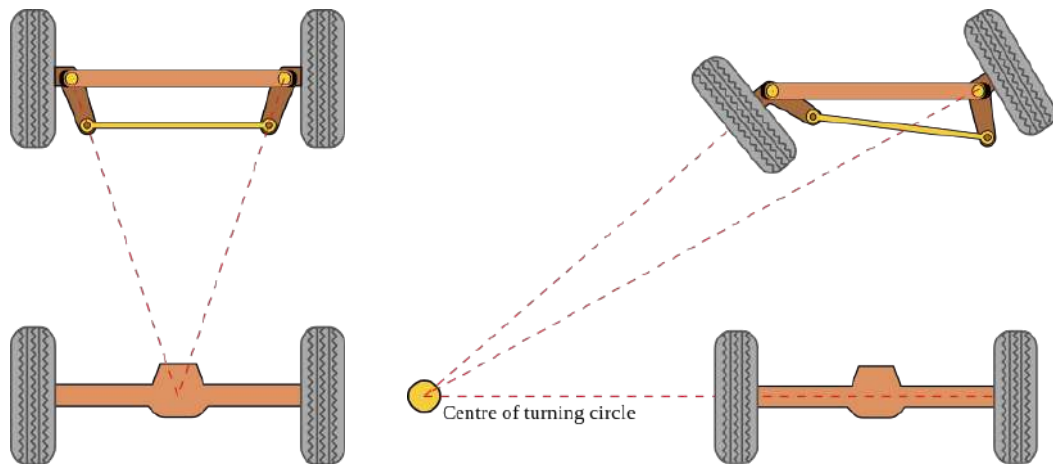


Figure 2.34: Ackermann Steering Compensation

Tilting system are generally applied to narrow three-wheeled vehicles to improve their manoeuvrability at high speeds and during large steering angles [32].

2.3.3 Lateral stability

Lateral stability (i.e., stability of a moving vehicle about the axis perpendicular to its direction of motion) refers to the tendency of the moving vehicle to go back to its original direction when disturbed from that direction [35].

2.3.3.1 Rollover

Rollover happens when the user uses control of the vehicles causing it to roll. This can cause safety hazards to the driver and other vehicles when speaking on a on-road

⁵Rudolph Ackermann, 1764-1834, Anglo-German bookseller, inventor, lithographer, publisher and businessman

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maneuvering.

It is important to prevent the vehicle from rolling-over, so the geometry of the vehicle has to be fit to prevent this phenomenon from happening.

Determining the acceleration threshold a_y [30] via

$$a_y = \frac{T \cdot g}{2H} \quad (2.11)$$

where H is the distance of the center of gravity from ground and T is the track width.

This indicates the importance of the geometry, which determines the center of the gravity, on the rollover of the vehicle.

2.3.3.2 Skid

Skidding happens when the roads are slippery. This phenomenon is an important factor that contributes to instability.

To prevent it from happening, the acceleration of the vehicle should be limited by the road/tire coefficient of friction [30] :

$$a_y \leq \mu \cdot g \quad (2.12)$$

Applying ABS brakes should also help preventing skidding.

The coefficient should be offered according to the following table :

Table 2.4: Road/tire friction coefficient for different types of roads

Road State	Road/tire friction coefficient
Dry	0.90
Wet	0.60
Snowy	0.20
Icy	0.05

2.3.4 Calculations and Modeling

For confidentiality, the calculation would not be fully mentioned in this work, however, all the mechanical work was inspired on **Dynamics of a Three-Wheel Vehicle with Tadpole Design** by Azadeh Zandieh [35], **Modelling and analysis of the dynamics of a tilting three-wheeled vehicle** by Johannes Edelmann, Manfred Plöchl and Peter Lugner [22] and **Studies on the wobble mode stability of a three-wheeled vehicle** by Venkata M Karanam and Ashitava Ghosal [25].

In those works, calculations of stability and analysis are found and even simulation using **MATLAB**. Please refer to the mentioned work for further details and to Chapter 3 for realisation of the vehicle.

2.4 Conclusion

In this chapter, different physical aspects both about pneumatics and mechanics were explained. The equation given can be used to model the vehicle on computer before realization.

This chapter represents the bases of the whole work from a mathematical and theoretical point of view. In fact this section underlined the background knowledge needed to tackle the project from a thermodynamic and mechanical perspective.

This section offered the minimum requirements, without going into details, to continue to the next parts of the project and to reach the realization phase.

However, more details and examples are covered in the references, considering them to be the corner stone of this section of the project.

Chapter 3

Electrical Part

3.1 Introduction

As any on-road vehicle should contain an electrical part, for light and signaling purposes. This section is dedicated to the calculation need for the realization, and the simulation of the given approaches.

In fact, different approaches and solutions are to be given and discussed from a feasibility and availability perspective.

3.2 Items Used

3.2.1 Mandatory

This part presents the items required like:

- **Headlights :** any vehicle need at least 2 headlights to lighten dark places like tunnels or to travel at night.
- **Taillight :** taillight is a requirement for all on-road vehicles.
- **Turn Signals:** like other lights, turns signals are a necessity when using roads to indicate the direction that the vehicle will turn to.

3.2.2 Optional

The following items are not a necessity but it was chosen to add them to the vehicle:

- **Electrical Horn :** to signal other drivers of threats or possible accidents.
- **LCD Display :** to indicate several parameters.
- **Proximity Sensor :** to prevent impacts when parking.

Finally, a Micro-controller is required to manage these items.

3.3 Energy Requirement

This table presents an estimation of the energy required by the items to function properly:

Table 3.1: Energy Requirement

Article	Power per unit(W)	Quantity	Consumption(Wh)
Headlight	7.5	2	7.5 (0.5h night)
Taillight	4	1	2 (0.5h night)
Turn Signals	0.08 (per 5s)	4(2in use)	8 (50 times)
Horn	0.1 (per signal)	1	1 (10 times)
LCD Display	0.1	1	0.1 (1h)
Microcontroller (Arduino Uno)	0.29	2	0.58 (1h)
Ultrasound Sensor	0.025	1	0.01 (0.25h)
Buzzer	0.05 (per signal)	1	0.5 (10 times)
Total			19.69
Losses (3%)			20

It can be said that at full use, which technically impossible, the electrical items require 20Wh.

3.3.1 Energy Source and Storage

To provide this energy, using a small solar panel was proposed. The smallest Solar Panel in available in the Tunisian market is **Vision 36M glass**, (Please refer to the appendix for full data sheet) providing 165 Wp. This solar panel has a weight equal to **25 kg** and covers an area of $155cm \times 71cm = 110058 cm^2$ equivalent to **1.1²** which is huge for the vehicle.



Figure 3.1: Vision 36M Glass Solar Panel

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Assuming in winter there is 6 hours of light and 10 hours in summer, this table is the result:

Table 3.2: Provided Energy

Nominal Power (Wp)	Winter (h)	Summer (h)
165	6	10
Total (Wh)	990	1650

It can be assumed from this table that the energy provide is relatively high. The dimension, however are considered to be big regarding the general form of the vehicle. An other alternative found is a **20 W** solar panel. What gives about 120 Wh in winter and 200 Wh in summer. The energy provided should be enough to cover the consumption and the dimension are adequate to be equipped on the vehicle, however, there are none to be found in Tunisia.

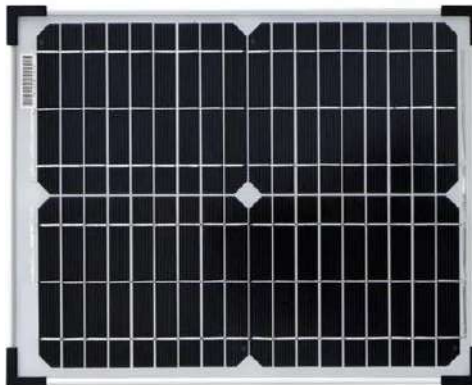


Figure 3.2: 20 W Solar Panel

This amount of energy can be stored in any kind of motorcycle battery. This **NP18-12R** battery was chosen. It offers :

- **Capacity** : 17.2 Ah
- **Voltage** : 12 V
- **Maximum Energy** : $17.2 \times 12 = 206.4$ Wh
- **Float Charge Voltage** : 13.5 v - 13.8 V



Figure 3.3: NP18-12R Battery

3.3.2 Encountered Problems

As the first solar panel's dimension are not adequate to the vehicle and the second panel's unavailability, a scrap alternative was required. The solution will be presented in the next section.

3.3.3 Realization

This section is divided into hardware and software realization, As time pressed the project and the budget was not enough, the choice was to work with alternative components just to explain the concept.

2 Arduino Uno¹ boards were used, therefore, the hardware control was split into two sections.

First section manages the turn signals and the horn. The board's setup is really simple as shown in the figure below.

¹a Microcontroller

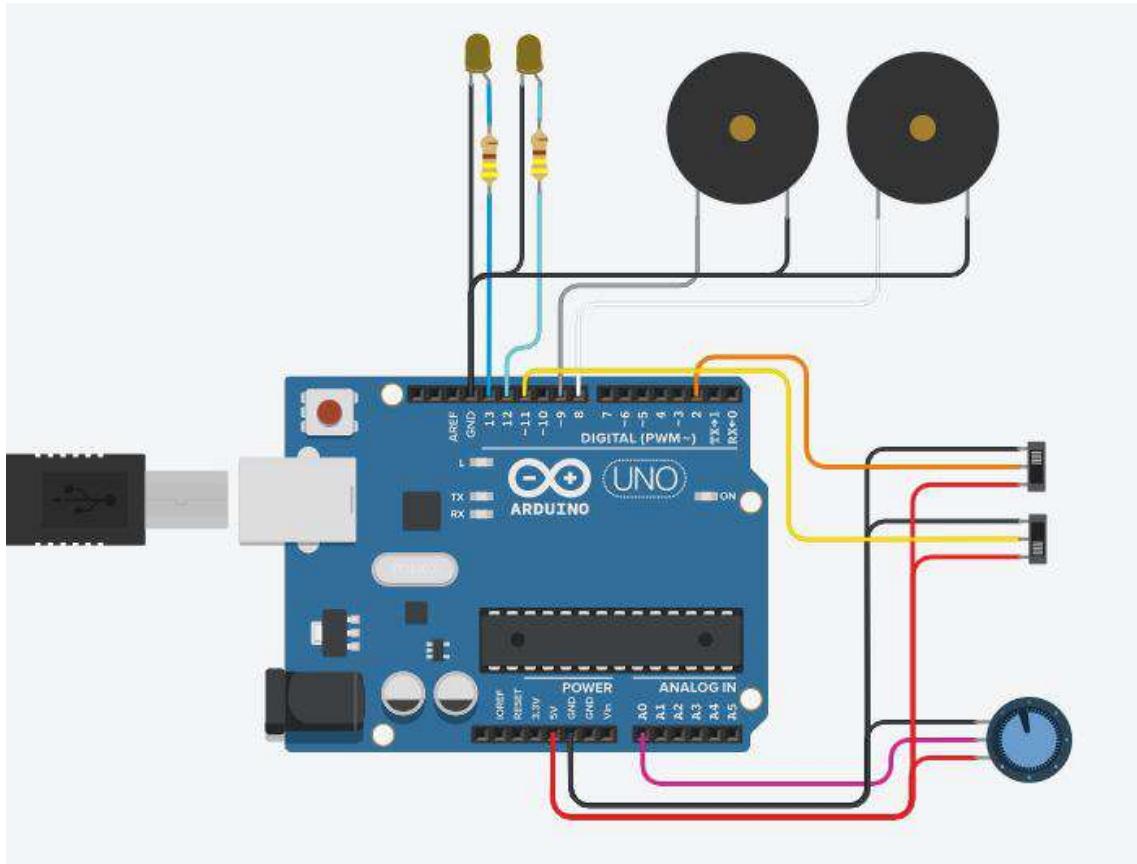


Figure 3.4: First Section

The circuit were first made via **TINKER CAD** for simulation, the compiled on a reel Arduino Uno board.



Figure 3.5: First Section - On board

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The algorithm uploaded on the Arduino Uno board is attached to the appendix. As shown, LEDs were used to simulate right and left turn lights.

The control of the turn lights is done via a potentiometer as if it indicates a certain values (below 50) the left LED starts blinking. If the value indicated is over 250, the right LED starts blinking

The range between 50 and 250 is considered to be a dead point where neither lights blink as all vehicles work.

The blinking is linked to a clicking sound made by a small buzzer.

If the user wants to four flashing of the turn signal, a simple click on the interrupter activates all LEDs to blink at the same time with a clicking sound.

The horn in the schematic is a piezoelectric component controlled by a push button.

From a programming point of view, these functions were created to assure that the system works properly.

- **Biip function** that controls the activation and the tone of the horn. This function only work by pressing the push button pinned to Pin 2. **Interruptions** were used.

```
void biip() {  
    tone(9, 300);  
}
```

Figure 3.6: Biip Function

- **Right and Left** for different cases

```
void Left() {  
    digitalWrite(13, HIGH);  
    tone(8, 1000, 2);  
    delay(500);  
    digitalWrite(13, LOW);  
    tone(8, 1000, 2);  
    delay(500);  
    noTone(8);  
}  
  
void Right() {  
    digitalWrite(12, HIGH);  
    tone(8, 1000, 2);  
    delay(500);  
    digitalWrite(12, LOW);  
    tone(8, 1000, 2);  
    delay(500);  
    noTone(8);  
}
```

Figure 3.7: Right and Left Functions

- **dbflash** for double flashing

```
void dbflash() {
    digitalWrite(12,HIGH);
    digitalWrite(13,HIGH);
    tone(8, 1000,2);
    delay(500);
    digitalWrite(12,LOW);
    digitalWrite(13,LOW);
    tone(8, 1000,2);
    delay(500);
    noTone(8);
}
```

Figure 3.8: dbflash Function

- **nothing** to rest the variables

These two figures represent the loop and the setup function used :

```
void setup() {
    // put your setup code here, to run once:
    pinMode(11,INPUT_PULLUP);
    pinMode(A0,INPUT);
    pinMode(12,OUTPUT);
    pinMode(13,OUTPUT);
    pinMode(2,INPUT_PULLUP);
    attachInterrupt(digitalPinToInterrupt(2), biip, LOW);

    void loop() {
        if (digitalRead(2)) {
            noTone(9);
        }
        x=digitalRead(11);
        a=analogRead(A0);
        if(x==0) {
            dbflash();
        }
        else if(a<50) {
            Right();
        }
        else if(a>250) {
            Left();
        }
        else{
            nothing();
        }
    }
}
```

Figure 3.9: Loop and setup Functions

The conditions made to chose the blinking turn signals as well as the interruption used to activate the horn are easily determined.

For the second section, an LCD was used to display as an option to display the direction

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of the vehicle. In case the vehicle is moving backward, the ultrasound sensor indicates the distance to the obstacles and send a buzzing sound to alert the user in cases of getting too close to the object. The distance chosen to signal the alert is 5 cm.

The schematics is done via **TINKER CAD** then tested on real hardware.

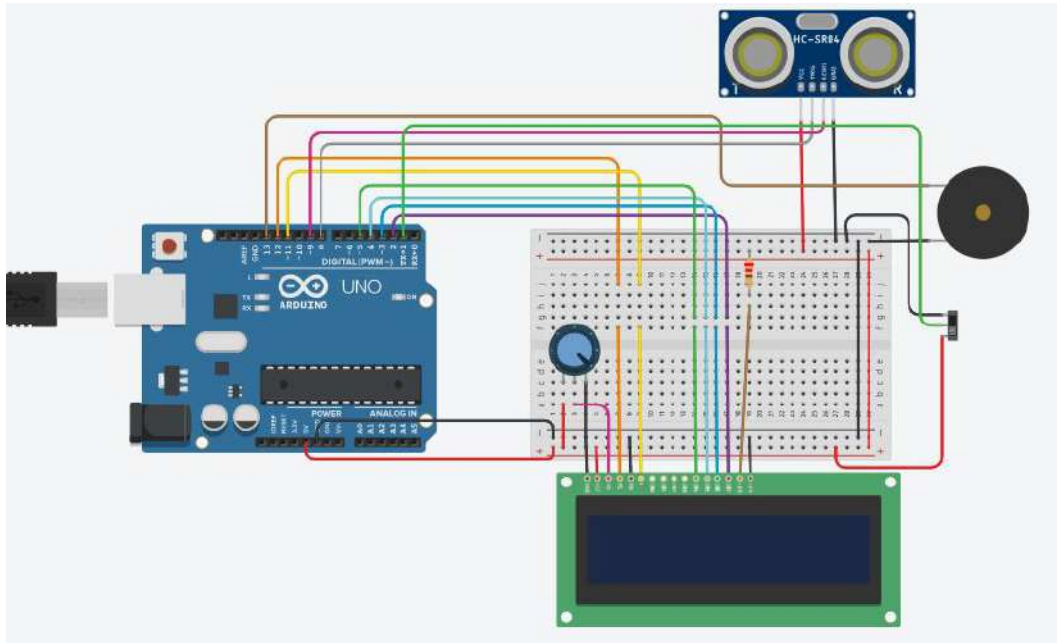


Figure 3.10: Second Section

The button here represents the Forward/Backward change sensor in the vehicle. Whenever the car is started, the LCD displays the Tunisian Flag then curves to the left to indicate the vehicle's name while blinking. Afterwards, The LCD displays "System On" for 2 seconds followed by the state of the direction lever.



Figure 3.11: Starting the Vehicle

- **Driving Forward** indicating the direction via arrows.
- **Driving Backward** showing the distance to the closest obstacle and beeping if it is within a range of 5 cm.



Figure 3.12: Forward/Backward Display

Even-though it looks like simple work but the flag display actually took to much effort. The algorithm is attached to the appendix.

3.3.4 Solution to Energy Source

After searching for enough time, a product that can satisfy the requirements was found. This product is called **SunBell-Smart**² a camping lantern with enough energy production to cover the needs.



Figure 3.13: SunBell-Smart

Full details will be joined to appendix.
These are some characteristics :

²BRIGHT - Norwegian Comapnay - <https://bright-products.com/product/sunbell-smart/>

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- **Battery Size** : 7.5 Wh
- **Battery Life Span** : 1 500 charge cycles
- **Energy Provided** : 1.8Wp
- **Estimated Runtime** : 7 hours

The light in the lantern can be used as headlights and taillight and their size is extremely small.



Figure 3.14: SunBell - Battery and Solar panel

Assuming it would need 3 articles from this products. This system as a whole was put to test, and it turns out that it works fine. However, due to lack of time and budget, this part should be completed in the next updates of this project.

3.4 Conclusion

As shown in this chapter, the calculation led to the 20 W solar panel to be the adequate fit. However, for its unavailability, the work should continue with the Sun-Bell panel for

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simulation of the proposed features.

Finally, the tasks turned out to work properly, letting the electrical part of this project be limited to this level and open for future upgrades depending on the availability of the needed products.

At the end of the next chapter, more ideas will be offered to optimise this part and the electrical autonomy of the vehicle as a whole.

Chapter 4

Prototyping

4.1 Introduction

This chapter focuses on the work done to make the prototype, as well as the different solution and problems encountered. This means, the previously mentioned formulas is used during this part in addition to most approaches done, failed or succeeded, chronologically ordered following the real-life prototyping.

The offered budget for this work was : **2 000 TND**

4.2 Motor

4.2.1 First Solution

The first suggested solution was designing a pneumatic air motor. This work referred to **Dr. Peter Beater's** *Pneumatic Drives*, in *Modeling and Control of Pneumatic Vane Motors* [13], **Mr. Mohamed Mehdi Filali's** Course in *Advanced Control Technologies* [16] and Chapter 2.

So, using the formula :

$$T = \frac{15.9 \cdot V \Delta p}{1000} \quad (4.1)$$

Where T : Torque *N.m*

V : Resultant of the Volume *cm³*

Δp : Pressure Difference *bar*

This formula can determine the Torque of vane engine.

Referring to the second chapter's formulas, the dimensions of the cylinder that can be used were calculated

However, for confidentiality matter, this work provides only the schemes regarding the engine.

Unfortunately the concept can not be applied due to the complexity of the design. In

Chapter 4. Prototyping

addition, the quantity of compressed air required is huge. That's why other approaches were suggested as alternatives.

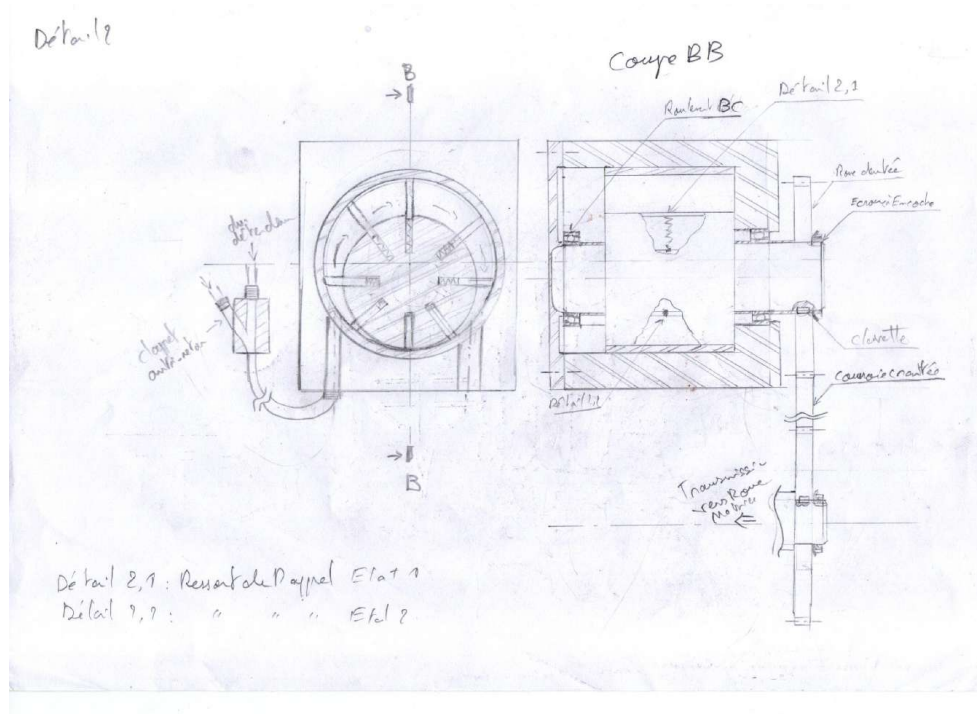


Figure 4.1: Vane Motor Design (no scale)

The full design is attached to the appendix

4.2.2 Second Solution

Considering that the First Solution was too complex to be made locally, this approach refers to the available Compressed air-engine manufacturers.

4.2.2.1 Calculations

The Conditions chosen are that the vehicle would not overcome **180 kg** including a user of **80 kg**.

Considering that the vehicle is a three-wheeler, this should make the gravitational force divided by 3 due the triangle shape.

$$F = m \cdot g = 1800N \quad (4.2)$$

Where g is the gravitational acceleration (N/kg). Let it be equal to 10 to simplify calculation.

By this point, every wheel should have a Force equal to 600 N. Now, to ease the calculations, the center of mass was considered to be equally far from all wheels.

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The wheels' diameter is 0.6 m

This table should cover all results needed to chose the motor in three different cases, please refer to the previous chapter for used formulas :

Parameter	Option 1	Option 2	Option 3
Vehicle Speed (km/h)	40	50	60
Linear Velocity (m/s)	12	14	17
Acceleration (m/s ²)	1	1	1
Rotation Speed (rpm)	382.17	445.86	541.4
Torque (N.m)	18	18	18
Power (W)	720	840	1020
Estimated Needed Power (kW)	1	1	1.5

Table 4.1: Estimated Needed Power (increased)

This table covers the range of air motors using Atlas Copo's search tool :

Reference	Speed (rpm)	Torque (N.m)	Power(kW)	Consumption (L/min)	Weight (kg)
LZL05-BF322-A-012	389	30.24	1.23	36.1	14
LZL35-BF322-A-012	386	123.99	5.01	129.2	34

Table 4.2: The only Matching Engines Available in Market

All details are included in appendix.

4.2.2.2 Problems Encountered

Unfortunately, after contacting Atlas Copo's  Branch In Tunisia, there are no available Air Engines and any type would need to be imported. After searching for all types or air engines in Tunisia, there are none to be found in all manufacturers.

The only Air motor found was in scrap and it costed **1000 TND**

An other drawback for this solution is, as shown in the table above, the motors consume a big quantity of air making it impossible to be used for a vehicle.

In addition, the weight and dimensions of this type of engine is considered relatively huge. The solution proposed was using an **air wrench**. Their prices are within acceptable ranges and the modifications are quite possible.

¹Founded in 1873, Swedish multinational industrial company



Figure 4.2: Air Wrench

However, the performance of these guns are inadequate regarding the project's needs, therefore, a train of mechanical gears is required to reduce the speed and rise the torque. None of the workshop contacted accepted to make such items within reasonable prices, plus, getting it from scrap would probably cost more.

In conclusion, This solution had to be dropped due to non-availability of the needed products and the budget's constraints.

4.2.3 Third Solution

This approach uses the system invented by **Leonardo Da Vinci**² with some adjustments.

The idea is called **The Ratchet Wheel with Slanted Pawl**.

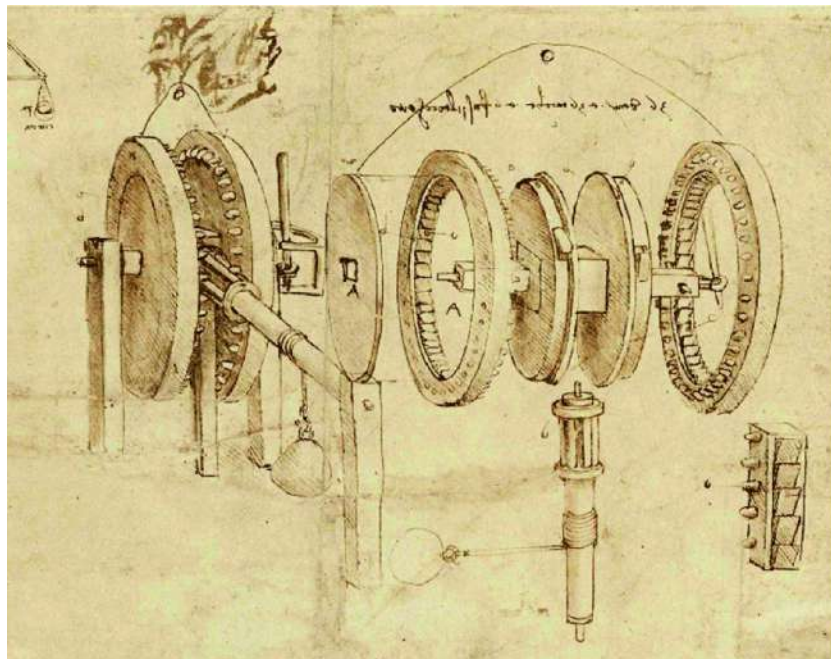


Figure 4.3: Leonarod Da Vinci's Ratchet Wheel

²1452-1519, Italian polymath

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The wheel teeth has right-angled profile. The pawl is placed on a slant leaves the wheel to rotate freely counterclockwise but not clockwise. Adding two wheels at each side, the machine gives a quasi-continuous rotation at it's output.

Placing a Cylinder's rod on the lever, that gives the linear movement, guarantees a rotation that can be used to drive the wheel.

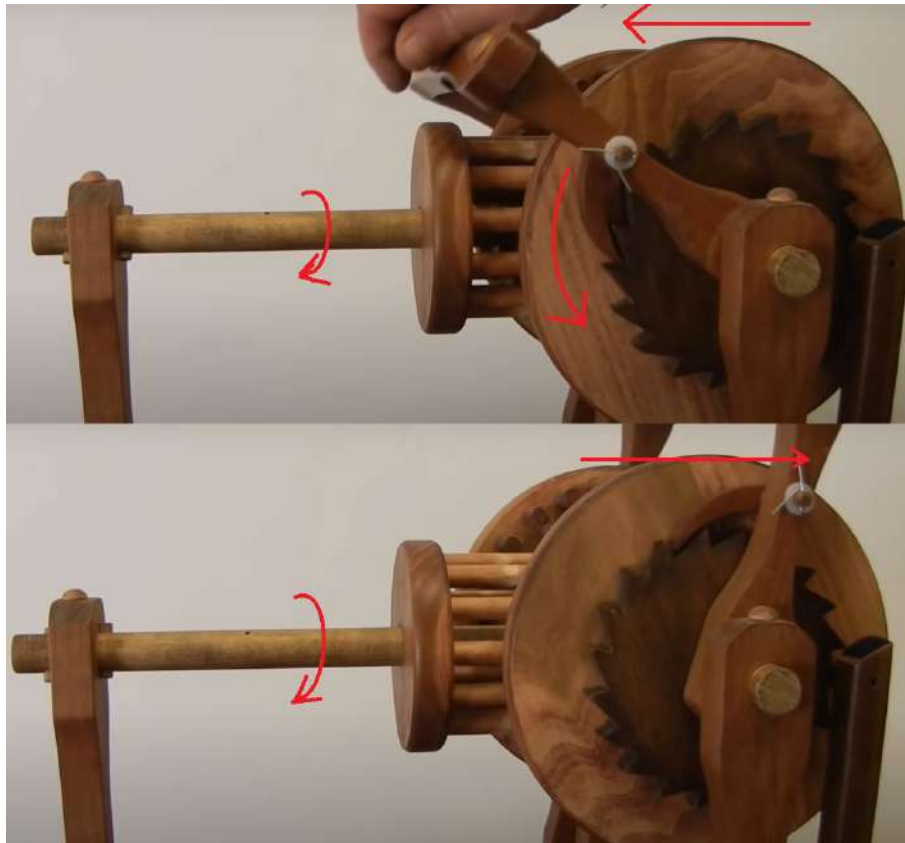


Figure 4.4: Ratchet Wheel Phases

The idea is so simple, yet, it had to encounter many constraints :

- Material that will be used on the profiles should be strong enough to support the forces. So, a whole study should be conducted.
- In case the study is done, a whole budget should be destined to the testing only of this part.
- There are no parts even in scrap that could resemble this idea so it had to be made.

Unfortunately, the idea was dropped to find other alternatives.

4.2.4 Forth Solution

4.2.4.1 Description of the Prototype Engine

The principle is simple and similar of that of an old steam engine.

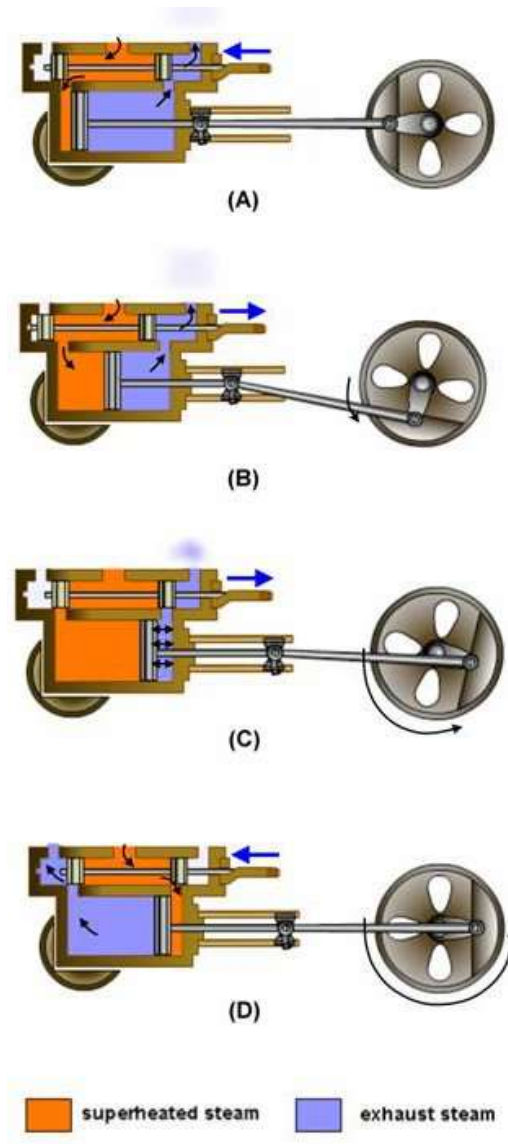


Figure 4.5: Steam Engine Phases

For explanation of the phases, The first phase (A) represents the expansion ; The heated steam flows through the engine inlet pushing the piston forward, therefore, turning the wheel.

In the second phase (B), the exhaust steam exits the engine due to the force applied by the heated air from phase (A).

Afterward, the piston reaches its full expansion letting all the exhaust air out. By

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this time, the wheel reaches almost 180 degrees turn which corresponds to the phase (C).

Finally, due to the stored kinetic energy in the wheel (Counterweight), the rotation in (D) should continue at the same rate and pushing the valve' rod to change the stem direction, inverting the cylinders flow in/out ways.

This procedure continues inverting every time the steam ways change creating the rotary motion in the wheel.

Relying on this technique and the references from Chapter 2, imitation of the function of an old steam engines was done by creating the same components.

This figure is a representation of a steam engine with its legend.

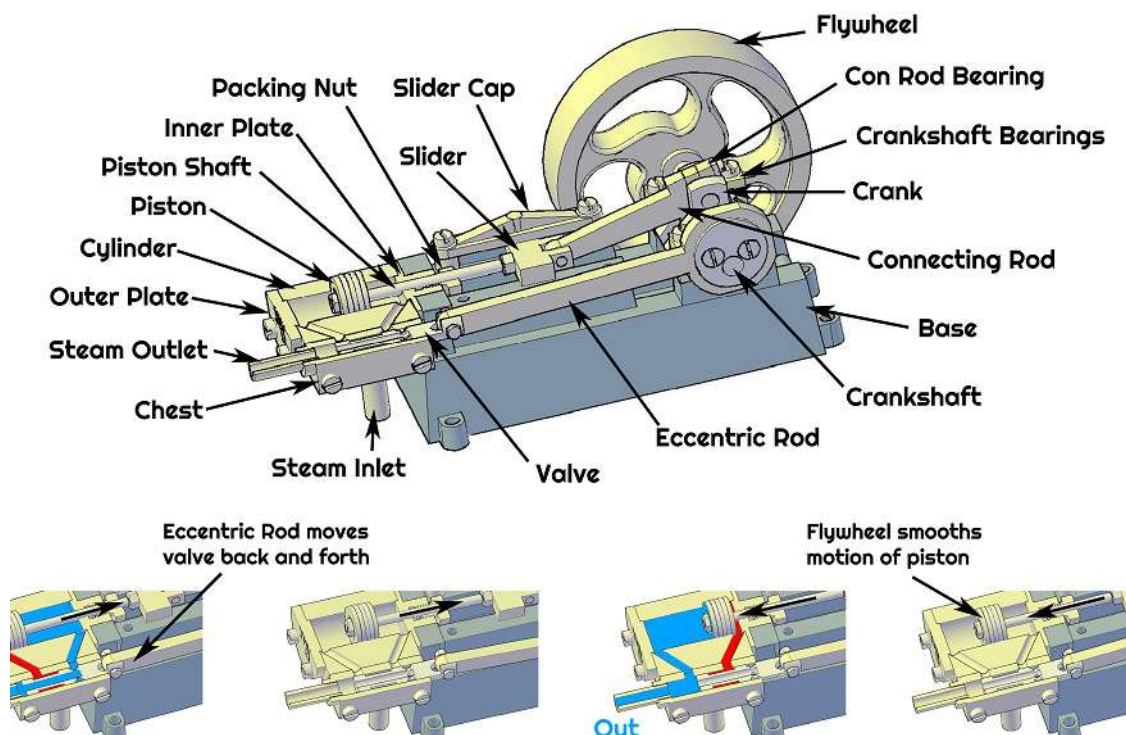


Figure 4.6: Steam Engine with Legend

4.2.4.2 Design and Application

To ease the work, a representation of the functioning engine was created via SMC eTOOLS.

In the following, the reader finds a pneumatic schematics of this circuit.

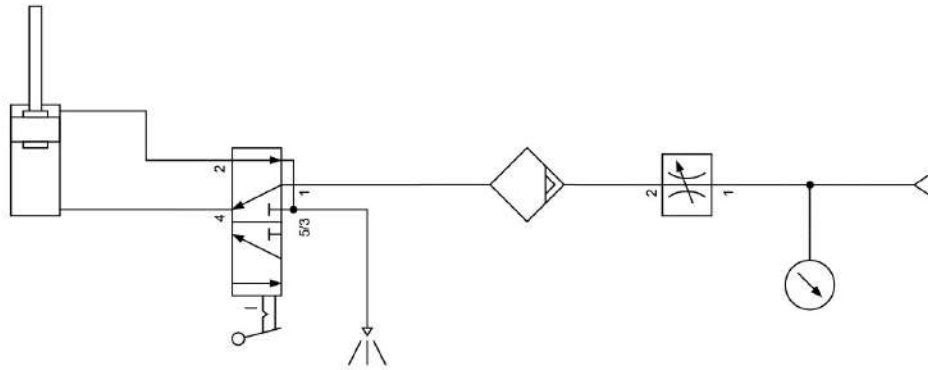


Figure 4.7: Pneumatic Schematics of the Fourth Engine

For more details, please refer to the appendix.

First, a small pneumatic cylinder was used reaching a rotation speed **180 rpm**.



Figure 4.8: Fourth Motor (realized)

As shown in the figure above, (1) represents the cylinder used. It's a DSNU-20-260-PPV-A FESTO air cylinder.

- **Estimated Price :** 145 TND
- **Piston Diameter :** 20 Ø

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- **Piston Stroke :** 260 mm
- **Speed at 6 bar (with stick-slip-free operation, horizontal, without load) :** 10 ... 100 mm/s (or 0.01 ... 0.1 m/s)
- **Force at 6 bar :** *Advancing* 189 N and *retracting* 158 N
- **Weight (after adding 26 mm stroke) :** 205.5 g
- **Maximum Lateral Force (full stroke) :** 45 N

All the details will be included in the appendix.

The element (2) represents the piston rod or shaft with a length of 260 mm.

The element (3) represents the concerting rod with a length of 285 mm.

The element (4) represents the Flying wheel.

The element (5) represents the counterweight.

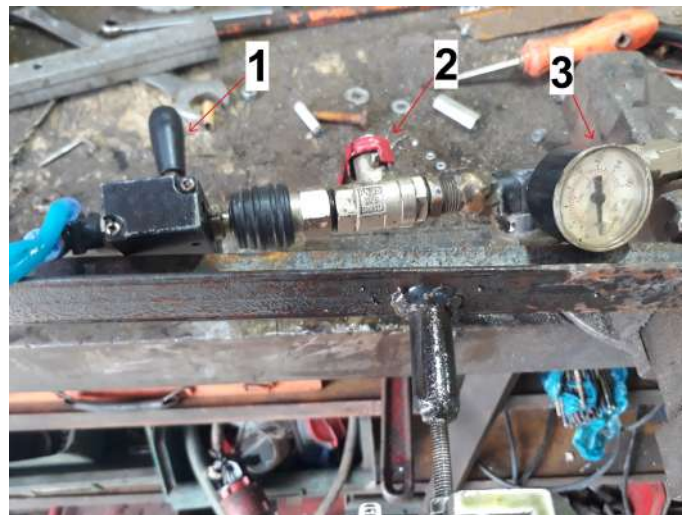


Figure 4.9: Engine Inlet

To command the speed of the engine, adding a flow control valve (3) is mandatory. This valve contains an indicator gauge. It should be enough to show the pressure flowing in the piston.

Followed by a shut off valve labeled (2). If the valve is off, the engine will not run, else it starts to rotate according to the level of the opening of the valve (2).

To manipulate the rotation of the wheel, a directional valve with a lever 5/2 as indicated labeled in the figure above by (1) was used.

- **Estimated Price :** 300 TND
- **Pressure Range :** $0 \rightarrow 10$ bar

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4.2.4.3 Cost Estimation

The following table represents the cost estimation of this solution and the application.

Article	Quantity	Estimated Price (TND/Unit)	Price (Scrap + maintenance) (TND/Unit)	Total/ Article
Air Cylinder	1	145 (FESTO)	80	80
Directional Valve 5/2	1	300 (FESTO)	50	50
Shut off Valve	1	30	5	5
Flow Control Valve	1	62 (FESTO)	25	25
Lines (6mm)	1m	7	-	-
Fittings (6mm)	5	4.5	2	10
Total		566.5 (5×Fitting)		165

Table 4.3: Cost Estimation

4.2.4.4 Problems Encountered

- **Availability**

One of the major problems encountered during this phase is finding the right equipment to make the engine, for instance, trying contacting several pneumatic systems providers in Tunisia, none had the required articles.

For example, after contacting *HP Automatisme*, information was received that the required cylinder or even closer models were not available in Tunisia. Therefore, it would need to be imported from France which would take at least 2 months in normal conditions.

- **Technical Difficulties**

After equipping the engine to the vehicle, the cylinder has encountered two major problems leading to the destruction of its rod, therefore, it would not work as intended;

1. *Rod Bearing Failure* : this problem occurred due to the fact the maximum point of the stroke was behind the cylinder. This happens because of the uncertainty in man made machines.
2. *Rod Buckling* : this problem occurred due to the long stroke relative to the rod diameter (Rod bearing Failure). This drove the Rod to bend and block the system.



Figure 4.10: Rod Buckling and Bearing Failure



Figure 4.11: Rod Buckling

After loosening the first cylinder, changing the stroke and rod diameter was a necessity to make the system more rigid.

4.2.5 Fifth Solution

4.2.5.1 Design and Application

Changing the cylinder is mandatory at this stage, that's why another type was tested, without changing the set up of the pneumatic system.

This approach uses a **BOSCH REXROTH MECMAN PNEUMATIK 168-04-1600-1** air cylinder with the reference below:



Figure 4.12: 168-04-1600-1 Cylinder reference

- **Estimated Price :** 821,92 TND

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- **Piston Diameter :** 40 Ø
- **Piston Stroke :** 160 mm
- **Force at 6 bar :** *Advancing* 756 N and *retracting* 636 N
- **Weight (after adding 26 mm stroke) :** 806.4 g

For more details, please refer to the appendix.

Using this cylinder, the reached a speed was at **120 rpm**. However, by decreasing the speed of the engine, the torque rises. Therefore, the weight of the whole vehicle can rise.

4.2.5.2 Cost Estimation

The only thing that changed between the first engine and the second is the price of the cylinder :

Article	Quantity	Estimated Price (TND/Unit)	Price (Scrap + maintenance) (TND/Unit)	Total/ Article
Air Cylinder	1	821,92 (BOSCH)	15	15
Total		1225.42 (5×Fitting)		95

Table 4.4: Cost Estimation

4.2.5.3 Problems Encountered

- **Technical Difficulties**

The maintenance of such a cylinder was really difficult due to the old product bought. Some parts needed to be replaced several times increasing its cost.

This cylinder need twice the amount of compressed air used by the first type, decreasing significantly the autonomy of the vehicle.

4.3 Chassis

During this project, many designs were taken in consideration. All designs proposed in this work are three-wheeled. This fact is explained by the complexity of the differential system used in four-wheel vehicles. Therefore, using three wheels, with two driven, will be the easiest to make.

Please refer to Chapter 2 for calculations will not be included in this work.

4.3.1 First Design

The first design was created basing on the vehicle's mechanical needs. Therefore, it was considered complicated to be made by hands. The following figure shows the sketch of first design:

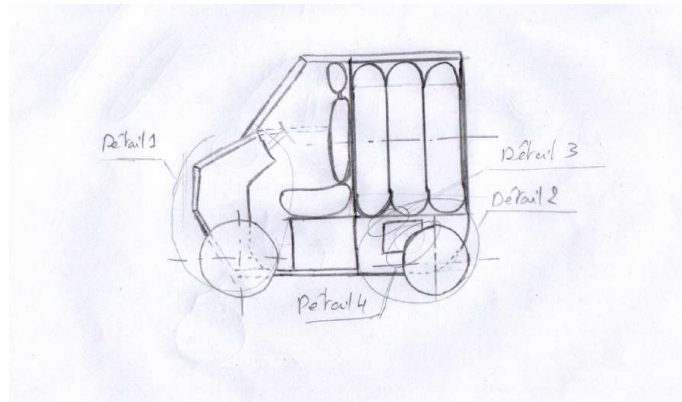


Figure 4.13: Draft of the First Design

For full schemes, please refer to the appendix.
The estimation of the price was really high that this idea of design was deemed to be rejected.

4.3.2 Second Design

The second proposed design was taking the old skeleton of a buggy and modifying it to suit the project

This solution is the most accurate in term of prototyping because the buggy skeleton is generally made to suit any kind of mechanism.

With such a solution offers a significant amount of gained time to focus on other parts of the project. However, the prices are really high regarding the project's budget. After contacting several scrap sellers, the cost of this solution without any maintenance ranges between :

- 1800 TND without any accessories (maintenance needed)
- 6000 TND with full accessories (no maintenance needed)



Figure 4.14: Buggy Skeleton (Similar Design)

Please refer to appendix for the schematics.

4.3.3 Third Design

The third design was a race Three-wheeled vehicle of a senior year mechanical engineer group from an American University [3].

This project was done in 2011, so, as a solution, this part was based on their papers



Figure 4.15: Third Design [3]

The figure above shows the work done by the previously mentioned team. To make this chassis, a company was contacted to do this work. It has been sad that this chassis would take approximately one month with a relatively high cost. After negotiating, the agreement settled on :

- 1500 TND for raw materials
- 1000 TND estimated extra fees for work

This solution was dropped due to the high cost.

4.3.4 Fourth Design

This design is considered to be the simplest. The following paragraphs detail the work and design of each section of the chassis.

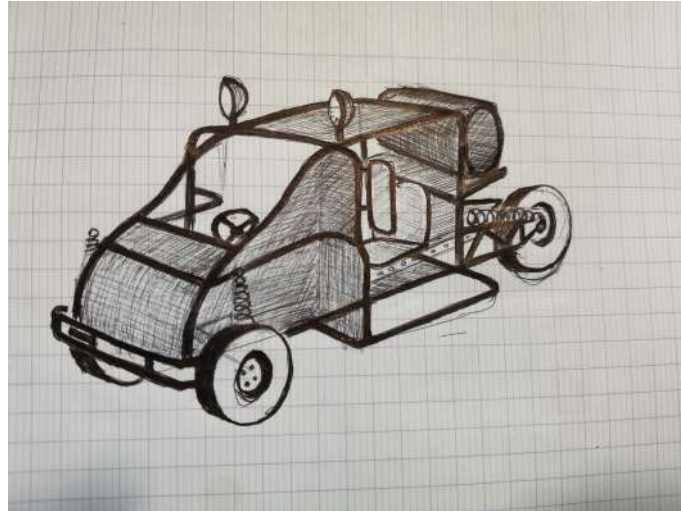


Figure 4.16: Draft of the Fourth Design

4.3.4.1 The general Form

The general form of the chassis is inspired by the three-wheeled vehicles designed by most manufacturers.

The choice was to split the chassis into two parts:

- **The Cockpit :** where the user will be sitting and controlling the vehicle.
- **The Engine Chamber :** where the engine is mounted

These two parts might move freely for a certain angle and fixed by two joints like shown in the figure below :

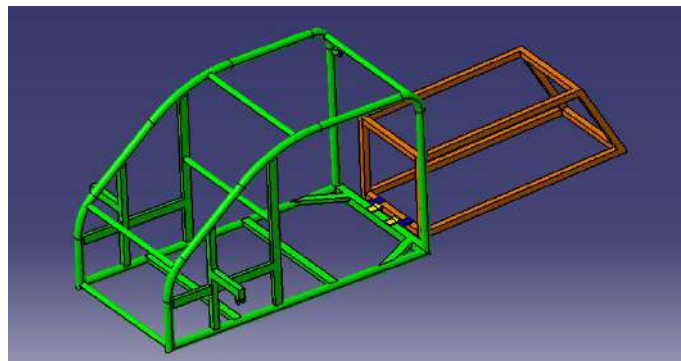


Figure 4.17: The Two parts of the Chassis Via CATIA

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Splitting the Engine Chamber from the Cockpit would reduce significantly the amount of vibration made by the motor while running.

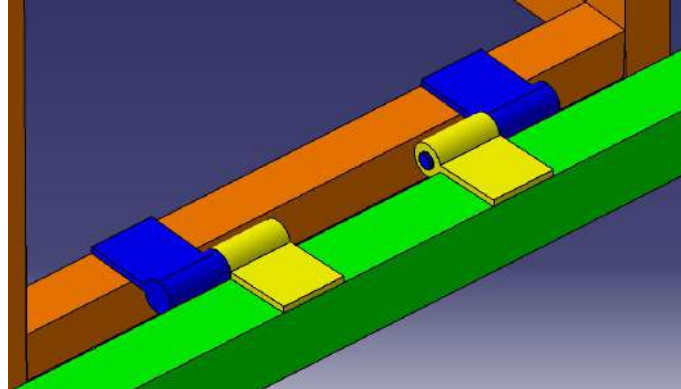


Figure 4.18: Modeling the Joints Via CATIA

This 3D Model shows the joints used to link the chassis' parts together while maintaining a degree of rotation used later to absorb the vibration by the dampers. The next step is applying the 3D models to reality. The application was the hardest part of the project since any wrong intervention might affect the overall results.

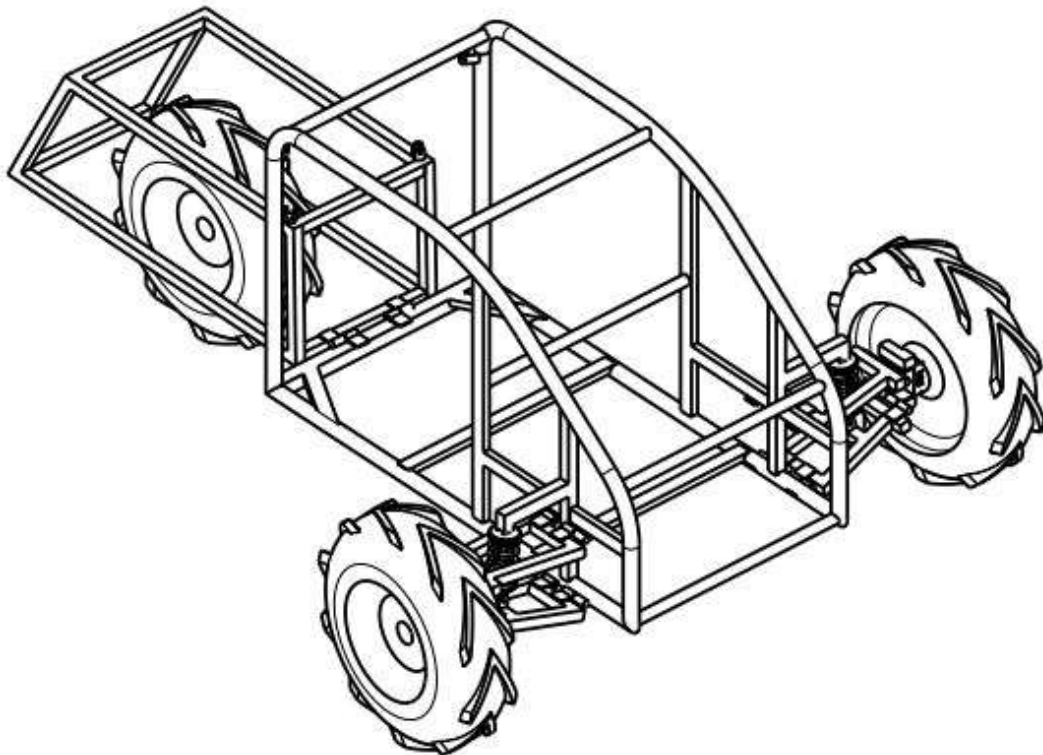


Figure 4.19: Isometric View (Scale 1:25)

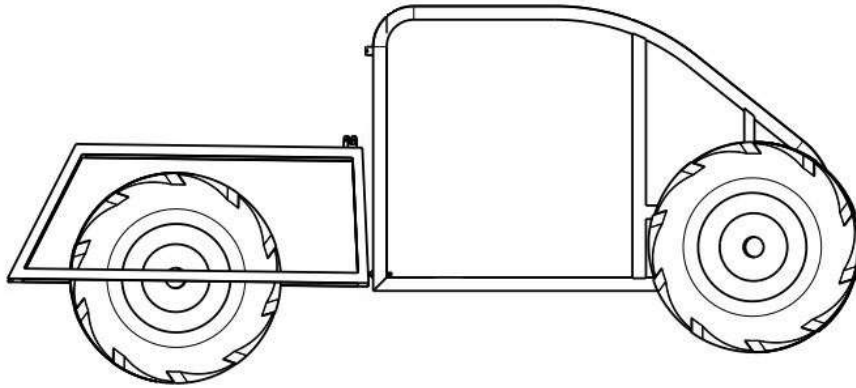


Figure 4.20: Side View (Scale 1:20)

These schematics are joined in the appendix with full details.

These are the steps taken to make the chassis:

- **Sizing :** The most important phase of prototyping is sizing, Before any procedure, it is crucial to map the dimensions of the cuts or the exact spot to weld. To do so, precut plastic bars were used to comprehend the real size of the vehicle



Figure 4.21: Calculating the sizes

- **Cutting :** The second phase is to cut the iron bars using a *disc grinding wheel* manually.



Figure 4.22: Cutting Iron Bars Using a Disc Grinding Wheel



Figure 4.23: Cut Iron Bars for Engine Chamber

- **Welding :** To join the bars together, welding all the irons to reach the needed form is crucial.



Figure 4.24: Cut Iron Bars for Engine Chamber

4.4 Vehicle suspension

The suspension system of a vehicle is considered to be the mechanical components that connect the wheels to the frame or body. Its role is to absorb all the shocks and vibrations.

The prototype represents a 3 wheel vehicle, two front and one rear.

4.4.1 Front Suspension

The idea of a front suspension comes from the general procedure used to make any front wheels suspension.

4.4.1.1 Modeling

The work refers to **CATIA** to model the necessary form like shown in the figure below:

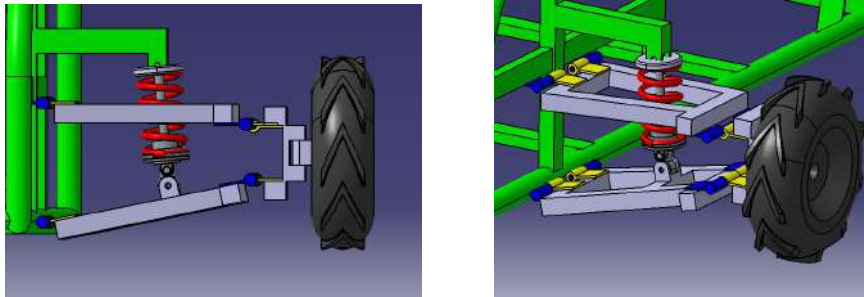


Figure 4.25: Front Suspension Via CATIA (Front and Perspective Views)

The next figure shows the position of the shock absorber in the suspension skeleton.

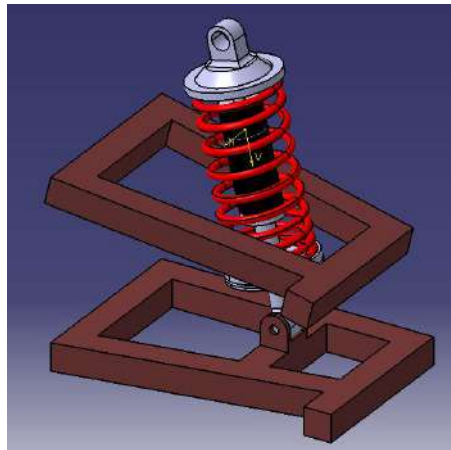


Figure 4.26: Shock Absorber and Front Suspension Via CATIA

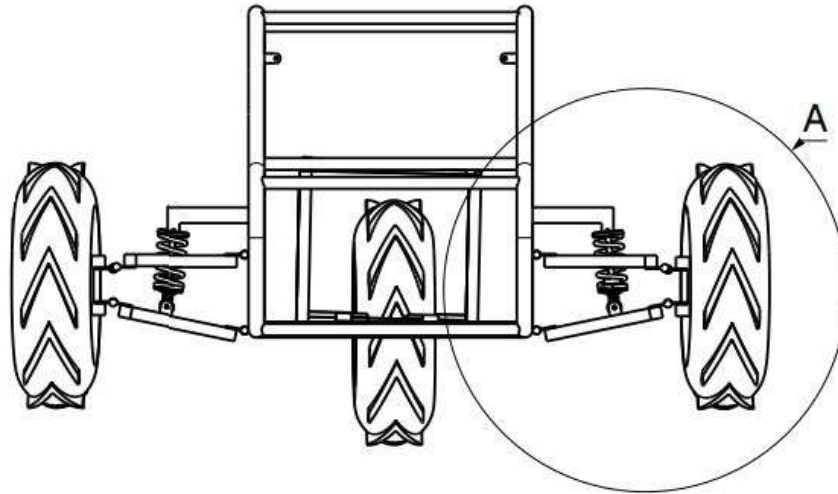


Figure 4.27: Front View (Scale 1:20)

The figure above is attached to the appendix.
All dimensions are on the schematic with a 1:20 scale.

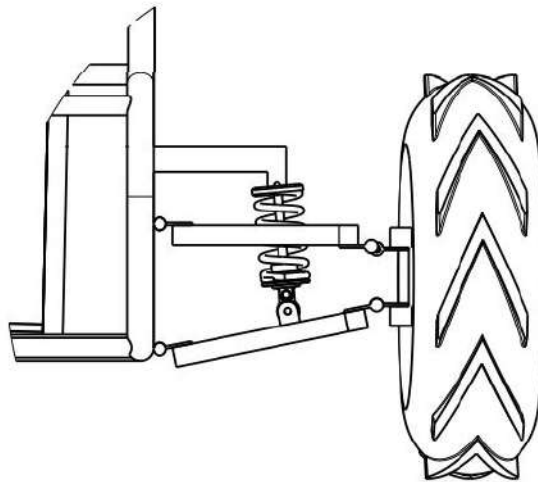


Figure 4.28: Front Suspension (Scale 1:10)

4.4.2 Realization

The following figures represents the real suspension after welding and mounting the wheels .



Figure 4.29: Front Suspension

4.4.3 Rear Suspension

The same concept was applied to the rear suspension. Its purpose was both to absorb the vibration from ground and engine, and to limit the rotation of the cockpit regarding the engine chamber.

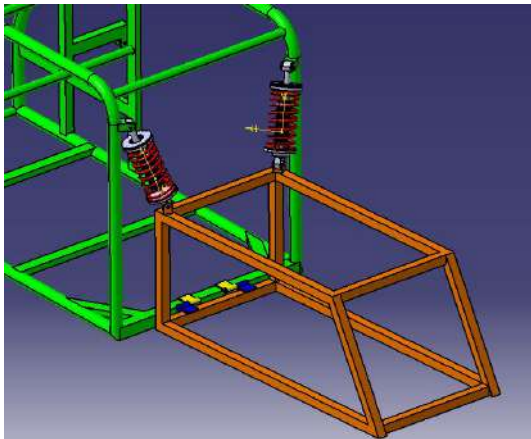


Figure 4.30: Rear Suspension(Modeled and realized)

4.5 Direction

For the front wheels direction, this work relied on a used an old **4L** Direction wheel mounted on a Universal joint, adding, at the end of the joint, a rack and pinon. The following figures shows the steps taken to apply the concept of this type of direction.



Figure 4.31: Universal Joint

This joint made it possible for the transformation of the torque needed to turn the wheels from the user's position to a tilted pinion.

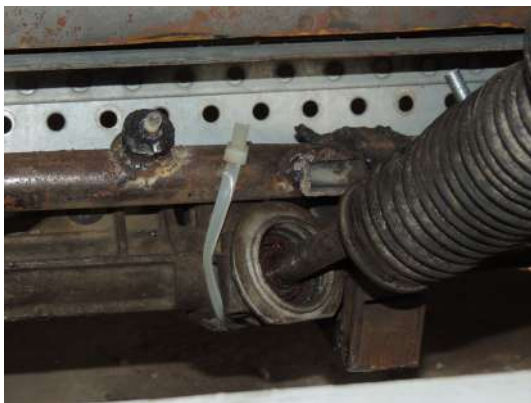


Figure 4.32: Rack and Pinon

As shown in the figure, the joints should be lubricated at all time to ensure the good transmission of the torque.

4.6 Transmission Chain

The transmission chain is the key element to transmit the force given by the cylinder to the driving wheel.

For this purpose, many ideas were proposed

- **Clutch :** Using a flying wheel and a clutch might be the best solution to drive the vehicle. In the early stage, the agreement was settled at using a Motorcycle clutch to this matter. One of the solution proposed was a centrifugal clutch like shown in the figure :

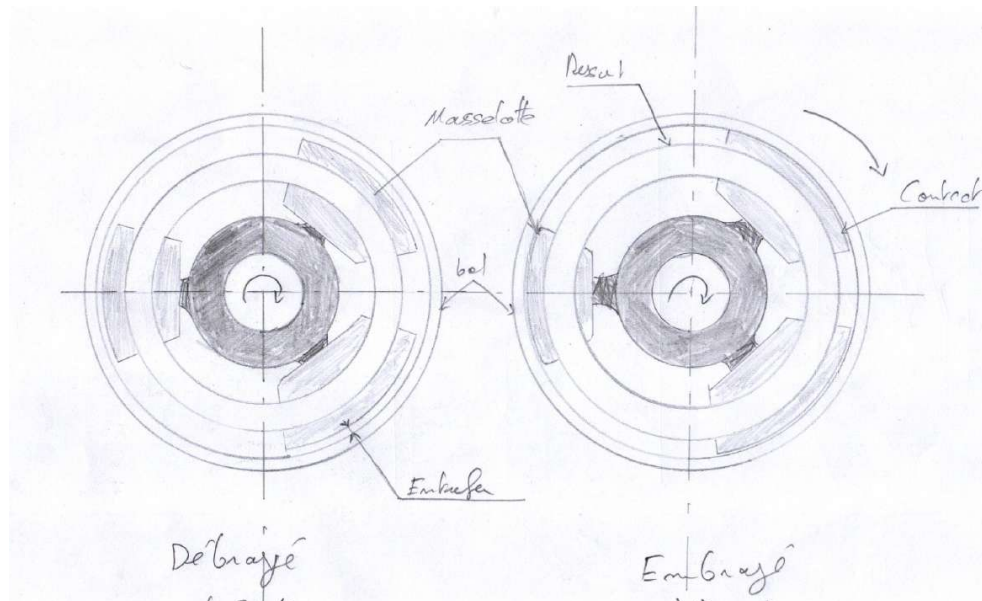


Figure 4.33: Scheme of the Centrifugal Clutch

Please refer to appendix for further details. This solution was dropped due to financial shortage.

- **Toothed Belt :** This type of drive is the most accurate as it does not need maintenance and its silence

This solution was dropped due to lack of customized belts length.

- **Sprocket and Chain :** This solution was adopted during this work due to its simplicity and availability. Chains are also adjustable in length. However, Maintenance was a big issue and it chains kept getting out of direction or dropping due to the length of transmission

Solution : Double Chain Drive : This technique shortens the length of chains and prevents its dropping.

In the following figure, the transmission chain is presented via CATIA 3D modeling.

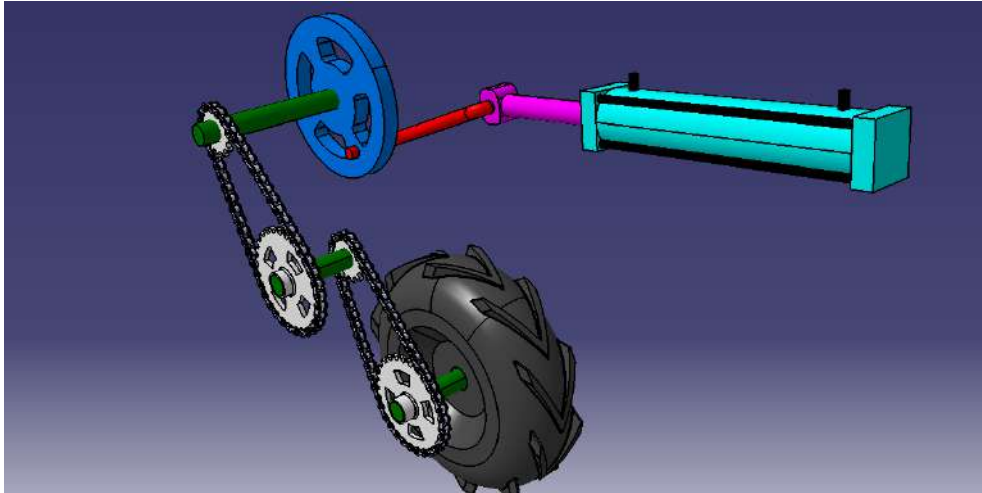


Figure 4.34: Transmission Chain Via CATIA

4.6.1 Realization

The realisation of this part of the project was the hardest as it needed special machines to make the transmission rods.

The work was done at **MECANORM** workshop giving access to use the necessary machines like:

- Milling Machine
- Metal Lathe
- Metal Saw



Figure 4.35: MECANORM Logo

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The raw material was scavenged from old metal cylinders.

4.6.1.1 Using Metal Lathe

Metal Lathe to lower was used the cylinders diameter and length.



Figure 4.36: Longitudinal and Face Turning

The lathe was used for tapping, to to screw the driving wheel.



Figure 4.37: Tapping (Manual and using Machine)

4.6.1.2 Using Milling Machine

To create rectangle form, the milling machine was used on the cylinders later to be equipped by the pinions for the transmission.



Figure 4.38: Using Milling Machine

4.6.2 Sawing

To cut Metal, there are two type of sawing machines

- Disc Grinding Wheel (Acceptable Surface State)
- Automatic Saw (Better Surface State)

Both machine were put to work depending on the type of surface needed.



Figure 4.39: Sawing Machines

All raw materials provided, the cost of the work is:

- Price : **185 TND**

This procedure took 2 weeks worth of work, to obtain the final rods.



Figure 4.40: Final Transmission Rod for Driving Wheel

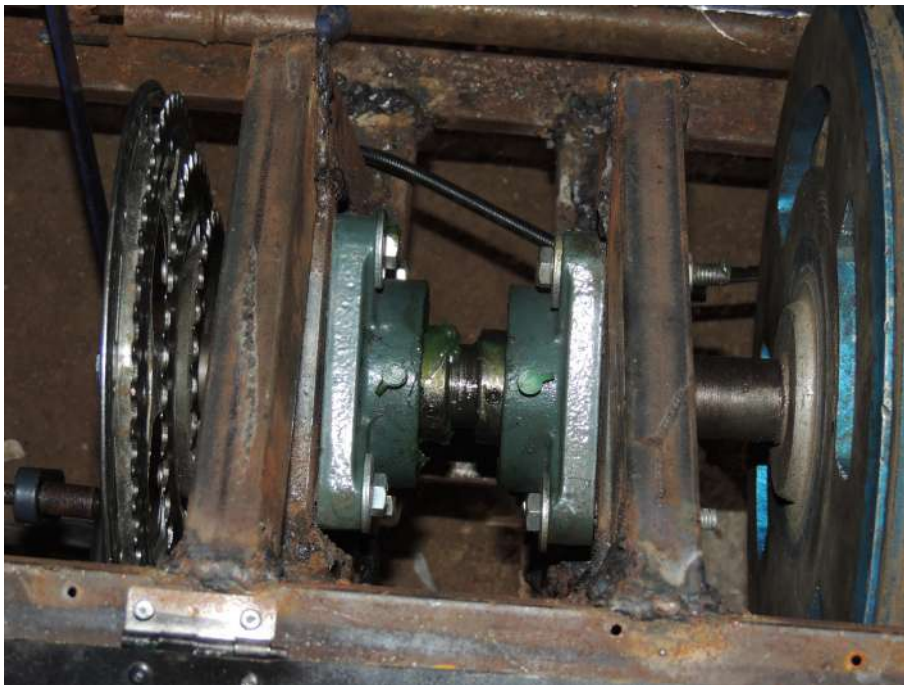


Figure 4.41: Final Transmission Rod for Flying Wheel

The pinion chosen has 3 levels.
This feature allows to change the torque and speed of the vehicle at will.

Two 3-level pinions were mounted in revers allowing the possibility to change the torque and speed ratio transferred.

First level has a $1/2$ Ratio regarding torque, the seconds has 1 ratio value and the third position has 2 ratio allowing to have twice the torque gave by the cylinder.

After Mounting Both Rods in their Places in the Engine Chamber, the work proceeded by adding the chain as long as ball bearings shown in Green in all figures.

The final stage looked like the figure below :



Figure 4.42: Engine Chamber Fully Mounted

4.7 Vehicle Control

This section presents the mechanisms proposed to assure any kind of motion in the vehicle as a whole, i.e. moving forward and backward, different speeds and braking.

4.7.1 Speed Control

As shown previously, in the engine prototyping phase, the speed control is guaranteed by the flow control valve and the shut-off valve. Along the project, some valves required replacement several times due to a number of losses and inadequate reductions (unavailable).

A lot of **Teflon** was put to use to cover the uncertainties.



Figure 4.43: Using Teflon

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The used flow control valve uses a Locking system and has a Gauge to indicate the pressure.



Figure 4.44: Flow Control Valve with Gauge

This item plays a double role:

- The first role is indication of the container's pressure. This allows the user to know when to refill the tank holding the pressurized air.
- The second role is to control the flow of air, therefore, controlling the pressure reaching the engine. For security purpose, the handle must be locked at 8 bars, i.e. only the designers would have access to this parameter.

The following item is a the shut-off valve :



Figure 4.45: Shut-off Valve

This valve works an accelerator pedal in normal cars. The user is able to open the valve as needed controlling how much air should reach the engine, meaning, controlling the speed of the motor.

4.7.2 Forward/Backward Control

Controlling the Forward/Backward direction of the vehicle was not any easy task. Many variables had to be known to choose the way the user wants the vehicle to follow. Crankshaft system mounted to the flying wheel and linked to two rods a 90 degrees phase shift to the directional valve needed to be made.

The idea as it presented in the figure below is a Standard Stephenson Valve Gear. This design was considered to be complicated and needed to much financial investment to make it work properly. So it was deemed to be left for future upgrades.

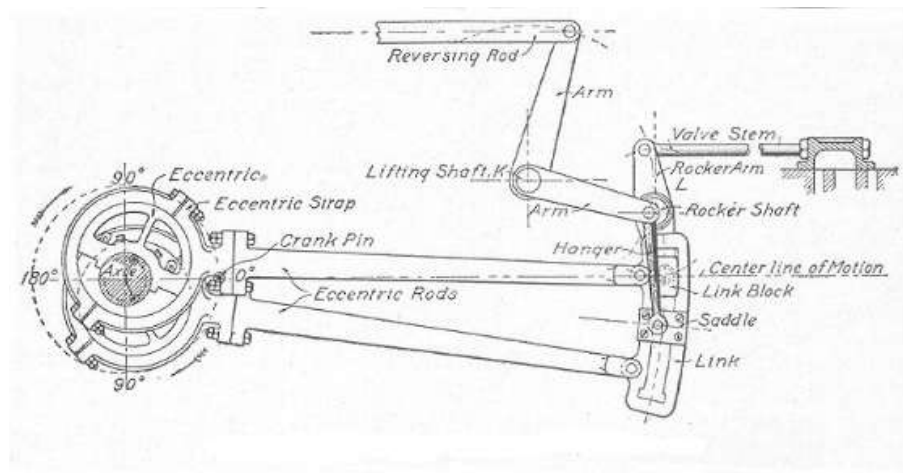


Figure 4.46: Standard Stephenson Valve Gear

This work was a much simpler technique using a handle equipped next to the user. This lever is attached to the lever of the directional valve that any time the user of the vehicle wants to change the direction, a push or pull action is needed.

To prevent any uncertainties done by the handle, a lock position was made in which the user would not be able to change the direction until the vehicle stops.



Figure 4.47: Lever Control Forward/Backward Attached to Directional Valve

The end of the Handle's rod is linked to the directional valve via a metal rope. That

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way, acting on the handle affects the valve's lever, however, while the handle is at lock position, no change would be felt from the valve change, i.e. a one-way, non-reversible action.

The figure below presents the handle at it's lock position :

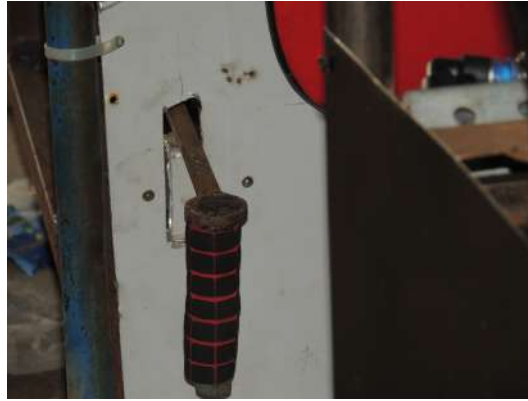


Figure 4.48: Handle at Lock Position

The handle would always face downwards until the user pulls or push it taking, then, a 90 degrees turn for a free action to take place. After letting go of the handle, it would automatically takes its original place to be locked.

4.7.3 Brake

The brake system in the vehicle is a centrifugal brake. Which means, no stopping action will accrue unless the shoes lock on the wheel. This figure explains the system used in the vehicle:

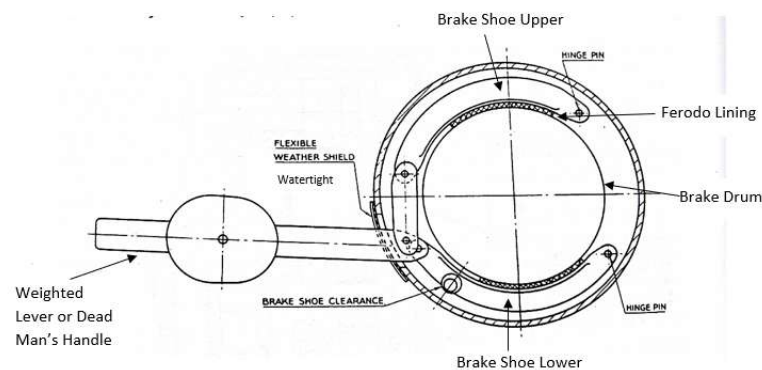


Figure 4.49: Brake System Schemes

Full schemes are provided in the appendix.



Figure 4.50: Brake System Mounted on the Driving Wheel

The dead man's handle is fixed to a metallic rope in a plastic sheathing, linked to a car handle with a lock button.



Figure 4.51: Brake Handle (Two Position)

The figure on the left represents the brake at its loose state so the vehicle may move freely. The other figure shows the brake handle up meaning the vehicle is stopped and would not move.

4.8 Storage

This section is dedicated to the storing systems proposed during the project.

4.8.1 Engine Air Consumption

To calculate the engine's consumption, the full volume of air needed for each stroke was required.

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To do so, summing the empty cylinder volume for advancing position and for retracting position was calculated via :

$$V_{stroke} = V_{advancing} + V_{retracting} \quad (4.3)$$

$$V_{advancing} = \pi \cdot D^2 \cdot h \quad (4.4)$$

$$V_{retracting} = \pi \cdot \frac{D^2 - d^2}{4} \cdot h \quad (4.5)$$

The engine is capable of running at a speed of **120 rpm** i.e. 120 stroke per min meaning 120 cycle per min.

Knowing the height h (160mm), the external diameter D (40mm) and the rod diameter d (16mm), a single stroke should be equivalent to 0.32 m. The pressure consider is 6 bar.

$$Q = V_{Stroke} \cdot n(p + 1) \quad (4.6)$$

All calculation done, the consumption should be equal $Q = 45.4$ l/min (0.75 l/s) So, this engine will need 45.4 liters of air every minute at a pressure of 6 bar. Referring to the formulas used in section 2, calculating the need tank to hold this quantity is possible.
For 10 bar, the engine consumes 75.7 l/min.

The provided pressure in gas stations is 10 bars, using this pressure, considering that the vehicle would run for 30 minutes before recharging, it would take 2271 Liters.

4.8.2 Storage Unit's Volume

Then , the vehicle would require a volume of 2.271 cubic meter which might be a problem.

The required solution is to go higher in pressure, taking that all manufactured CAV function within a pressure range of 100 bar to 300 bar.

The other alternative is to make the tank big which is considered to be a problem.

4.8.3 Realization

There was no tanks that would support more than 15 bar of pressure available and no high compressed air providers, a small air tank was chosen for prototyping purpose.

The only available solution is using the butane gas tank. Converting it from the domestic use to the vehicle was not an easy task as it needed to be completely emptied using water then welding a new opening using welding torch.



Figure 4.52: Emptying the Tank with Water



Figure 4.53: Welded Opening and Shut-Off Valve

Adding a shut-off valve would assure an other level of security. When refilling the tank, the user should close the valve and dismounted easily the tube. Then, it should be easy to link the compressor tube to the tank opening the valve again for refilling.

Once full, the user should close the valve and mount the tubes again.



Figure 4.54: Refilling Injector

The tank is fixed on the front of the vehicle and covered.
Other solutions were proposed but none made it to the stage of realization, mentioning the idea of using air chambers as storage. The idea was abandoned because air chambers are considered to be bad storage unit without their wheels.

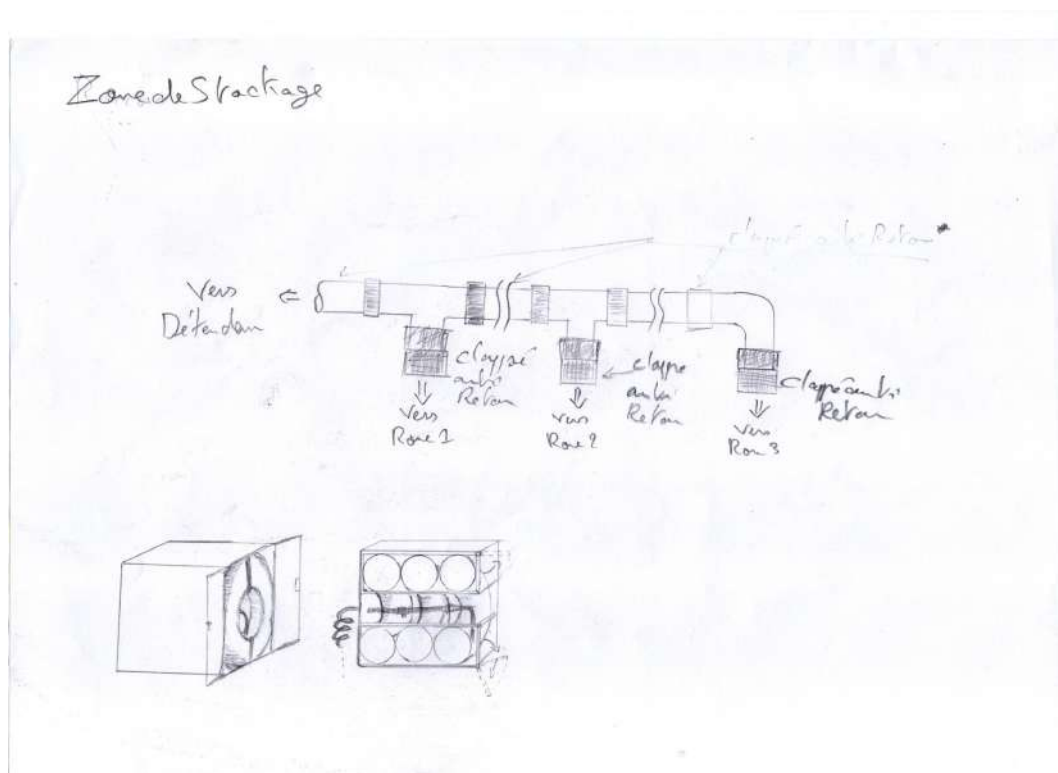


Figure 4.55: Proposed solution

4.9 Final Touches

At this point, it is assumed that the vehicle is ready to be used. However, some final touches are required

4.9.1 Cover

To cover the skeleton iron sheets (1 mm) were used :



Figure 4.56: Covering the Vehicle

First step was taking the needed measures, and cutting the iron sheets using a grinding wheel. Then, different hole both on the sheets and the skeleton were drilled and fix them to each-other using screws.

For the engine chamber, this system was made, the the user can lock to do the needed maintenance in the chamber and unlocked later to close it.



Figure 4.57: Opening of the Engine Chamber

4.9.2 Paint

choice was linked to the National Engineering School of Tunis' logo color.



Figure 4.58: ENIT's Logo

The two colors are:

- **Yellow** : Tangerine Yellow # ffc000
- **Blue** : Night Blue # 0e1b81



Figure 4.59: Paint Colors



Figure 4.60: Final Prototype

4.10 In field Testing

4.10.1 Capability

Conditions ;

User : 80 kg

Tank Pressure : 10 bar

Flow Pressure : 8 bar

Surface : 0 degrees (flat)

4.10.1.1 Storage

- **Free Load**

The testing was done first on a suspended vehicle i.e without the driving wheel touching ground.

For a full tank at 10 bar pressure, the tank could run for 2 minutes and 45 seconds before the pressure reach it's critical limit and no longer has enough force to rotate the wheel.

- **Empty Vehicle** (No Conductor)

The result has shown that the vehicle can drive with no conductor for 2 minutes before pressure drops.



Figure 4.61: Testing the Vehicle with no User

- **Conductor** (80 kg)

The vehicle can be driven for 1 min before it can no longer fight back the torque.



Figure 4.62: Testing the Vehicle on Road

4.10.1.2 Speed

The maximum speed reached on road is 10 km/h. For free load, the vehicle can reach 26 km/h.

4.10.2 Problems Encountered Solution

4.10.2.1 Working Space

The Working space was one of the major problems encountered. Due to Coronavirus Out Break, traveling between cities was not possible. Therefore, establishing a temporary workshop was mandatory.

As a first working place, access to work in the **Tunisian Red Crescent's Local Committee of Menzel Bouzelfa's** Headquarters was given.



Figure 4.63: Tunisian Red Cross - Menzel Bouzelfa's Local Committee's Logo

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The area provided was considered enough to make a small workshop but problems started escalating as soon the welding process began. The problem was the circuit breaker in the building could not support more than 25 A, so it kept opening every time the welding was used for more than 30 seconds.



Figure 4.64: Tunisian Red Cross's Headquarters

The temporary Solution was to lower the Current in the welding machine's rods and not to use it for more than the given time.

Later, and as the vehicle started getting bigger, changing the place was mandatory. Negotiation to get a work space within the city's perimeters were made, for a period of a 6 weeks.



Figure 4.65: Second Work Space

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The area was big enough to give the possibility to established identified work areas within the place applying the **5S Methodology**³.



Figure 4.66: Drilling and Cutting Station

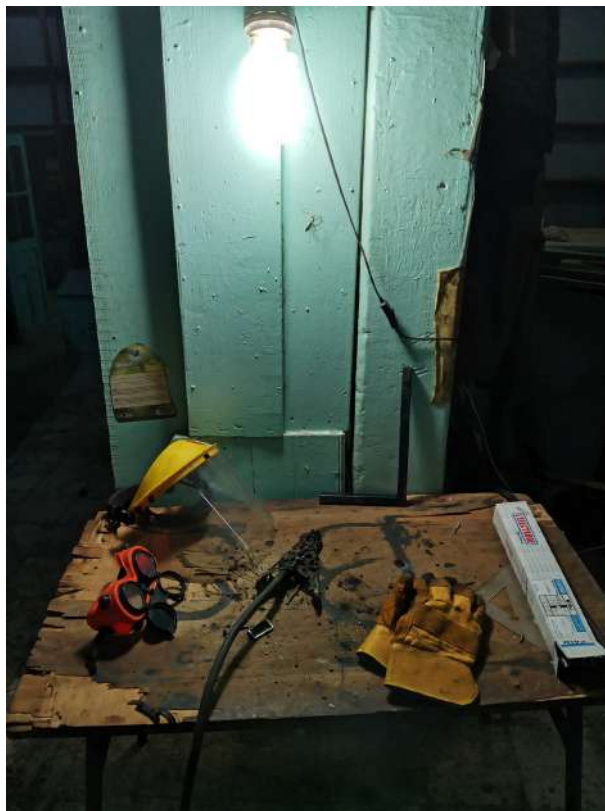


Figure 4.67: Welding Station

³workplace organization method (Japan)



Figure 4.68: Assembling Station

After finishing the prototype, the vehicle had to be moved to the **Tunisian Red Crescent's Local Committee of Menzel Bouzelfa's** warehouse for painting and testing.



Figure 4.69: Third Workspace

4.10.2.2 Lack of References

As this application is considered to be new, the lack of references was a major problem. The amount of time invested in searching and reading books, calculation and simulating was huge.

Generally, all the books available on the internet were expensive. In addition, there were no experts in this domain that are willing to help.

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However, some help was offered from **Mr Mohamed Mahdi Fellali** regarding ideas form designing the engine.

4.10.2.3 Lack of Infield Experience

As the team was never composed of non-related to this project, the advancement struggled a lot to make the needed and required work and even trying for more than 3 approaches to solve the same problems.

In addition, at the beginning of realizing the prototype, an extra hand needed to be hired to do the welding. Thing that costed a lot on the budget.

The only solution was to learn how to weld and use several machine in a short time to cut the expenses of the extra hand.

4.10.2.4 Lack of Equipment

The major problem encountered was the lack of equipment. Many tasks could have been easily done and in a short period of time, were either skipped or took longer than expected just because the machines needed to do the work were not available. For instance, Borrowing an **Air Compressor** for all the testing was the only solution, the same for the **welding machine, drill and grinding wheel**. Some times, renting the machines to ease the work, for example, after some of the equipment stopped working.

4.10.2.5 Budget Shortage

The budget offered was a big constraint because it had to be split int **Testing, Prototyping, Labour, Equipment, Maintenance... etc.** So, Minimize the expenses by relying on scrap parts and negotiating the prices to reach the maximum with what was offered was the procedure taken.

4.10.2.6 Labour Shortage

To do the work of realization, a team was required. To reach this goal, the simplest way was to hire labour, which is costly , or seeking help from Friends.

The helped was offered by **Firas Khlif** Mechanical Engineer, **Oussema Bedoui**, student with the preparatory school of Nabeul **Lugi Tucci** and **Sami Hammami**.

The common link is they are all members of The red crescent and they choose willingly to volunteer with in this project hand labour to whom the project owes its realization.

4.10.2.7 Time Shortage

The initial time given o do this work was 4 months. This period of time is insufficient to do all this project's parts. This lead to drop some ideas along the way because it would take to much time to realize likewise importing materials. In addition, asking for an extension was the only way to complete the minimum requirements for the work.

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4.10.2.8 Unavailability of Materials

Much of the materials required was just unavailable in the Tunisian market and needed to be imported to cost even more than the usual price.

As a solution, scavenging parts from scrap is deemed the only effective way to get the items needed, some times even manufacturing some parts to get going within the project.

Scavenging parts from old was a hard work both in searching and fixing. sometimes every item retrieved is dismantled to the point of renovating of the way it functions.

Searching for the needed items required much time, up to a week, to find the exact part required. Eventually, this turned out to be an effective way to complete the work.

4.10.2.9 Coronavirus Outbreak Situation

The year 2020 was a rough year due to the Coronavirus Outbreak, during which the government took some sever decision affected the project. In fact, traveling between cities was not allowed and curfew with really strict time limits. Plus, the whole outbreak situation was not going well as the virus is rapidly spreading.

The solution to this restrictions was getting backed up by the **Tunisian Red Crescent** as it provided the necessary requests to grant some extension to the time restrictions and travel.



Figure 4.70: Meeting in the Tunisian Red Crescent's Headquarter in Menzel Bouzelfa

4.11 Breakdown of Expenses

The full report of the expenses is joined in the appendix.

4.12 Perspective

This work represents the first prototype for the idea which is **The use of compressed air for transportation**. This part is considered a success as the proposed concept deemed feasible and gave promising results. However, this prototype requires a lot of work to be

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fully commercialized.

For next updates, new concept are to be considered and applied

4.12.1 Mechanics

On the mechanical level, more test should be invested to analyze the rigidity of the chassis.

A crush test is considered to be mandatory.

Changing the material of the skeleton from iron to aluminum gives a significant amount of weight reduction, therefore, torque reduction.

4.12.2 Electrics

The next step to be taken in consideration is realizing the electrical part, as well as the energy autonomy of the vehicle.

Change mechanical systems to electrical ones, like directional valves and shut-off valves. It should, in the next phase be fully automatic and all the data gathered and controlled via a micro-controller. This idea would rise the accuracy of the engine, prevent air losses and optimize its use.

Adding more gadgets, in case a more efficient solar panel is used, might add a good quality of drive to the user.

4.12.3 Autonomy

The autonomy of the prototype is a big issue at this phase, therefore, adding the volume of storage and going high in pressure at least 100 bar is deemed to be a necessity.

Adding a filter and stabilizer should be considered during next updates. An other concept is making a closed-loop system and using the exhaust air to power the electrical part via an alternator.

Next update should contain an in-built compressor to refill the tank automatically the tank whenever it's empty. This type of Air Compressors requires a 24v Voltage and it is already used to refill tires.



Figure 4.71: Mini Air Compressor 24v 10 bar

The power required should be covered if other type of solar panels are used likewise the one manufactured by TOYOTA for their hybrid cars.



Figure 4.72: Solar Roof Panels for Toyota cars

These panels should be easy to import and can be made to fit the curves of any vehicle. They are almost transparent gain a big advantage on the overall design and color of the vehicle.

An other idea suggested was to make an air heater; as Air can rises its volume when heating (refer to Chapter 2), the idea is to use a second reservoir to heat air in it. Th air flows from the firs tank to the second, heating it, then using the hot air, this would offer a huge win in term of air storage.

4.12.4 Team

This project could you use more engineers in the following majors; **Mechanics** and **Energy** as well as a team of workers.

4.12.5 Budget

To upgrade this project, the budget should cover all expenses for all the parts needed.

4.13 Conclusion

This Section presented all the prototyping from calculation to modeling and simulation, finishing by realization and infield testing of the vehicle.

At last, this work is a proof of the feasibility of a concept once considered doomed and obsolete.

In fact, much of the solutions provided are feasible and can be even prototypes, but the approach chosen during this work is highly related to the budget shortage, unavailability of the needed materials, and other obstacles.

All conditions provided, this idea can be even simpler and more robust to be made and should be a fine alternative to the nowadays climate crisis.

General Conclusion

Regarding the omnipresence of the climate crisis that did affect a list of countries, for instance, China, Mexico, Brazil and Australia, and, in addition to the fact that fuel prices are getting higher, this project is considered to be one of the alternatives that should be applied to slow down the rapid reshape of the weather pattern and the environment in general.

This work is the foundation stone of other innovations, in the same context, yet to be published where the shortfalls would be overcome.

In this project, different aspects of the use of compressed-air were introduced and dealt with from a theoretical perspective leading to the design of the pneumatic part of the vehicle, first, later manipulating different actuators and proposing solution to problems encountered.

From a mechanical perspective, this project focuses mainly on the prototyping part, meaning, the procedures taken and obstacles surpassed.

Other aspects were taken in consideration likewise, electricity. This part was introduced to help the vehicle sustain a good amount of power to rise its autonomy, believing that in the next upgrade, the project would be fully autonomous on an electrical level.

This work treats, most importantly, the financial part of the project in terms of quality-price ratio, offering, the costs of most of the components, their availability in the market and efficiency.

However, the vehicle can use bigger Air tanks or higher pressure before it can be destined for industrialization, but the dilemma of the pressure-volume should be dealt with first.

A diverse team would do better version of the vehicle treating all the aspects detailed, mechanically speaking, and a bigger budget surely would do good to the overall results of the project.

None can deny the fact that the vehicle requires more adjustments and work, opening the gate for later innovation and upgrades to reach the full potential of the project's global Idea.

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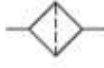
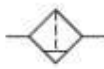
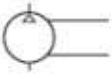
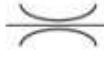
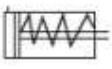
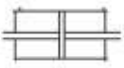
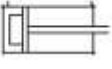
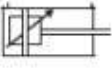
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Appendix 1 :

Pneumatic Actuators Symbols

Appendix

	Accumulator		Direction of Flow
	Air Dryer		Exhaust Line or Control Line
	Air Motor (One Direction Flow)		Filter
	Air Motor (Two Direction Flow)		Filter (Automatic Drain)
	Check Valve (Spring Loaded)		Filter (Manual Drain)
	Compressor		Fixed Restriction
	Cylinder (Spring Return)		Air Motor (Two Direction Flow)
	Cylinder Double Acting (Double Rod)		Lubricator
	Cylinder Double Acting (Single fixed cushion)		
	Cylinder Double Acting (Two adjustable cushions)		
	Differential Pressure		

Appendix B :

Minimum Piston Rod Diameter

MINIMUM PISTON ROD DIAMETER

Figures in body of chart are suggested minimum rod diameters, in inches.

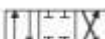
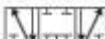
Load, Pounds	Exposed Length of Piston Rod, Inches / Rod Diameter, Inches									
	10"	20"	40"	60"	70"	80"	100"	120"		
1,000			3/4	1						
1,500			13/16	1-1/16						
2,000		5/8	7/8	1-1/8	1-1/4	1-3/8				
3,000		11/16	15/16	1-3/16	1-3/8	1-1/2				
4,000		3/4	1	1-1/4	1-7/16	1-9/16	1-7/8			
6,000	13/16	7/8	1-1/8	1-3/8	1-9/16	1-5/8	1-7/8			
8,000	15/16	1	1-3/16	1-1/2	1-5/8	1-3/4	2	2-1/4		
10,000	1	1-1/8	1-5/16	1-9/16	1-3/4	1-7/8	2-1/8	2-3/8		
15,000	1-3/16	1-1/4	1-7/16	1-3/4	1-3/4	2	2-1/4	2-1/2		
20,000	1-3/8	1-7/16	1-5/8	1-7/8	2	2-1/8	2-7/16	2-3/4		
30,000	1-11/16	1-3/4	1-7/8	2-1/8	2-1/4	2-3/8	2-11/16	3		
40,000	2	2	2-1/8	2-3/8	2-1/2	2-5/8	2-7/8	3-1/4		
60,000	2-3/8	2-7/16	2-1/2	2-3/4	2-3/4	2-7/8	3-1/4	3-1/2		
80,000	2-3/4	2-3/4	2-7/8	3	3	3-1/4	3-1/2	3-3/4		
100,000	3-1/8	3-1/8	3-1/4	3-3/8	3-1/2	3-1/2	3-3/4	4		
150,000	3-3/4	3-3/4	3-7/8	4	4	4-1/8	4-3/8	4-1/2		
200,000	4-3/8	4-3/8	4-3/8	4-1/2	4-3/4	4-3/4	4-7/8	5		
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Appendix C :

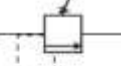
Common Pneumatic Symbols

Appendix

Directional Control Valve Symbols

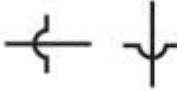
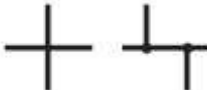
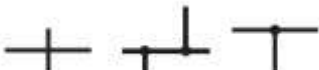
	2-position, 2-way, 2 ported
	2-position, 3-way, 3 ported
	2-position, 4-way, 4 ported
	2-position, 4-way, 5 ported
	3-position, 4-way, 4 ported Closed Center
	3-position, 4-way, 5 ported Closed Center
	3-position, 4-way, 5 ported Pressure Center
	3-position, 4-way, 5 ported Open Center

Simple Pneumatic Valves

	Check Valve
	Flow Control
	Relief Valve

Lines

	Main Line
	Pilot Line

	Lines Crossing
	Lines Joined
	Lines Joined

Actuator Symbols

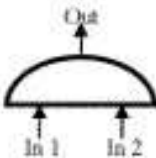
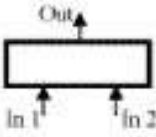
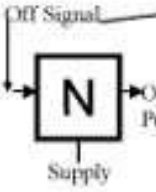
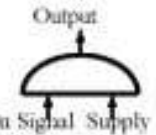
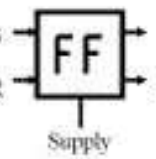
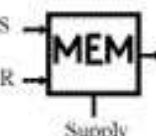
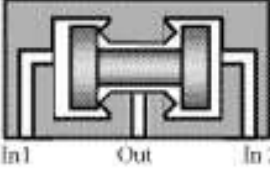
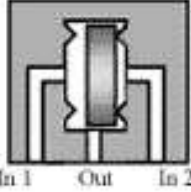
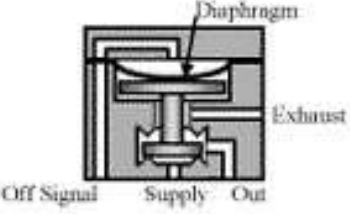
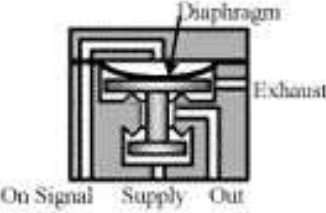
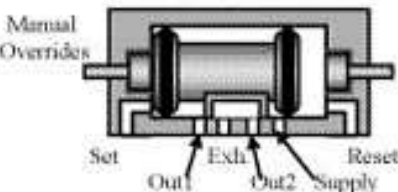
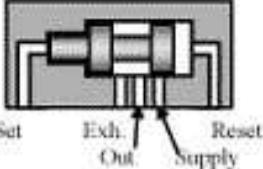
	Manual
	Push Button
	Lever
	Foot Operated
	Mechanical
	Spring
	Detent
	Solenoid
	Internal Pilot
	External Pilot
	Piloted Solenoid with Manual Override
	Lever Operated, Spring Return

Appendix D :

Air Logic (ISO)

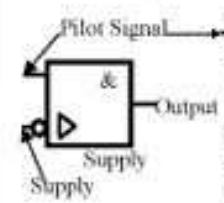
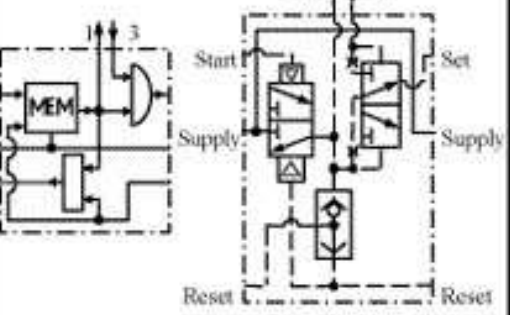
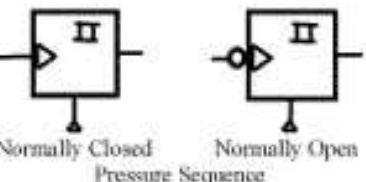
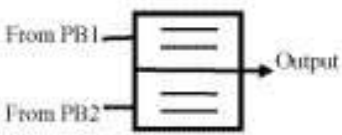
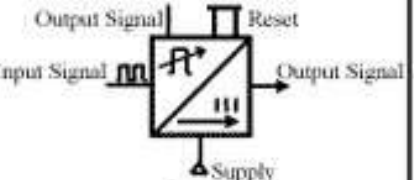
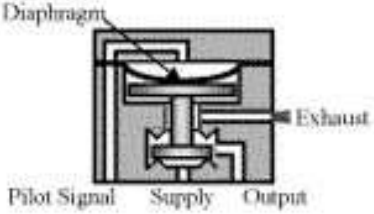
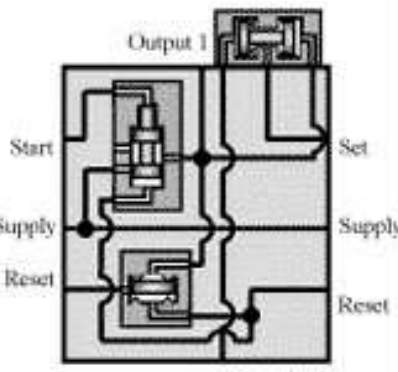
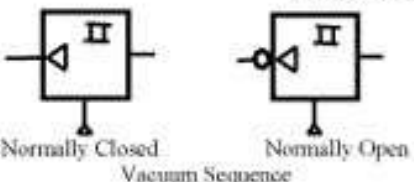
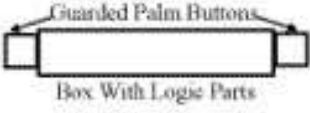
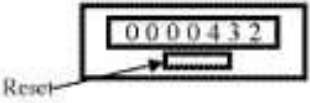
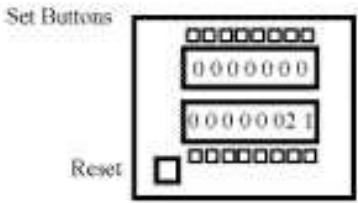
Appendix

CHAPTER 4 ISO Symbols & Glossary

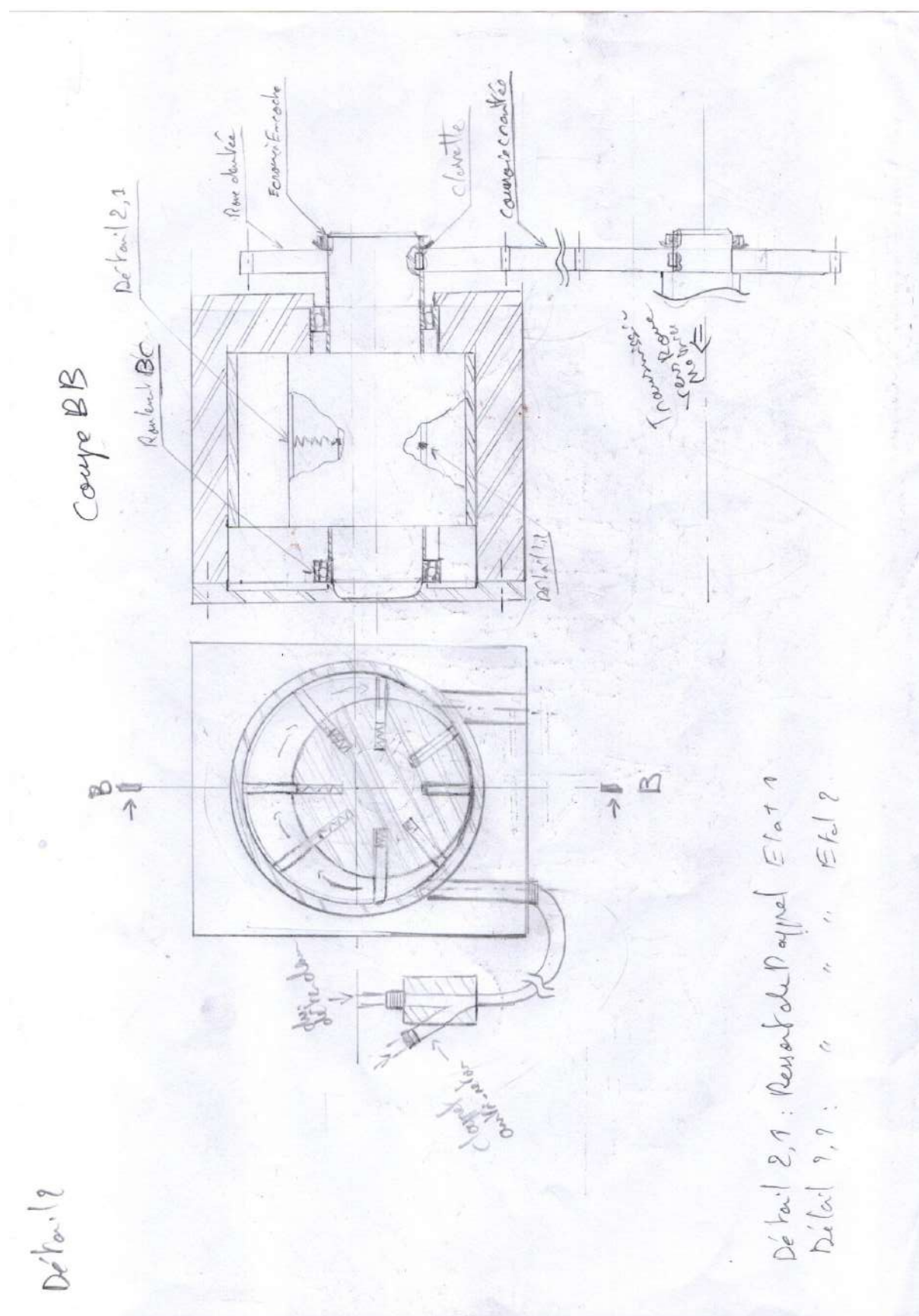
ISO Designation	Symbol	Picture Representation
Air Logic Chapter 19		
"AND" Element A Signal at "In 1" AND "In 2" Gives an Output. Output Always Lowest Of Input Signals "OR" Element A Signal at "In 1" OR "In 2" Gives an Output. Output Always Highest Input Signals "NOT" Element 3-Way Normally Open "Supply" Goes to Output Until an "OFF Signal" is Received. "YES" Element 3-Way Normally Closed Similar to "AND" But Can Choose "Output" Signal "FLIP-FLOP" Element 4-Way Double Pilot Operated Supplies and Exhaust's Either of Two Outputs "MEMORY" Element 3-Way Normally Closed Supplies or Exhaust's a Single Port. Can be "Reset" While "Set" Signal is on.	ANSI B93-38 1976 ISO Graphic Symbol      	     

ISO Designation	Symbol	Picture Representation
Air Logic (Cont'd.)	ANSI B93-38 1976 ISO Graphic Symbol	
"DIFFERENTIATOR" Element 3-Way Normally Open (Also Called a "ONE SHOT" Element or an "IMPULSE GENERATOR") A Fixed Signal in Gives a Fixed or Adjustable Timed Signal Out		
"ON DELAY TIMER" Element 3-Way Normally Closed An "Output" is Given After a "Pilot Signal" Has Been on for a Fixed or Adjustable Time Delay.		
"OFF DELAY TIMER" Element 3-Way Normally Open The "Output" is Lost After a "Pilot Signal" Has Been on for a Fixed or Adjustable Time Delay.		
"FREQUENCY GENERATOR" Element 3-Way Normally Open An "Input" Signal Gives a Fixed or Adjustable Output for One Time Period, Then it Goes Off for Two Time Periods.		

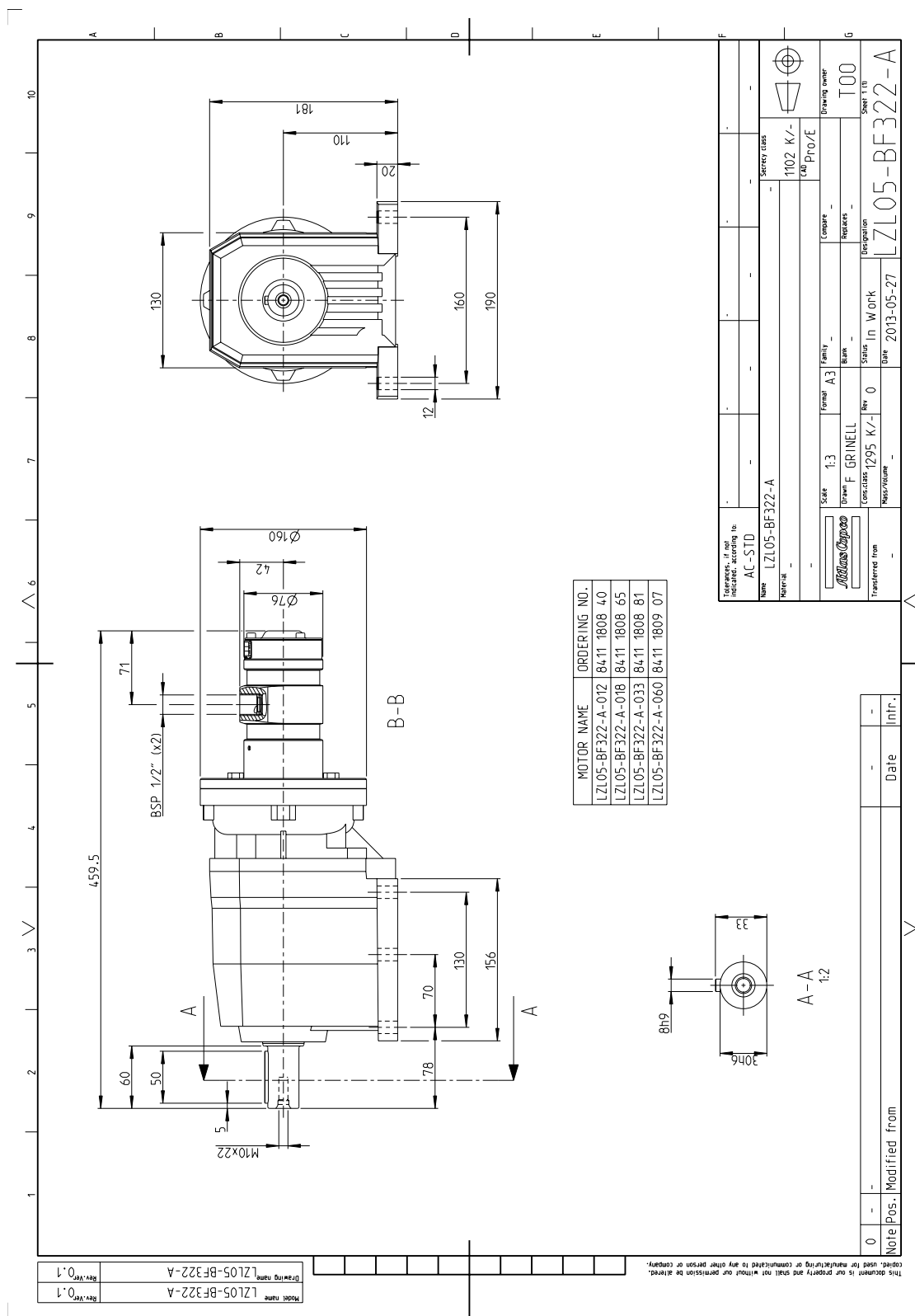
CHAPTER 4 ISO Symbols & Glossary

ISO Designation	Symbol	Picture Representation
Air Logic (Cont'd.) "PRESSURE DECAY SENSOR" Element 3-Way Normally Open Same as a "NOT" except it Takes a lower "Pilot Signal" to Shift Used for Indicating end of Stroke of Cylinders. "SEQUENCER" Element A Combination of an "AND", "OR" and a "MEMORY" Element in a Common Housing to Simplify Circuit Design. Used in Series To Make a Complete Operating System. "SEQUENCE" Element Gives or Drops a Signal at a pre-determined Adjustable Pressure. "ANTI TIEDOWN" Element Requires Two Inputs to Get an Output Counters: Counts Number of Cycles Counts Down From Pre-selected Number Then Gives an Output Pressure Indicator	Manufacturers Symbol  ISO Equivalent        	      

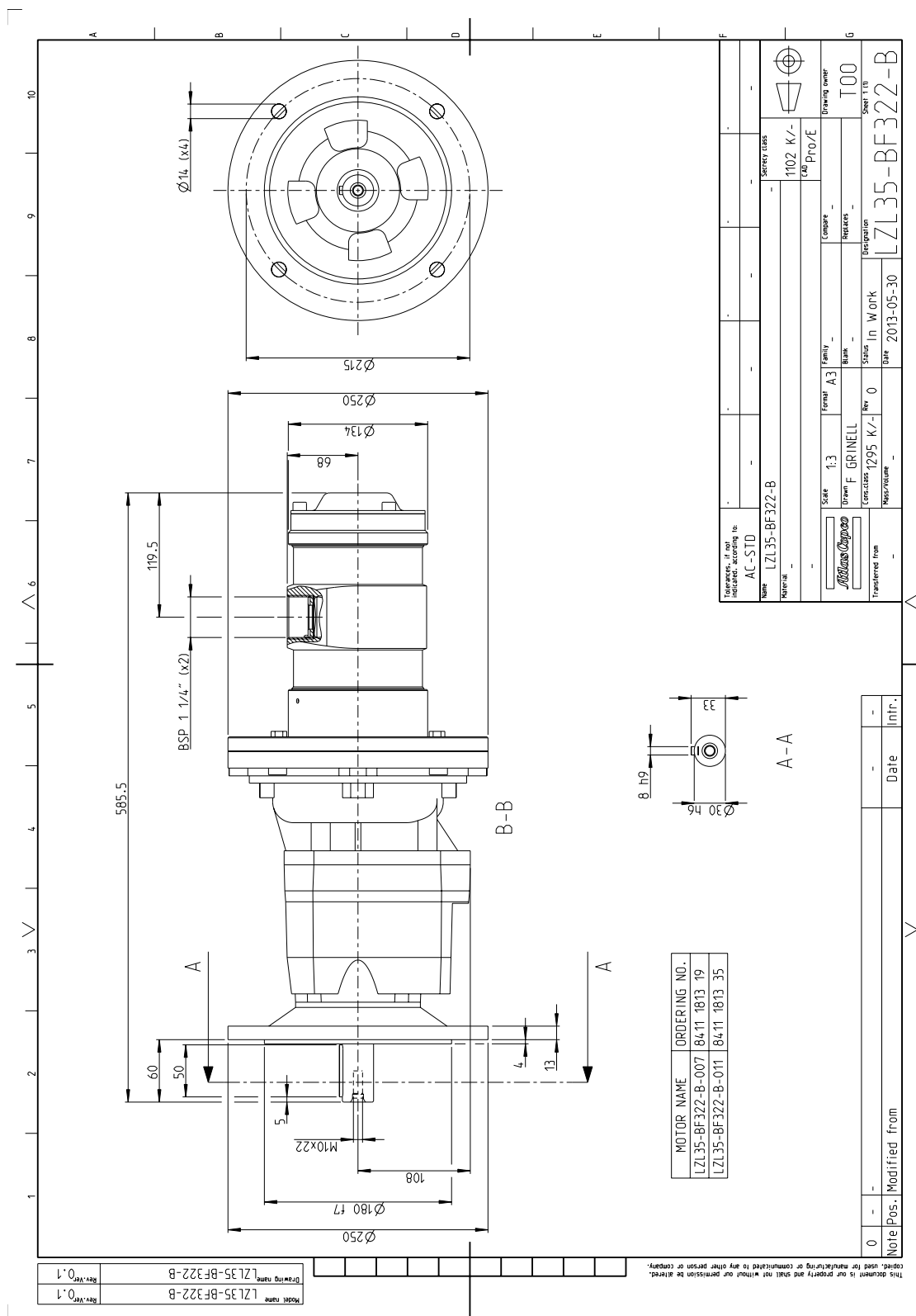
Appendix E :
Vane Motor Design (by Writer)



Appendix F :
LZL05-BF322-A-012



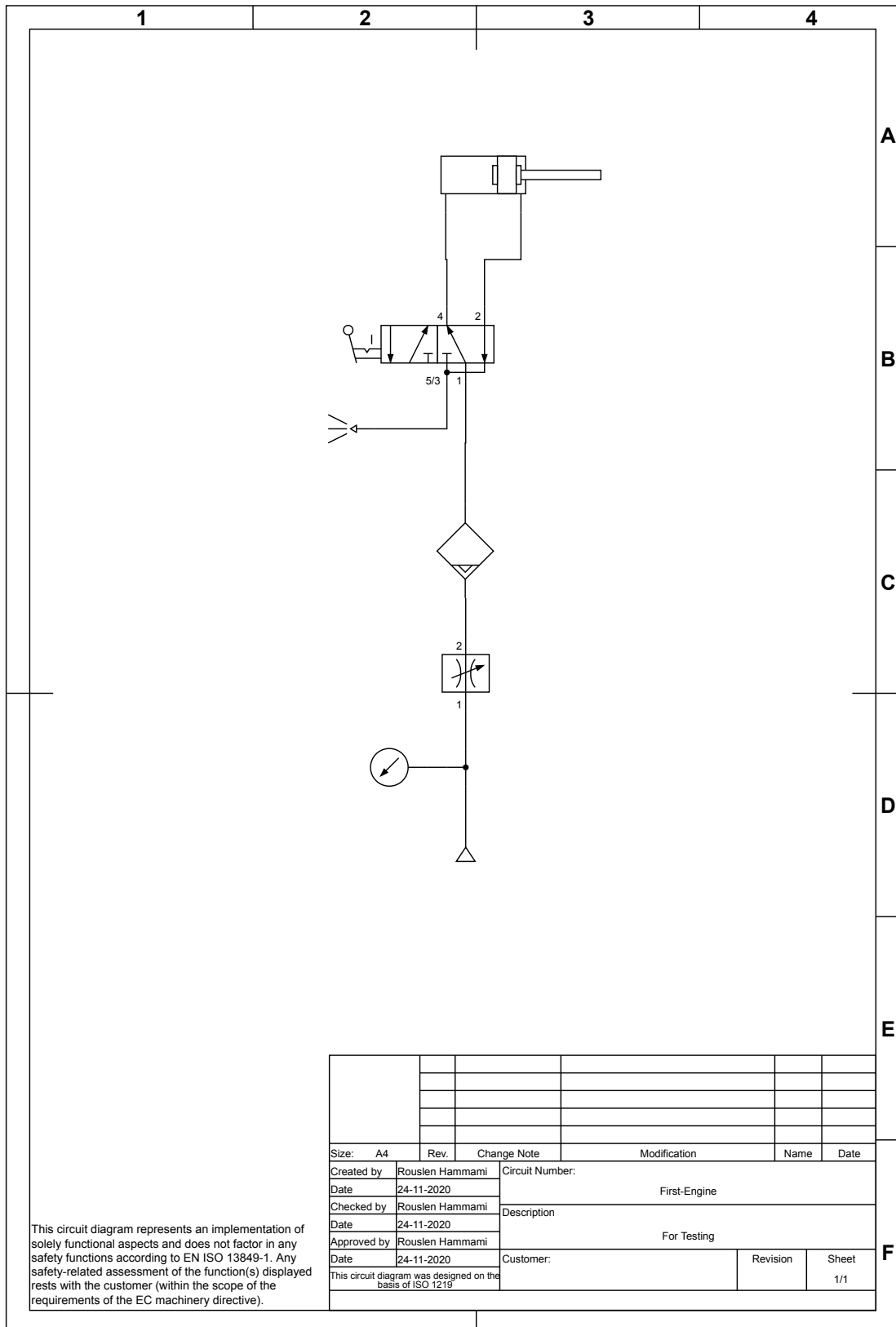
Appendix G :
LZL35-BF322-A-012



Appendix H :

**Pneumatic Schematics of the Fourth
Engine (by writer)**

Appendix



Appendix I :
Standard Cylinder DSNU (FESTO)

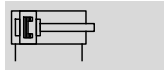
Appendix

Standard cylinders DSNU, ISO 6432

Technical data

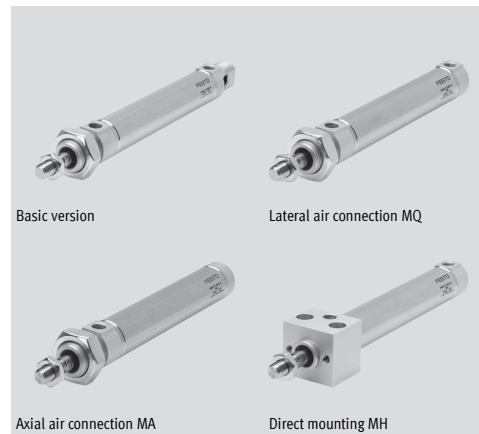
FESTO

Function



- $\varnothing$ - Diameter
8 ... 25 mm
-  - Stroke length
1 ... 500 mm

Variants
→ 17



General technical data		8	10	12	16	20	25
Piston $\varnothing$		8	10	12	16	20	25
Pneumatic connection		M5	M5	M5	M5	G $\frac{1}{8}$	G $\frac{1}{8}$
Piston rod thread		M4	M4	M6	M6	M8	M10x1.25
Constructional design	Piston						
	Piston rod						
	Cylinder barrel						
Cushioning	P	Flexible cushioning rings/pads at both ends					
	PPV	-			Adjustable cushioning at both ends		
	PPS	-			Self-adjusting cushioning at both ends		
Cushioning length	PPV [mm]	-			12	15	17
	PPS [mm]	-			12	15	17
Position sensing		Via proximity sensor					
Type of mounting		Direct mounting (MH variant only)					
		Via accessories					
Mounting position		Any					

- $\frac{1}{2}$ - Note: This product conforms to ISO 1179-1 and to ISO 228-1

Operating conditions		8	10	12	16	20	25
Piston $\varnothing$		8	10	12	16	20	25
Operating medium		Compressed air in accordance with ISO 8573-1:2010 [7:4:4]					
Note on operating/pilot medium		Operation with lubricated medium possible (in which case lubricated operation will always be required)					
Operating pressure	Basic version [bar]	1.5 ... 10 ¹⁾			1 ... 10		
	S10	-			1.5 ... 10		
	S11	-			0.45 ... 10		

1) With DSNU-12- ... PPV (pneumatic cushioning adjustable at both ends): 2 ... 10 bar

Ambient conditions		Basic version	S6	S10	S11	R3
Standard cylinder		Basic version	S6	S10	S11	R3
Ambient temperature ¹⁾	[°C]	-20 ... +80	0 ... +120	+5 ... +80		-20 ... +80
Corrosion resistance class CRC ²⁾		2	2	2	2	3
ATEX		Specified types → www.festo.com				

1) Note operating range of proximity sensors.

2) Corrosion resistance class 2 as per Festo standard 940 070

Components subject to moderate corrosion stress. Externally visible parts with primarily decorative surface requirements which are in direct contact with a normal industrial environment or media such as coolants or lubricating agents.

Corrosion resistance class 3 as per Festo standard 940 070

Components requiring higher corrosion resistance. External visible parts in direct contact with industrial atmospheres or media such as solvents and cleaning agents, with a predominantly functional requirement for the surface.

Appendix

Standard cylinders DSNU, ISO 6432

Technical data

FESTO

Speed [mm/s]			
Piston Ø	16	20	25
Speed with stick-slip-free operation, horizontal, without load, at 6 bar	10 ... 100		
Minimum speed, advancing	S11	2.7	5.3
Minimum speed, retracting	S11	3.2	4.7
			<1 ¹⁾
			<1 ¹⁾

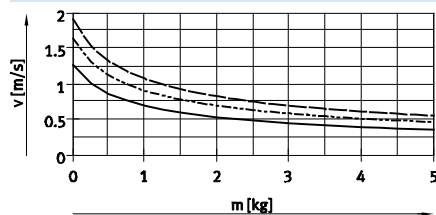
1) Measurements of less than 1 mm/s were not conducted

Force [N] and impact energy [J]						
Piston Ø	8	10	12	16	20	25
Theoretical force at 6 bar, advancing	30	47	68	121	189	295
Theoretical force at 6 bar, retracting	23	40	51	104	158	247
Max. impact energy at the end positions for flexible cushioning elements ¹⁾	0.03	0.05	0.07	0.15	0.20	0.30

1) The values are reduced by approx. 50% at an ambient temperature of 80 °C

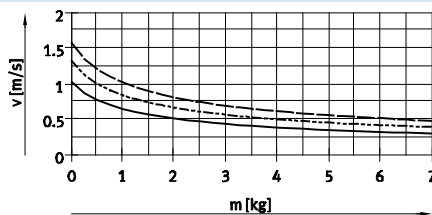
Average piston speed v as a function of applied load m in combination with PPS cushionings

Piston Ø 16



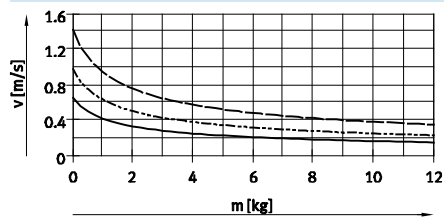
— DSNU-16-50
- - - DSNU-16-100
- · - DSNU-16-200

Piston Ø 20



— DSNU-20-50
- - - DSNU-20-100
- · - DSNU-20-200

Piston Ø 25



— DSNU-25-50
- - - DSNU-25-100
- · - DSNU-25-200

Note
Average piston speed
= stroke/movement time

Note
Design software for flexible cushioning elements → ProDrive
Additional graphs for PPS cushioning → www.festo.com
Design software for PPV cushioning → ProDrive

Appendix

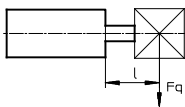
Standard cylinders DSNU, ISO 6432

Technical data

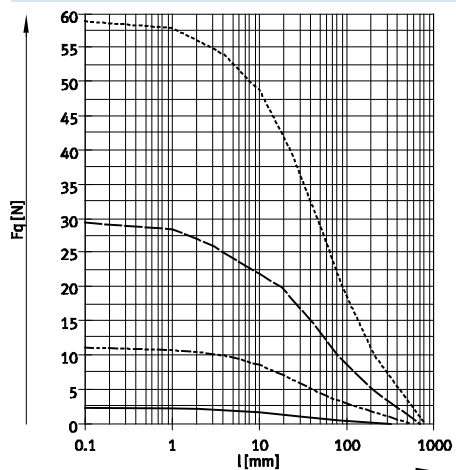
FESTO

Weight [g]						
Piston Ø	8	10	12	16	20	25
Product weight with 0 mm stroke	34.6	37.3	75	89.9	186.8	238
Additional weight per 10 mm stroke	2.4	2.7	4	4.6	7.2	11

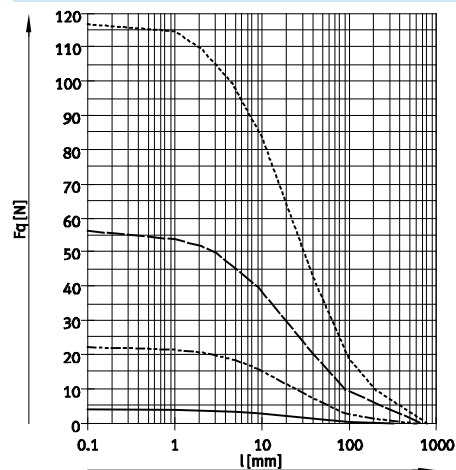
Max. lateral force F_q as a function of stroke length l



Basic version



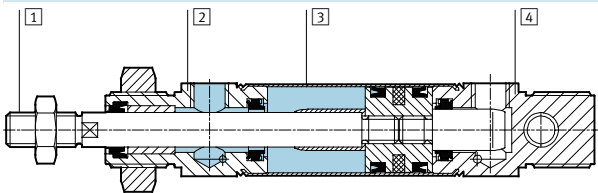
S2 – Through piston rod



- Ø 8/10
- - - Ø 12/16
- Ø 20
- · - · Ø 25

Materials

Sectional view



Standard cylinder	Basic version	R3	S6	S10	S11
1 Piston rod	High-alloy stainless steel				
2 Bearing cap	Anodised aluminium				
3 Cylinder barrel	High-alloy stainless steel				
4 End cap	Anodised aluminium				
– Seals	Polyurethane, nitrile rubber		Fluoro elastomer		
Note on materials	RoHS compliant				

Appendix J :
168-04-1600-1 Cylinder (BOSCH)

Piston rod cylinders → Standard cylinders

● ISO 15552, Series 168

Ø 25 - 100 mm; Ports: G 1/8 - G 1/2; double-acting; with magnetic piston; cushioning: pneumatic, adjustable; piston rod: external thread



Compressed
air connection

internal thread

Working pressure min./max.

1.5 bar / 10 bar

Ambient temperature min./max.

-20 °C / +70 °C

Medium temperature min./max.

-20 °C / +70 °C

Medium

Compressed air

Particle size max.

50 µm

Oil content of compressed air

0 mg/m³ - 5 mg/m³

Pressure for determining piston forces

6 bar

Materials:

Cylinder tube

Aluminum, anodized

Piston rod

Stainless steel

Front cover

Aluminum

End cover

Aluminum

Seal

Acrylonitrile Butadiene Rubber

Nut for piston rod

Steel, galvanized

Scraper

Polyurethane

Technical Remarks

- The pressure dew point must be at least 15 °C under ambient and medium temperature and may not exceed 3 °C.
- The oil content of air pressure must remain constant during the life cycle.
- Use only the approved oils from Bosch Rexroth, see chapter „Technical information“.

Piston Ø	[mm]	25	32	40	50	63
Retracting piston force	[N]	227	415	636	990	1680
Extending piston force	[N]	295	482	756	1176	1860
Cushioning length	[mm]	11	11.5	15	17	16.5
Weight	0 mm stroke	[kg]	0.27	0.45	0.76	1.1
	+10 mm stroke	[kg]	0.018	0.02	0.029	0.045
Stroke max.	[mm]	1500	1500	1600	1600	1600

Piston Ø	[mm]	80	100			
Retracting piston force	[N]	2718	4229			
Extending piston force	[N]	3000	4676			
Cushioning length	[mm]	19.5	19.5			
Weight	0 mm stroke	[kg]	2.5	3.7		
	+10 mm stroke	[kg]	0.078	0.1		
Stroke max.	[mm]	1700	1700			

- * Core products - preferably for new constructions, worldwide service
 - Service products - existing standard program, worldwide service
- Pneumatics catalog, online pdf, as of 2008-11-02, © Bosch Rexroth AG

Appendix

2

Bosch Rexroth AG | Pneumatics

Piston rod cylinders → Standard cylinders

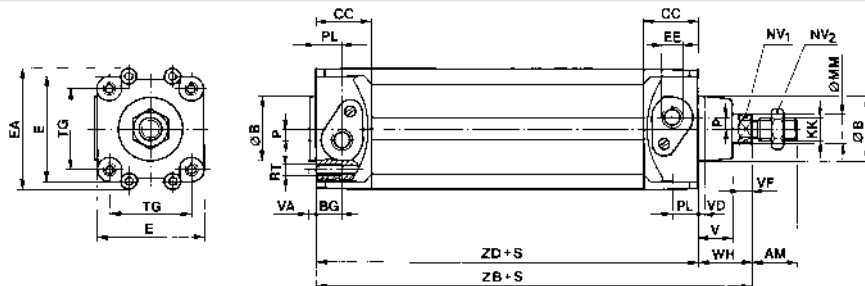
● ISO 15552, Series 168

Ø 25 - 100 mm; Ports: G 1/8 - G 1/2; double-acting; with magnetic piston; cushioning: pneumatic, adjustable;
piston rod: external thread

	Piston Ø Piston rod thread Ports	25 M10x1,25 G 1/8	32 M10x1,25 G 1/8	40 M12x1,25 G 1/4	50 M16x1,5 G 1/4	63 M16x1,5 G 3/8
	Stroke 25	1680202000	1680302000	1680402000	1680502000	1680602000
	50	1680205000	1680305000	1680405000	1680505000	1680605000
	80	1680208000	1680308000	1680408000	1680508000	1680608000
	100	1680210000	1680310000	1680410000	1680510000	1680610000
	125	1680212000	1680312000	1680412000	1680512000	1680612000
	160	1680216000	1680316000	1680416000	1680516000	1680616000
	200	1680220000	1680320000	1680420000	1680520000	1680620000
	250	1680225000	1680325000	1680425000	1680525000	1680625000
	320	-	1680332000	1680432000	1680532000	1680632000
	400	-	1680340000	1680440000	1680540000	1680640000
	500	-	1680350000	1680450000	1680550000	1680650000
	Piston Ø Piston rod thread Ports	80 M20x1,5 G 3/8	100 M20x1,5 G 1/2			
	Stroke 25	1680802010	1681002020			
	50	1680805010	1681005020			
	80	1680808010	1681008020			
	100	1680810010	1681010020			
	125	1680812010	1681012020			
	160	1680816010	1681016020			
	200	1680820010	1681020020			
	250	1680825010	1681025020			
	320	1680832010	1681032020			
	400	1680840010	1681040020			
	500	1680850010	1681050020			

other versions can be ordered from Bosch Rexroth sales offices

Dimensions



D168_012

S = stroke

Ø	AM	B e11	BG 1)	CC	E	EA	EE	KK	MM g8	NV1	NV2	P	PL
25	22	25	7.5	20	37	40	G 1/8	M10x1,25	12	10	17	5	9
32	22	30	16	28	45	49	G 1/8	M10x1,25	12	10	17	5	14
40	24	35	16	33	50	58	G 1/4	M12x1,25	16	13	19	7	15
50	32	40	16	31	62	69	G 1/4	M16x1,5	20	17	24	7	14
63	32	45	16	38	73	84	G 3/8	M16x1,5	20	17	24	8	18

* Core products - preferably for new constructions, worldwide service
● Service products - existing standard program, worldwide service
Pneumatics catalog, online pdf, as of 2008-11-02, © Bosch Rexroth AG

Piston rod cylinders → Standard cylinders

● ISO 15552, Series 168

Ø 25 - 100 mm; Ports: G 1/8 - G 1/2; double-acting; with magnetic piston; cushioning: pneumatic, adjustable;
piston rod: external thread

Ø	AM	B e11	BG 1)	CC	E	EA	EE	KK	MM g8	NV1	NV2	P	PL
80	40	45	16	38	92	104	G 3/8	M20x1,5	25	22	30	12	17
100	40	55	16	42	109	126	G 1/2	M20x1,5	25	22	30	10	21

Ø	RT	TG	V	VA	VD	VF	WH	ZB	ZD				
25	M5	27 ±0,2	16	–	4	6	24	98	74 ±0,4				
32	M6	32,5 ±0,2	15	3.5	4	6	26	120	94 ±0,4				
40	M6	38 ±0,2	19	3.5	4	6.5	30	135	105 ±0,4				
50	M8	46,5 ±0,2	24	4	4	8	37	143	106 ±0,4				
63	M8	56,5 ±0,2	24	4	4	8	37	158	121 ±0,4				
80	M10	72 ±0,2	30	4	4	10	46	174	128 ±0,8				
100	M10	89 ±0,2	35	4	4	10	51	189	138 ±0,8				

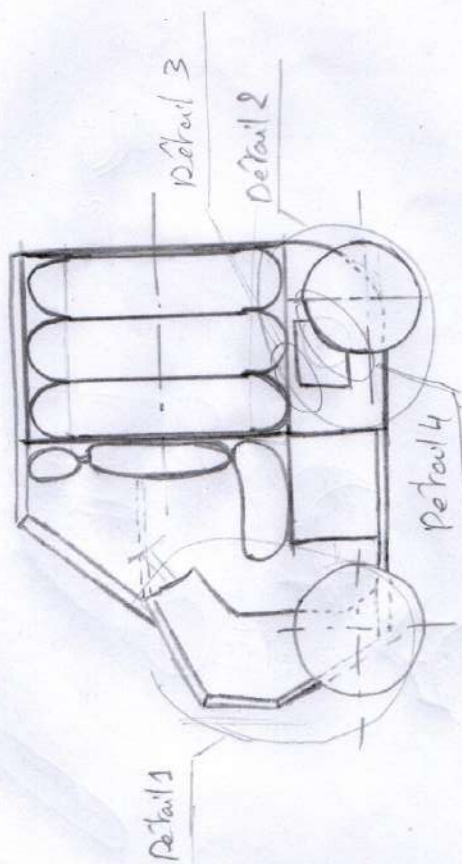
1) Min.

* Core products - preferably for new constructions, worldwide service
 ● Service products - existing standard program, worldwide service
 Pneumatics catalog, online pdf, as of 2008-11-02, © Bosch Rexroth AG

Appendix K :

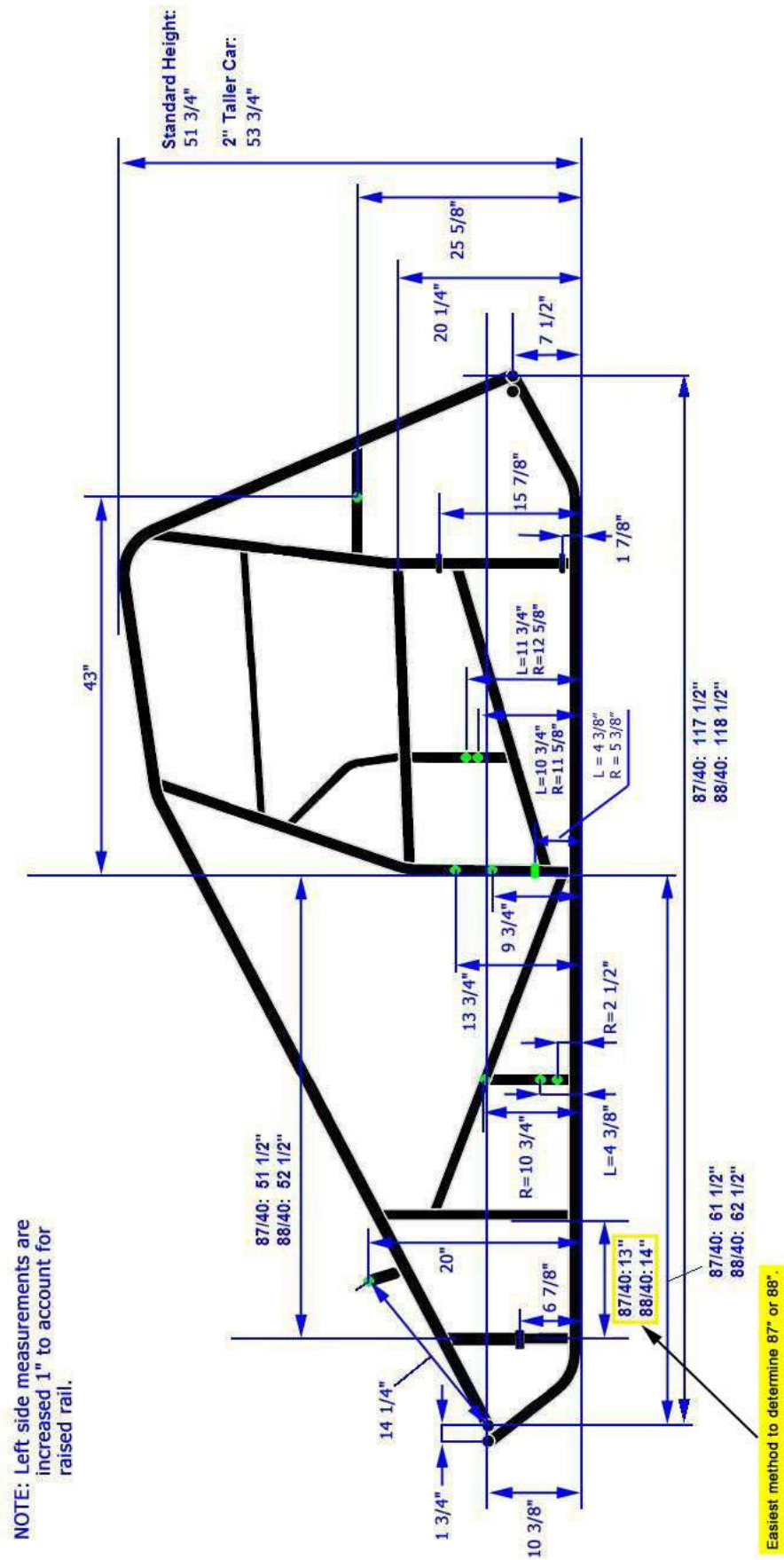
Chassis' First Design

Scheme Exploded



Appendix L :

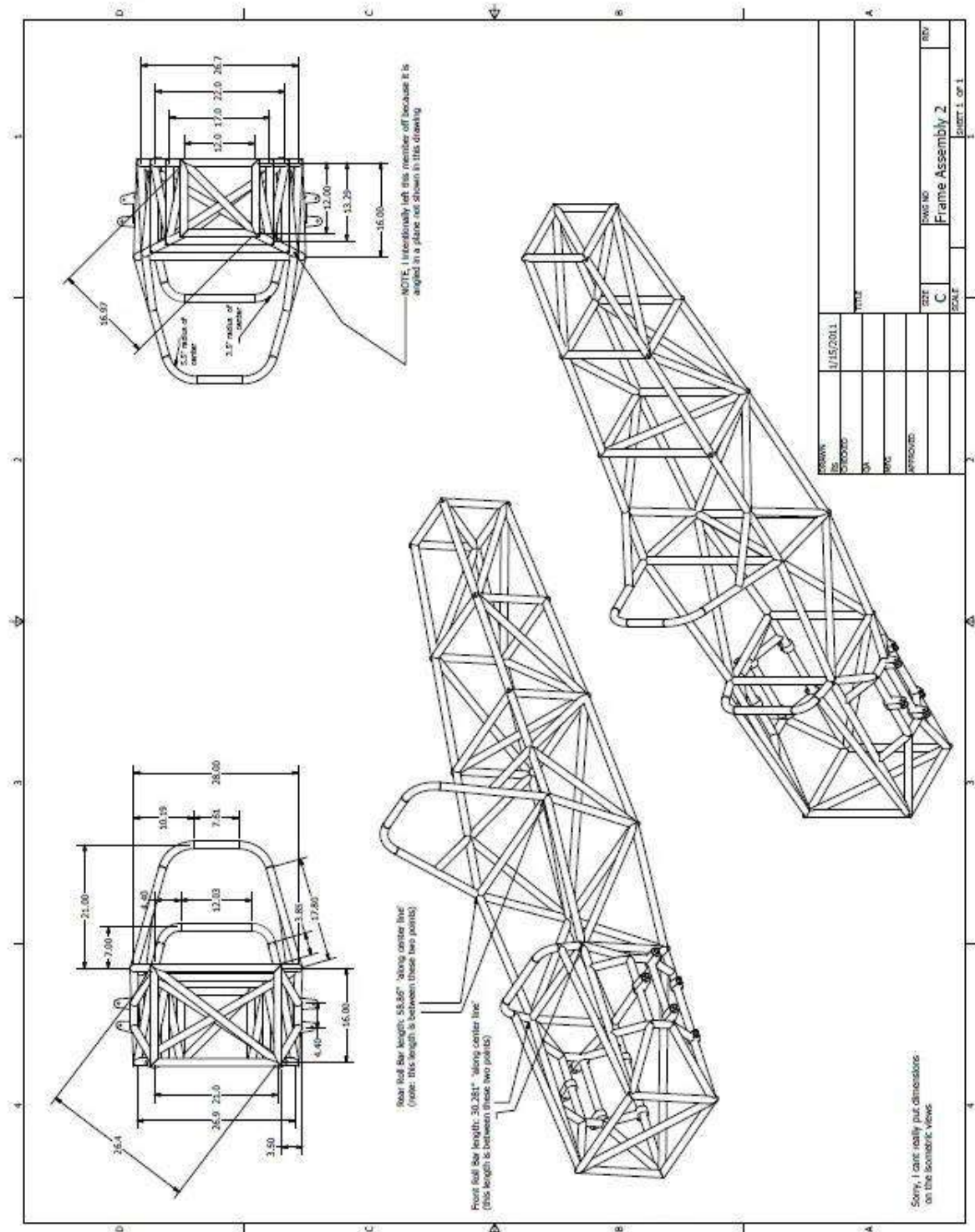
Buggy Chassis



Appendix L :

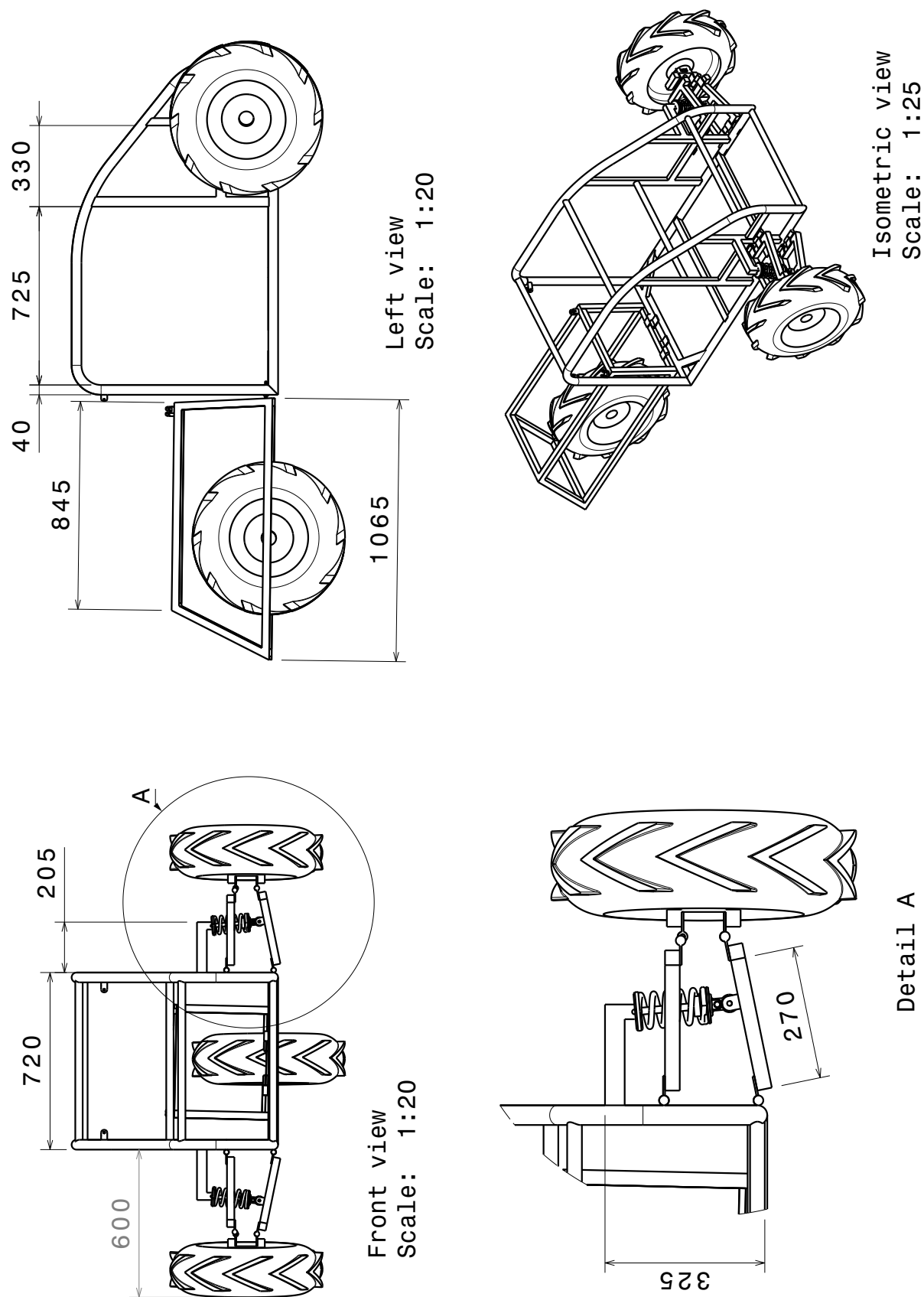
Chassis' Third Design

Appendix



Appendix M :

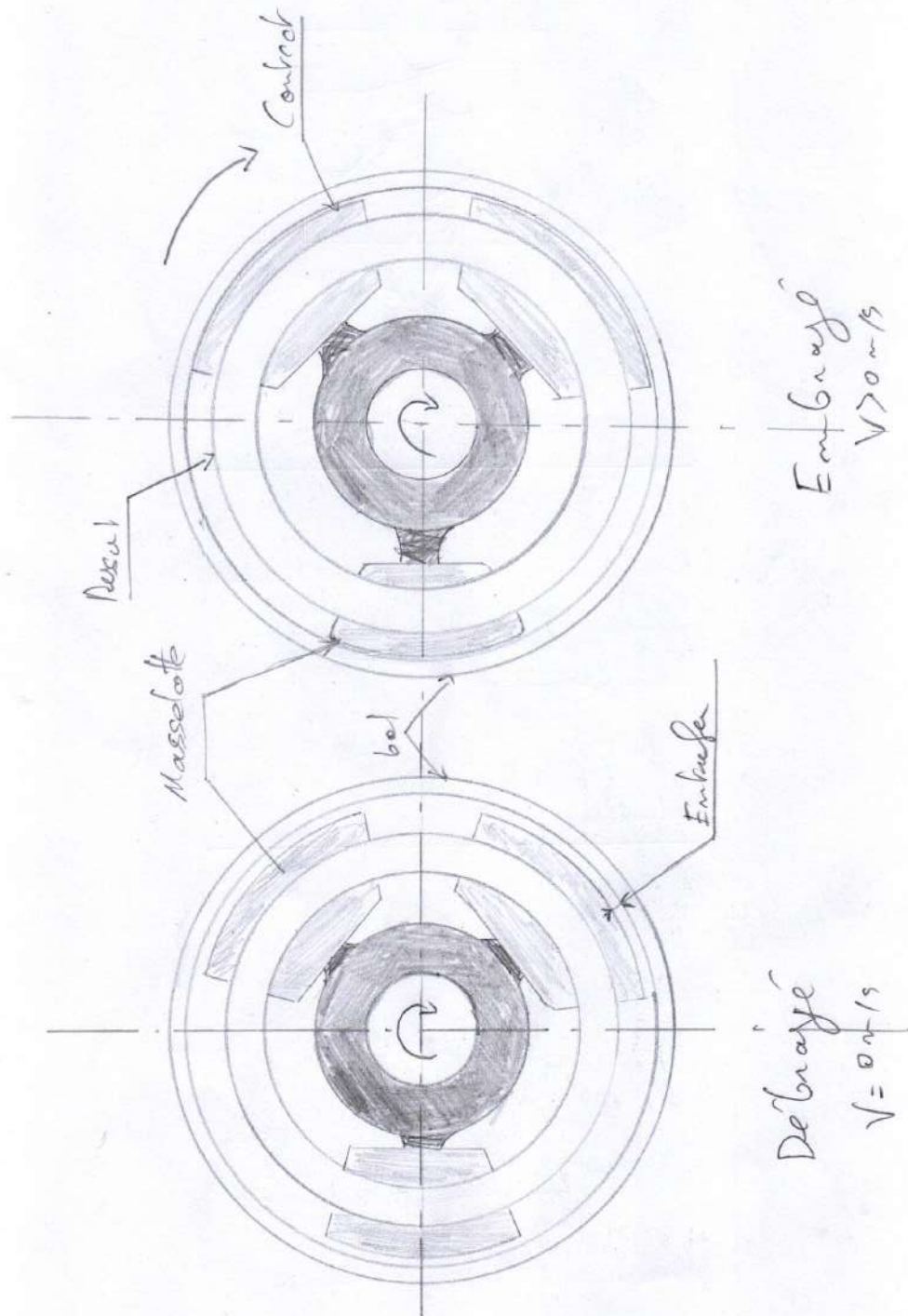
The realized Chassis



Appendix N :

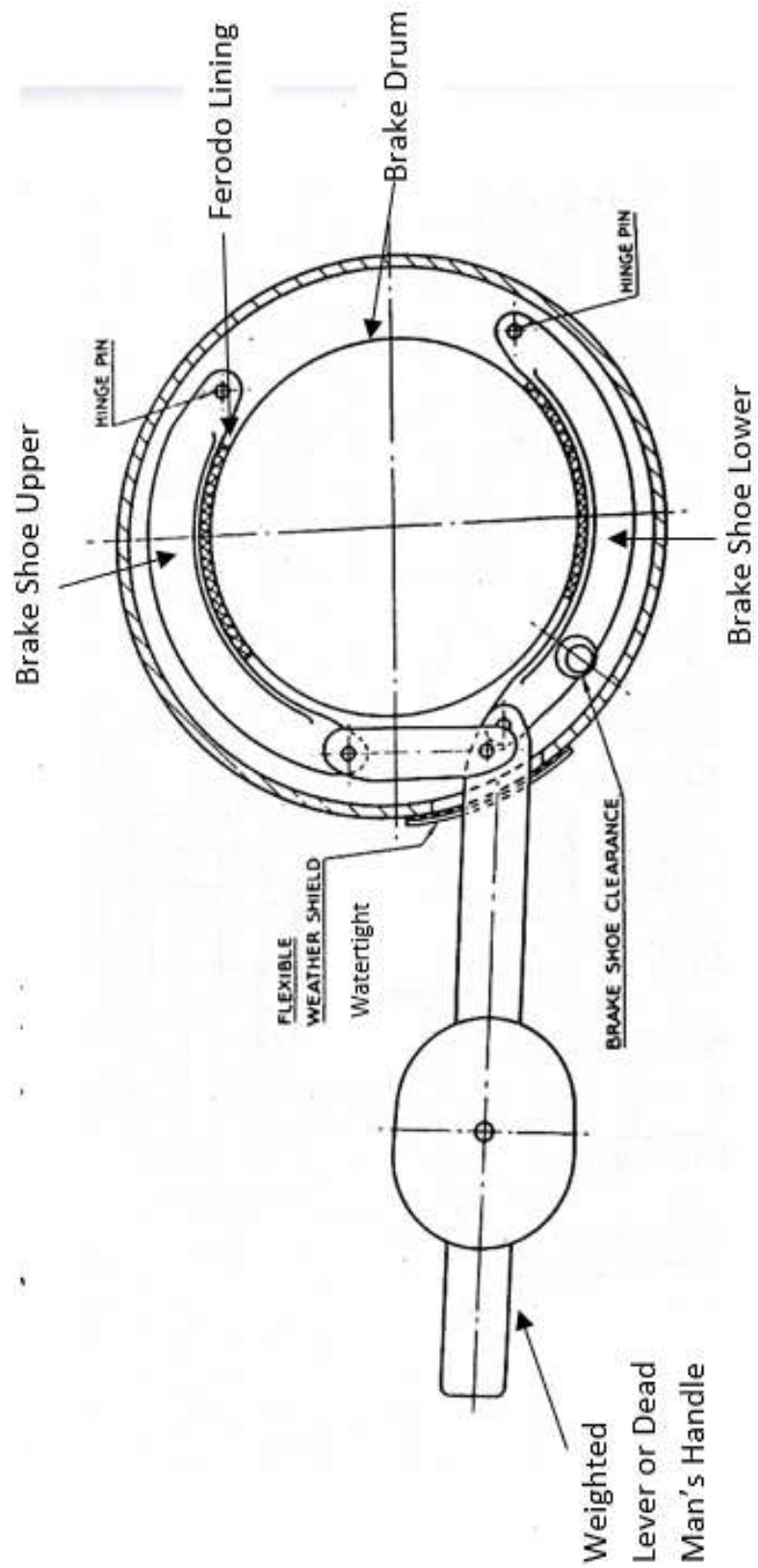
Centrifugal Clutch

Detail 4: Embrayage centrifuge



Appendix O :

Brake System



Appendix P :

Solar Panel



Technical data sheet

Vision 36M glass

Glass-glass module High yield and transparent

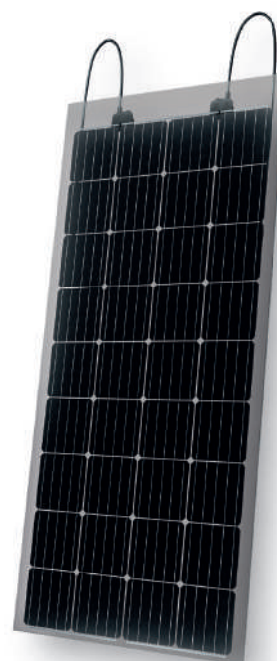
Thanks to their modern design SOLARWATT glass-glass modules deliver the highest long-term yields. They are robust and resilient, yet just as light as their glass-foil predecessors.

The high-performance solar cells are embedded almost indestructibly in the glass-glass composite and thus optimally protected against all weather effects and mechanical stress. SOLARWATT can therefore offer a 30-year warranty on performance and product quality.

The SOLARWATT FullCoverage insurance is included for 5 years and free of charge. It insures almost all risks and takes effect even if the modules do not produce electricity or deliver less than expected in the event of damage.

Product Quality

- National technical approval (AbZ)
- ammonia resistant
- intensive hailstorm resistant
- salt mist resistant
- 100 % plus-sorting
- 100 % PID protected



Service

FullCoverage insurance
included (up to 1,000 kWp*)

Simple returns policy
as per „Delivery terms for
SOLARWATT solar modules“

* country-specific deviations apply

30 Year Product Warranty
as per „Warranty conditions for SOLARWATT
solar modules“

30 Year Performance Warranty
on 87 % of nominal power as per „Warranty
conditions for SOLARWATT solar modules“

Subject to change | Errors excepted
AZ-TDB-PMS-0475 | This data sheet fulfills the requirements listed in IEC 61215-1-1 | REV 010 | 04/2019 | EN

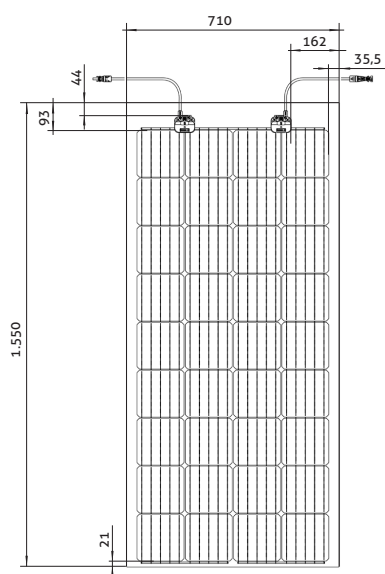
SOLARWATT GmbH | Maria-Reiche-Str. 2a | 01109 Dresden | Germany |
Tel. +49 351 8895-333 | Fax +49 351 8895-100 | www.solarwatt.com
Certified acc. to DIN EN ISO 9001, 14001, 50001 | BS OHSAS 18001:2007

Appendix

Technical data sheet Vision 36M glass



Dimensions



General data

Module technology	Glass-glass laminate
Covering material	Partially tempered high transparent float glass with anti-reflective finish, 4 mm
Encapsulation	EVA-solar cells-EVA, transparent
Backing material	Partially tempered float glass, 4 mm
Transparent areas	appr. 20 %
Solar cells	36 monocrystalline high power solar cells
Cell dimensions	157 x 157 mm
L x W x H	1550 ⁺² x 710 ⁺² x 9 ⁺¹ (without junction box) mm
Height of junction box	22 mm
Weight	ca. 25 kg
Connection technology	Cables 2 x 0,45 m/4 mm ² Hirschmann HC4-connector
Bypass diodes	2
Max. system voltage	1,000 V
IP rating	IP67
Protection class	II (acc. to IEC 61140)
Fire class	C (acc. to IEC 61730), E (acc. to EN 13501)
Certified mechanical ratings as per IEC 61215	Suction load up to 2,400 Pa (test load 3,600 Pa) Pressure load up to 5,400 Pa (test load 8,100 Pa)
Qualifications	IEC 61215 IEC 61730 IEC 61701 IEC 62804 National technical approval (AbZ)

Electrical data (STC)

STC (Standard Test Conditions): Irradiation intensity 1,000 W/m², spectral distribution AM 1,5 | Temperature 25 ± 2 °C, in accordance to EN 60904-3

Nominal power P_{max}	165 Wp	170 Wp
Nominal voltage V_{MP}	19,2 V	19,3 V
Nominal current I_{MP}	8,71 A	8,90 A
Open circuit voltage V_{OC}	23,4 V	23,6 V
Short circuit current I_{SC}	9,02 A	9,17 A
Module efficiency	15,2 %	15,7 %

Measurement tolerances: P_{max} ± 5 %; V_{OC} ± 10 %; I_{SC} ± 10 %, I_{MP} ± 10 %

Reverse-current power rating I_r : 20 A, operating modules with an external power source is only permissible if using a phase fuse with a tripping current of ≤ 20 A.

Electrical data (NMOT and weak light)

NMOT (Nominal Module Operation Temperature): Irradiation intensity 800 W/m², spectral distribution AM 1,5, Temperature 20 °C

Weak light conditions: Irradiation intensity 200 W/m², Temperature 25 °C, Wind speed 1 m/s, load operation

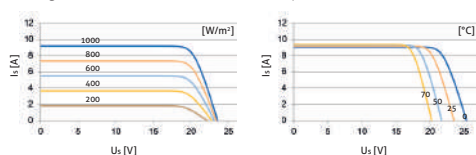
Nominal power $P_{max @ NMOT}$	122 W	126 W
Nominal power $P_{max @ 200 W/m^2}$	32,8 W	33,7 W

Measurement tolerances: P_{max} ± 5 %; V_{OC} ± 10 %; I_{SC} ± 10 %, I_{MP} ± 10 %

Reduction of module efficiency when irradiance is reduced from 1000 W/m² to 200 W/m² (at 25 °C): 4 ± 2 % (relative) / -0,6 ± 0,3 % (absolute).

Characteristic lines (Performance Class 170 Wp)

Voltage characteristic line at different temperatures and irradiances



Thermal Features

Operating temperature range	-40 ... +85 °C
Ambient temperature range	-40 ... +45 °C
Temperature coefficient P_{max}	-0,41 %/K
Temperature coefficient V_{OC}	-0,31 %/K
Temperature coefficient I_{SC}	0,05 %/K
NMOT	44 °C

Appendix Q :

First Arduino Board Program

Appendix

```
int x=0;
int a=0;
void setup() {
  // put your setup code here, to run once:
  pinMode(11,INPUT_PULLUP);
  pinMode(A0,INPUT);
  pinMode(12,OUTPUT);
  pinMode(13,OUTPUT);
  pinMode(2,INPUT_PULLUP);
  attachInterrupt(digitalPinToInterrupt(2), biip, LOW);
  Serial.begin(9600);
}

void loop() {
  if (digitalRead(2)){
    noTone(9);
  }
  x=digitalRead(11);
  a=analogRead(A0);
  if(x==0){
    dbslash();
  }
  else if(a<50){
    Right();
  }
  else if(a>250){
    Left();
  }
  else{
    nothing();
  }
}

void biip(){
  tone(9, 300);
}

void dbslash(){
  digitalWrite(12,HIGH);
  digitalWrite(13,HIGH);
  tone(8, 1000,2);
  delay(500);
  digitalWrite(12,LOW);
  digitalWrite(13,LOW);
  tone(8, 1000,2);
  delay(500);
  noTone(8);
}

void Right(){
  digitalWrite(12,HIGH);
  tone(8, 1000,2);
  delay(500);
  digitalWrite(12,LOW);
  tone(8, 1000,2);
  delay(500);
  noTone(8);
}

void Left(){
  digitalWrite(13,HIGH);
  tone(8, 1000,2);
  delay(500);
  digitalWrite(13,LOW);
  tone(8, 1000,2);
  delay(500);
  noTone(8);
}

void nothing(){
  digitalWrite(12,LOW);
  digitalWrite(13,LOW);
}
```

Appendix R :

Second Arduino Board Program

Appendix

```
#include <LiquidCrystal.h>

LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
int c;
int trig = 8;
int echo = 9;
long lecture_echo;
long cm;
byte R00[8] = {
  B11111,
  B11111,
  B11111,
  B11111,
  B11111,
  B11111,
  B11111,
  B11111,
};
byte R11[8] = {
  B11111,
  B11110,
  B11000,
  B10001,
  B10011,
  B00110,
  B00110,
  B01100,
};
byte R12[8] = {
  B11111,
  B00000,
  B01100,
  B11111,
  B10000,
  B00000,
  B00101,
  B01111,
};
byte R13[8] = {
  B11111,
  B11111,
  B00111,
  B00011,
  B00011,
  B10001,
  B10001,
  B00001,
};
byte R21[8] = {
  B01100,
  B00110,
  B00110,
  B10011,
  B10001,
  B11000,
  B11110,
  B11111,
};
byte R22[8] = {
  B00110,
  B00101,
  B00000,
  B10000,
  B11111,
  B01100,
  B00000,
  B11111,
};
byte R23[8] = {
  B00001,
  B00001,
  B00001,
  B10011,
  B00011,
  B00111,
  B11111,
  B11111,
};
byte f1[8] = {
  B00100,
  B01110,
  B11111,
  B01110,
  B01110,
  B01110,
  B01110,
  B01110,
};
```

```

    B01110,
    B01110,
};

void setup() {
    pinMode(trig, OUTPUT);
    pinMode(c, INPUT);
    digitalWrite(trig, LOW);
    pinMode(echo, INPUT);
    lcd.createChar(0, R11);
    lcd.createChar(1, R12);
    lcd.createChar(2, R13);
    lcd.createChar(3, R21);
    lcd.createChar(4, R22);
    lcd.createChar(5, R23);
    lcd.createChar(6, R00);
    lcd.createChar(7, f1);
    lcd.begin(16, 2);
    lcd.setCursor(6, 0);
    lcd.write(byte(6));
    lcd.write(byte(0));
    lcd.write(byte(1));
    lcd.write(byte(2));
    lcd.write(byte(6));
    lcd.setCursor(6, 2);
    lcd.write(byte(6));
    lcd.write(byte(3));
    lcd.write(byte(4));
    lcd.write(byte(5));
    lcd.write(byte(6));
    delay(2000);
    lcd.clear();

    lcd.setCursor(5, 0);
    lcd.write(byte(6));
    lcd.write(byte(0));
    lcd.write(byte(1));
    lcd.write(byte(2));
    lcd.write(byte(6));
    lcd.setCursor(5, 2);
    lcd.write(byte(6));

    lcd.write(byte(3));
    lcd.write(byte(4));
    lcd.write(byte(5));
    lcd.write(byte(6));
    delay(20);
    lcd.clear();

    lcd.setCursor(4, 0);
    lcd.write(byte(6));
    lcd.write(byte(0));
    lcd.write(byte(1));
    lcd.write(byte(2));
    lcd.write(byte(6));
    lcd.setCursor(4, 2);
    lcd.write(byte(6));
    lcd.write(byte(3));
    lcd.write(byte(4));
    lcd.write(byte(5));
    lcd.write(byte(6));
    delay(20);
    lcd.clear();

    lcd.setCursor(3, 0);
    lcd.write(byte(6));
    lcd.write(byte(0));
    lcd.write(byte(1));
    lcd.write(byte(2));
    lcd.write(byte(6));
    lcd.setCursor(6, 2);
    lcd.write(byte(6));
    lcd.write(byte(3));
    lcd.write(byte(4));
    lcd.write(byte(5));
    lcd.write(byte(6));
    delay(20);
    lcd.clear();

    lcd.setCursor(2, 0);
    lcd.write(byte(6));
    lcd.write(byte(0));
    lcd.write(byte(1));
    lcd.write(byte(2));

```

```
lcd.write(byte(6));
lcd.setCursor(5, 2);
lcd.write(byte(6));
lcd.write(byte(3));
lcd.write(byte(4));
lcd.write(byte(5));
lcd.write(byte(6));
delay(20);
lcd.clear();

lcd.setCursor(1, 0);
lcd.write(byte(6));
lcd.write(byte(0));
lcd.write(byte(1));
lcd.write(byte(2));
lcd.write(byte(6));
lcd.setCursor(1, 2);
lcd.write(byte(6));
lcd.write(byte(3));
lcd.write(byte(4));
lcd.write(byte(5));
lcd.write(byte(6));
delay(20);
lcd.clear();

lcd.setCursor(0, 0);
lcd.write(byte(6));
lcd.write(byte(0));
lcd.write(byte(1));
lcd.write(byte(2));
lcd.write(byte(6));
lcd.write(" Air");
lcd.setCursor(0, 2);
lcd.write(byte(6));
lcd.write(byte(3));
lcd.write(byte(4));
lcd.write(byte(5));
lcd.write(byte(6));
lcd.write(" Nomad");
lcd.setCursor(12, 0);
delay(1000);
lcd.setCursor(7, 0);

lcd.write("Air");
lcd.setCursor(6, 1);
lcd.write("Nomad");
delay(500);
lcd.setCursor(7, 0);
lcd.write(" ");
lcd.setCursor(6, 1);
lcd.write(" ");
delay(500);
lcd.setCursor(7, 0);
lcd.write("Air");
lcd.setCursor(6, 1);
lcd.write("Nomad");
delay(500);
lcd.setCursor(7, 0);
lcd.write(" ");
lcd.setCursor(6, 1);
lcd.write(" ");
delay(500);
lcd.setCursor(7, 0);
lcd.write("Air");
lcd.setCursor(6, 1);
lcd.write("Nomad");
delay(500);
lcd.setCursor(7, 0);
lcd.write(" ");
lcd.setCursor(6, 1);
lcd.write(" ");
delay(500);
lcd.clear();
lcd.setCursor(5, 0);
lcd.write("System");
```

Appendix

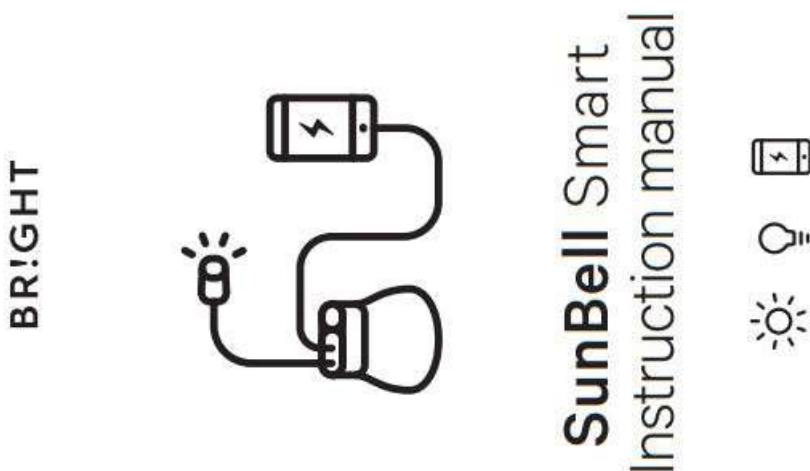
```
lcd.setCursor(7, 1);
lcd.write("On");
delay(1000);
lcd.clear();
}

void loop() {
  c=digitalRead(1);
  if(c)
  {
    digitalWrite(trig, HIGH);
    delayMicroseconds(10);
    digitalWrite(trig, LOW);
    lecture_echo = pulseIn(echo,HIGH);
    cm = lecture_echo /58;
    while(cm<5)
    {tone(13, 1000,250);
    lcd.setCursor (0,0);
    lcd.print ("Driving Backward");
    lcd.setCursor (3,1);
    lcd.print (cm);
    lcd.setCursor (14,1);
    lcd.print ("CM");
    delay(500);
    cm = lecture_echo /58;}
    noTone(13);
    lcd.setCursor (0,0);
    lcd.print ("Driving Backward");
    lcd.setCursor (0,1);
    lcd.print ("Distance :");
    lcd.setCursor (11,1);
    lcd.print (cm);
    lcd.setCursor (14,1);
    lcd.print ("CM");
    delay(500);
    lcd.clear();}
  else{
    lcd.setCursor(0,0);
    lcd.print ("Driving Forward");
    lcd.setCursor (3,1);
    lcd.print ("Drive Safe");
    lcd.setCursor(0,1);

    lcd.write(byte(7));
    lcd.setCursor(15,1);
    lcd.write(byte(7));
    delay (2000);
    lcd.clear();}
}
```

Appendix S :

Second Arduino Board Program



10. Technical data

Battery
7.5 Wh Li-ion NMC battery (2,200 mAh at 3.7 V)
4+ year lifespan (1,500 charge cycles)

The battery is made to last 1500 cycles or 4+ years and must be replaced by a qualified agent.

The light source of this luminaire is not replaceable; when the light source reaches its end of life the whole luminaire shall be replaced.

Materials
ABS, PC, PP, Steel, Aluminium

LED type
Samsung

11. Warranty

Limited Product Warranty for SunBell Smart

BRIGHT PRODUCTS AS warrants that the product conforms to the technical specifications set out in the Factory in a timely working condition, free of visible defects. BRIGHT PRODUCTS AS warrants the product to be free from defects in material and workmanship, under normal use and operation, for a period of two (2) years after the date of sale/handover to consumer.

"The product" in this warranty is defined as the "SunBell Smart" solar lamp, consisting of the SunBell LED lamp, the SunBell battery unit, the SunBell LED lamp & battery unit, the SunBell Solar panel with cable.

The warranty is limited to the free replacement of the defective main part(s) or replacement of the entire product during the two years of warranty. Defective parts/products replaced under this warranty shall become the property of BRIGHT PRODUCTS AS.

The above stated Warranty is subject to the following conditions:

Products are sold by authorized BRIGHT PRODUCTS AS representatives. The products are sold by authorized BRIGHT PRODUCTS AS representatives with BRIGHT PRODUCTS operating instructions, maintenance instructions and specifications. Use of only BRIGHT PRODUCTS AS authorized parts and accessories. The warranty shall not apply to damages resulting from normal wear and tear of product use, nor shall it apply to alterations or misuse by the purchaser. Parts of the product, which would be generally considered as expendable service items during normal use, such as cardboard box/packaging items, instruction manual etc., are not covered by the warranty.

The USB mobile phone charger is made to support most commonly available mobile phones based on their current/known specifications for charging current and voltage. The warranty does not cover replacement of the entire product or the USB mobile phone charger if the mobile phone used has specifications that does not comply with the charging current or specifications for the SunBell and the USB charger.

How to make a warranty claim:

Warranty claims must be made directly to the point of purchase with a valid proof of purchase. If the warranty claim is found to be valid upon point of inspection, a replacement product or part will be handed out or ordered for hand out at a later date. If there is doubt about the validity of the warranty claim, the product may be kept by the seller to be inspected by BRIGHT PRODUCTS AS. BRIGHT PRODUCTS AS reserves the right to request the return of the product for inspection and testing. If needed, we recommend that you ship your returned products to point of purchase via a trackable shipment method.

3. Light modes

Full battery run time

1 st press	2 nd press	3 rd press
120 h+	30 h+	7 h+
5 lm	20 lm	120 lm

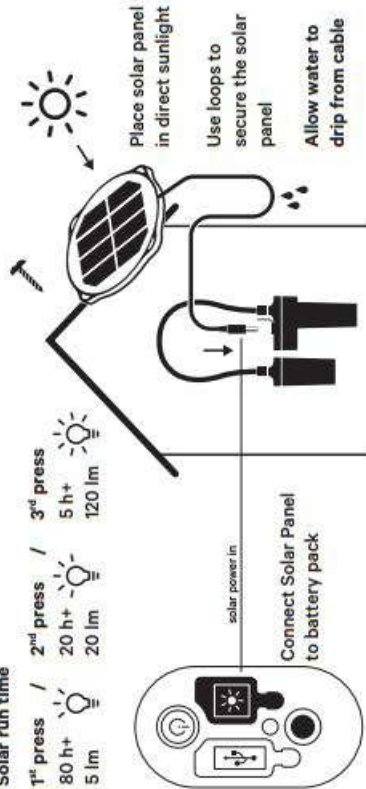
Solar run time

1 st press	2 nd press	3 rd press
80 h+	20 h+	5 h+
5 lm	20 lm	120 lm

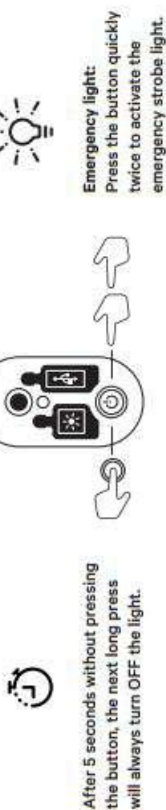
4. Charge it

10 h+	12 h+	15 h+

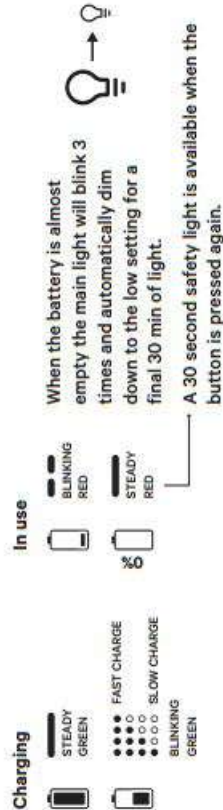
Depending on sun conditions



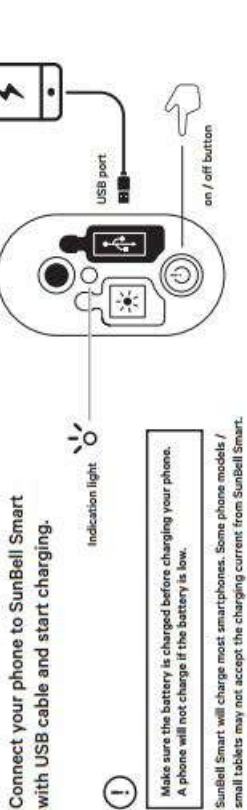
5. Additional light functions



6. Battery indication



7. Phone charging

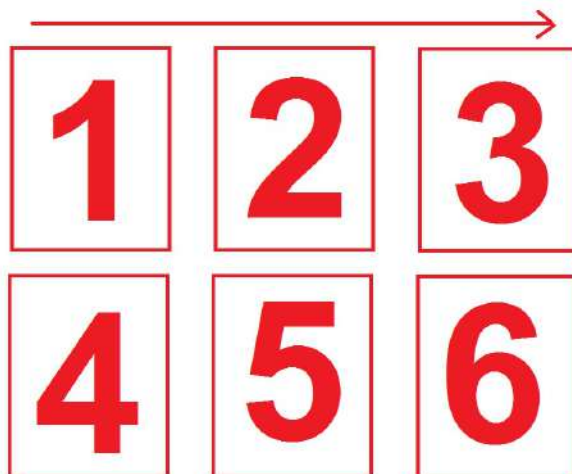


Appendix T :

Accountability

Appendix

Instructions :Due to its huge size, the table is divide into 6 parts. The parts are read following the rows which means that page 1, 2 and 3 represent the extension of the same row followed by 4, 5 and 6 which contain the end rows of the table.



Appendix

Budget = 2000,000			
List of Materials (EN/F			
Date	N° bill	Ref (according to bill)	Unit Price incl tax (TCC)
12/8/2020	F°21703	disque tranconner acier2	3.178
1/6/2020	F°	reguleteur de presseur ENC ER2000-025	62.000
1/6/2020	F°00415	rouleau teflon PTFE	0.546
1/6/2020	F°00415	vanne spherique passage normale 1/2	7.529
1/6/2020	F°00415	reduction collerette laiton 1/2	1.509
1/6/2020	F°00415	mamelon reduit CHR 1/2	5.146
1/6/2020	F°00415	buse coupleur rapide 1/2	4.226
12/13/2020	F°	disque tranconner acier2	3.515
12/18/2020	F°22398	coupleur rapide 3/8	9.200
12/18/2020	F°22398	buse coupleur rapide 3/8	3.910
12/18/2020	F°22398	reduction special laiton 3/8	4.912
1/2/2021	F°00108	boulon T.HEXAG	0.171
1/2/2021	F°00108	rondelle 08*27	0.060
1/4/2021	F°24	tube RECT 40*20*1,5	23.499
1/5/2021	F°00362	meche HSS pointe q 12MM	5.848
1/5/2021	F°00362	foret rectifie au cobalt HSS	10.765
1/5/2021	F°00362	boulon T.HEXAG	0.803
1/5/2021	F°00362	rondelle 12*27	0.109
1/5/2021	F°00362	rondelle 08*27	0.109
1/6/2021	F°415	rouleau teflon PTFE 12MM	0.643
1/6/2021	F°415	vanne spherique passage normale egale 1/2 FEM	9.295
1/6/2021	F°415	ecrou tetine laiton droit 1/2 FEM	6.896
1/6/2021	F°415	joint caoutchouc 15*21	0.148
1/6/2021	F°415	raccord rapide droit 1/4	4.587
1/6/2021	F°415	raccord rapide droit 3/8	6.340
1/6/2021	F°415	raccord rapide droit 1/2	6.293
1/6/2021	F°415	coupleur rapide 3/8 MAL	11.359
1/6/2021	F°415	manchon CHR egale 3/8 FEM	5.639
1/6/2021	F°415	buse coupleur rapide DN 08	3.752
1/6/2021	F°	roulement UCF	25.000
1/11/2021	F°00743	collier serre cable plastique	0.302
1/12/2021	F°00790	graisse 100G TUBE MULTI	2.501
1/19/2021	F°1025	vanne spherique passage normale egale 1/2	9.295
1/19/2021	F°1025	mamelon laiton froid reduit 5/8	19.992
1/19/2021	F°1025	reduction collerette laiton 1/2_3/8	1.862
2/24/2021	F°3116	peiture jaune	10.597
2/24/2021	F°3116	peiture bleu fonce	9.421
3/11/2021	F°04111	peiture bleu fonce	9.197
3/11/2021	F°04111	peiture jaune	10.397
3/11/2021	F°04111	peiture noir brillant	9.256
N/A	F°	Crank	19.500
N/A	F°	Chain	6.700
N/A	N/A	Speed Change	6.750
N/A	N/A	Low cost paint	4.500
N/A	N/A	U Form Iron	4.000
N/A	N/A	Damper	15.000
N/A	N/A	Wheel front	35.000

Appendix

R)			
Price excl tax (HT)	VAT(TVA)	Quantity	Amount incl tax (TTC)
2.671	0.604	1	3.178
52.101	11.780	1	62.000
0.459	0.104	2	1.092
6.327	1.431	1	7.529
1.268	0.287	1	1.509
4.324	0.978	1	5.146
3.551	0.803	1	4.226
2.954	0.668	2	7.030
7.731	1.748	1	9.200
3.286	0.743	1	3.910
4.128	0.933	1	4.912
0.144	0.032	5	0.855
0.050	0.011	5	0.300
19.747	4.465	1	23.499
4.914	1.111	1	5.848
9.046	2.045	1	10.765
0.675	0.153	6	4.818
0.092	0.021	12	1.308
0.092	0.021	8	0.872
0.540	0.122	2	1.286
7.811	1.766	1	9.295
5.795	1.310	1	6.896
0.124	0.028	1	0.148
3.855	0.872	2	9.174
5.328	1.205	2	12.680
5.288	1.196	2	12.586
9.545	2.158	2	22.718
4.739	1.071	1	5.639
3.153	0.713	1	3.752
21.008	4.750	2	50.000
0.254	0.057	20	6.040
2.102	0.475	2	5.002
7.811	1.766	1	9.295
16.800	3.798	1	19.992
1.565	0.354	1	1.862
8.905	2.013	1	10.597
7.917	1.790	2	18.842
7.729	1.747	2	18.394
8.737	1.975	1	10.397
7.778	1.759	1	9.256
16.387	3.705	1	19.500
5.630	1.273	2	13.400
-	-	2	13.500
-	-	8	36.000
-	-	5	20.000
-	-	4	60.000
-	-	2	70.000

Appendix

[illegible]

Appendix

N/A	N/A	wheel rear	90.000
N/A	N/A	driving wheel	20.000
N/A	N/A	Alucoubond 1*1,5	140.000
N/A	N/A	hinge joint door	5.000
N/A	N/A	Lock	10.000
N/A	N/A	crank	15.000
N/A	N/A	irond rod full	20.000
N/A	N/A	iron (scrap)	-
N/A	N/A	universal joint	20.000
N/A	N/A	iron sheet	49.000
N/A	N/A	car mirror	20.000
N/A	N/A	brake system	30.000
N/A	N/A	welding electrodes	20.000
N/A	N/A	screw	-
N/A	N/A	cables	0.302
N/A	N/A	Rack	50.000
N/A	N/A	directional valve	50.000
N/A	N/A	tank	80.000
N/A	N/A	flow control Valve	25.000
N/A	N/A	Shutt off valve	5.000
N/A	N/A	fitting	2.000
N/A	N/A	cylinder	80.000
N/A	N/A	cylinder	15.000
N/A	N/A	cylinder	60.000
N/A	N/A	Renting Equipment	-
N/A	N/A	Labour	-
Total =			

Appendix

-	-	1	90.000
-	-	1	20.000
-	-	1	140.000
-	-	24	120.000
-	-	2	20.000
-	-	1	15.000
-	-	1	20.000
-	-	-	150.000
-	-	1	20.000
-	-	2	98.000
-	-	1	20.000
-	-	1	30.000
-	-	3	60.000
-	-	-	6.000
-	-	20	6.040
-	-	1	50.000
-	-	2	100.000
-	-	1	80.000
-	-	1	25.000
-	-	1	5.000
-	-	5	10.000
-	-	1	80.000
-	-	4	60.000
-	-	1	60.000
-	-	-	60.000
-	-	-	120.000
			2099.288

Appendix

200.000	200.000	Depends on Type : Amazon
50.000	50.000	Depends on Type : Amazon
	140.000	
	120.000	
	20.000	
60.000	60.000	Alibaba
	20.000	
	150.000	
100.000	100.000	Alibaba
	98.000	
50.000	50.000	Auto Shop
150.000	150.000	Auto Shop
	60.000	
	6.000	
	6.040	
200.000	200.000	Alibaba
300.000	300.000	Amazon
100.000	100.000	Repair Shop
62.000	62.000	Pneumarics Shop
7.500	7.500	Hardware Store
4.500	22.500	Hardware Store
145.000	145.000	Pneumatic Hardware providers Online
821.920	821.920	Pneumatic Hardware providers Online
220.000	220.000	Pneumatic Hardware providers Online
	840.000	3 months work 10TND per day
	1350.000	3 months work minimum wage 450TND
Estimated Budget =	6283.208	Note : This is only an estimation